

IDEA for Teachers: Tutorial

Kolibri, Nextcloud, Wikipedia offline, Files Disk and Backup Disk

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Part A: For teachers

1. What IDEA is

IDEA (Initiative for Digital Education in Africa) brings offline computing to rural schools. A small computer in the school, the **IDEA box**, holds ready-made lessons in Kolibri, a place for your own files in Nextcloud, and the whole of Wikipedia. Teachers and learners open them from any classroom computer, laptop or phone on the school Wi-Fi, without internet. This tutorial shows you, step by step, how to use them in your lessons.

> **No internet needed.** Everything in this guide works inside the school. Nothing you do is sent to the internet.

1.1 Words used in this guide

Word	Meaning
IDEA box	The small Raspberry Pi computer in the school that runs the apps. Each box has its own name, on its sticker. The kit's three boxes are **sunny-robin**, **happy-zebra** and **kind-panda**. (The screenshots come from a demo box with another name.)
Console	The web page that lists the apps and opens them.
Demo apps	**Kolibri Demo**, **Nextcloud Demo** and **Wikipedia (Demo)**: practice copies with example content, on every IDEA box. Use them to follow this tutorial (1.3).
IDEA shortcut	The one shortcut on a classroom computer's desktop. It finds the IDEA box and opens the Console (chapter 2).
System Disk	The IDEA box's own disk. The apps live on it.
IDEA Disk	A USB disk or stick that the Console has prepared. It can become a **Files Disk** or a **Backup Disk**.
Spare disk	An IDEA Disk that is still empty.
Files Disk	An IDEA Disk for your own files. It appears as the folder **School Files** in Nextcloud (chapter 8).
Backup Disk	An IDEA Disk that keeps copies of the apps' data (chapter 9).
Operator	The person at the school who looks after the IDEA box (Part B).

1.2 The apps

App	What you use it for	Chapters
Kolibri	Ready-made lessons, videos and exercises (on the kit, for example CK-12, PhET, OpenStax and African Storybook). You make classes, lessons and quizzes and see each learner's progress.	3, 4

App	What you use it for	Chapters
Nextcloud	Your own files: share worksheets with a class, write documents together in **Nextcloud Office**, collect homework.	5, 6, 8
Wikipedia	The encyclopedia, offline: the full English Wikipedia, and a Simple English one in the Demo.	7

1.3 How to use this guide

- **Part A (chapters 1 to 9)** is for teachers.
- **Part B (chapters 10 to 12)** is for the school operator.
- Each chapter has numbered steps with screenshots. The screenshots come from the IDEA demo box and use the demo school **IDEA Demo School** with class **Grade 5** and the demo accounts in the appendix (`admin`, `teacher1`, `teacher2`, `learner1` to `learner3`; teachers and the admin have a password: the account name followed by `911!`; learners sign in in Kolibri with only their username). Your school's names will be different.
- **Practise in the Demo apps.** Every IDEA box of the kit has six apps: the school's own **Kolibri**, **Nextcloud** and **Wikipedia**, and a Demo copy of each. Follow this tutorial in **Kolibri Demo**, **Nextcloud Demo** and **Wikipedia (Demo)**: they hold the demo school, the demo accounts and the example content. The school's own apps stay clean for real lessons.

2. Getting connected and opening an app

All apps open from one web page: the **Console**. How you get to the Console depends on the computer you use.

2.1 Connect to the Wi-Fi

The classroom computers are already connected. Do this only on your own laptop, tablet or phone.

The IDEA router at your school makes two Wi-Fi networks. Their names and first passwords are the same at every IDEA school:

Network	Password (until your school changes it)	Who joins it	What it reaches
IDEA	`idea911!`	Students, and the classroom computers	Only the IDEA box: the Console and its apps. No internet.
IDEA-Teachers	`idea-teachers911!`	Teachers only	The IDEA box, and the internet. Internet is **on**; your IDEA contact can switch it off when the head teacher asks.

- Give students only the **IDEA** password. Keep the **IDEA-Teachers** password among the teachers.
- The classroom computers stay on **IDEA**. For a teacher's classroom computer on **IDEA-Teachers**, ask your IDEA contact.

To connect:

1. Open the Wi-Fi settings on your device.
2. Choose **IDEA-Teachers** (teachers) or **IDEA** (students).
3. Type the password and join.

2.2 Open the Console

There are two ways to open the Console. Use the one that fits the device in front of you.

2.2.1 On a classroom computer

The classroom computers have **one** shortcut on the desktop, named **IDEA**. There are no separate icons for Kolibri, Nextcloud or the other apps.

1. Double-click **IDEA** on the desktop. You can also find it in the app menu. It opens in Chrome.

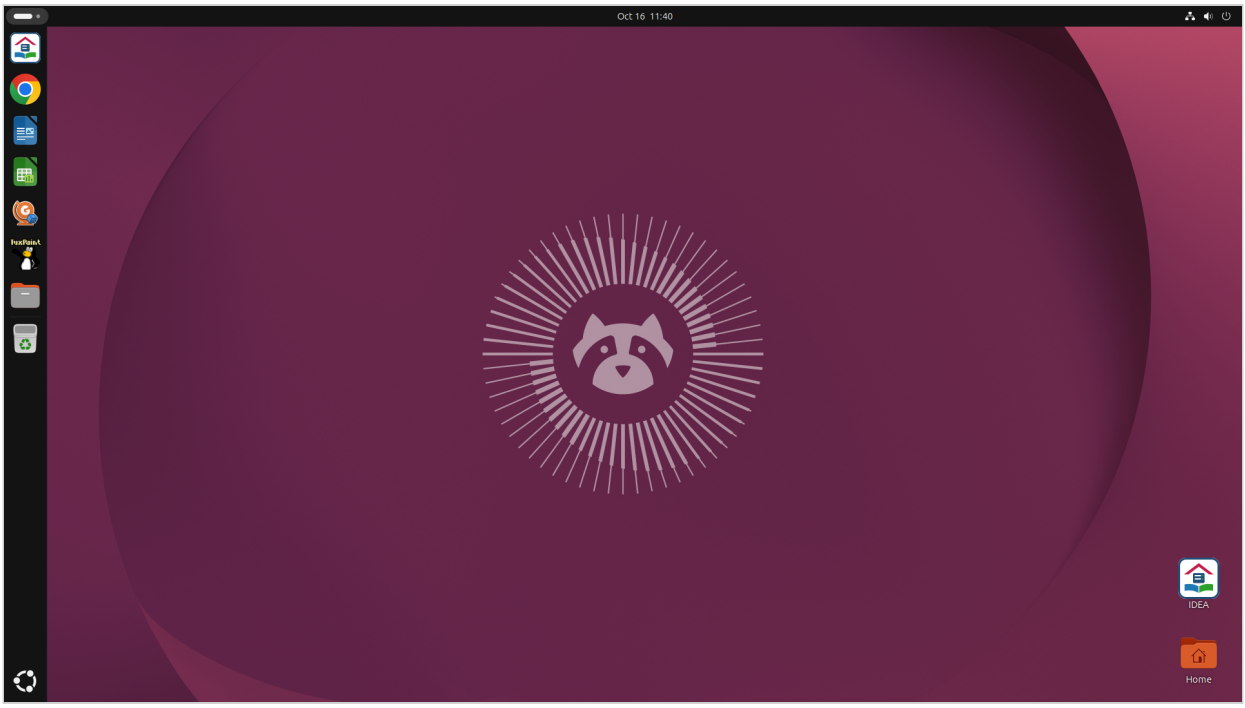


Figure 1. The classroom computer desktop with the one IDEA shortcut

2. A start page titled **IDEA** appears for a moment and says **Looking for the school server…**.

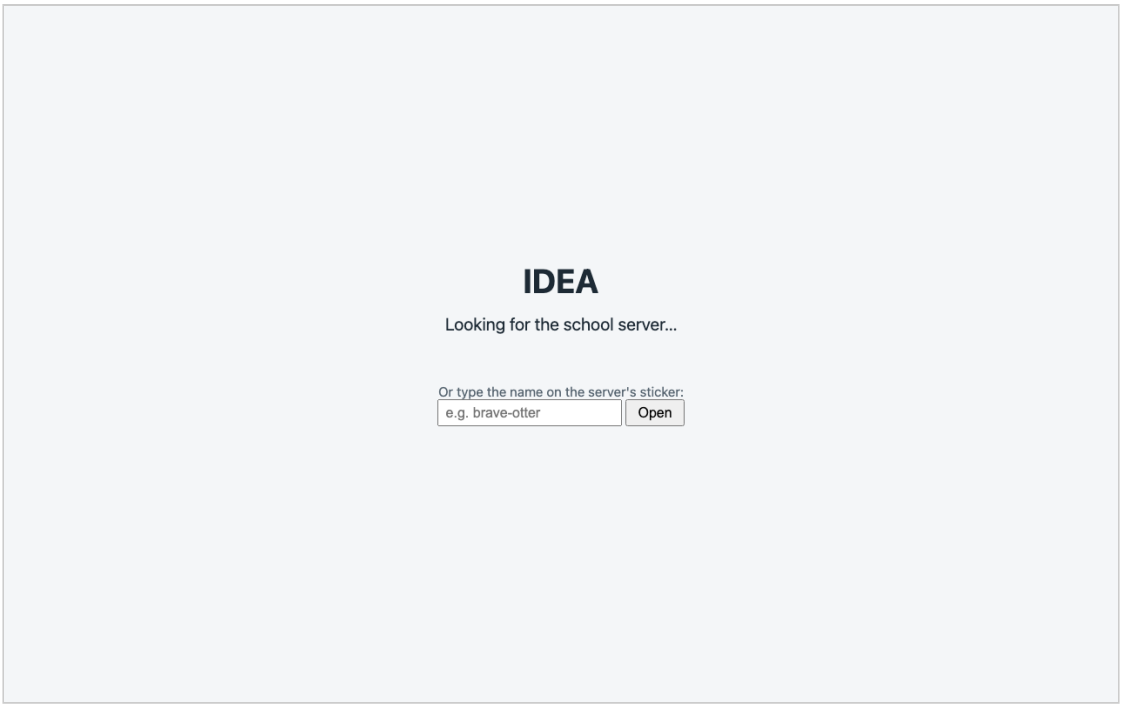


Figure 2. The IDEA start page: Looking for the school server…

3. The Console opens by itself.

If the start page says **No school server was found**, check that the computer is connected to the school network. Then type the first line of the box's sticker, for example `brave-otter`, in the field **Or type the name on the server's sticker**. Use small letters, with `-` between the words. Press **Open**. The start page remembers that IDEA box.

2.2.2 On your own laptop, tablet or phone

1. Connect to the Wi-Fi (2.1).
2. Open the browser.
3. Type `http://idea.local` in the address bar and press Enter.
4. The Console opens.

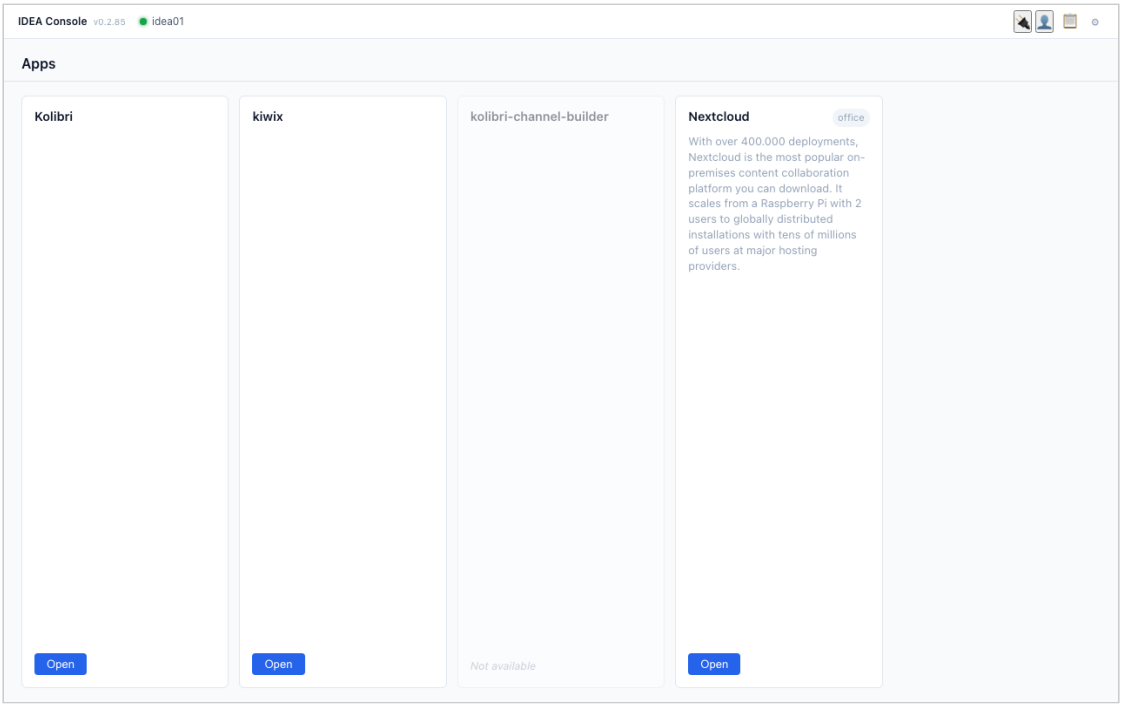


Figure 3. The Console in the browser

5. Bookmark it, so you can open it again with one click. In Chrome, press **Ctrl+D** (on a Mac ⌘+**D**), then **Done**.

If the Console does not open, type the address on the second line of the box's sticker instead, for example `http://brave-otter.local`.

> **Two IDEA boxes at your school?** The Console shows the apps of all of them. Each card says on which box the app runs, for example **on kind-panda**.

2.3 First steps in the Console

1. The Console opens on the **Apps** page: one card per app, each with a short description and a blue **Open** button. The green dot at the top shows the Console is connected to the IDEA box.

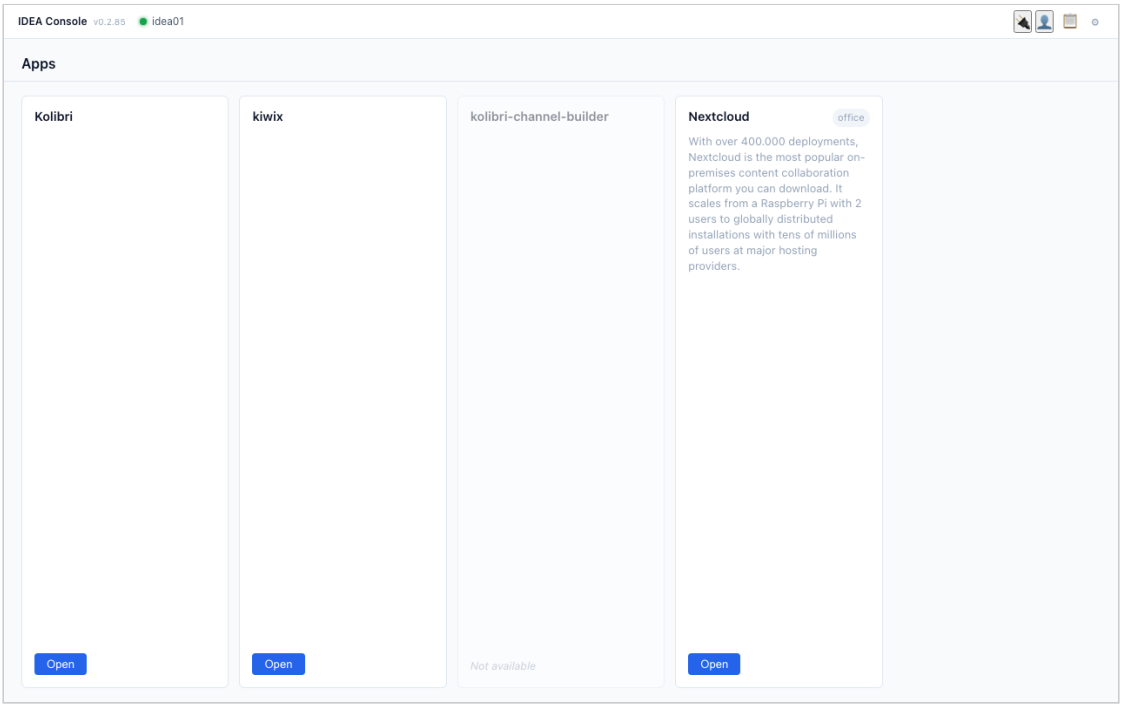


Figure 4. Console Apps page (two of the cards)

2. Click the blue **Open** button on the card of the app you need. The app opens in a new tab. Here it is Kolibri's sign-in page.

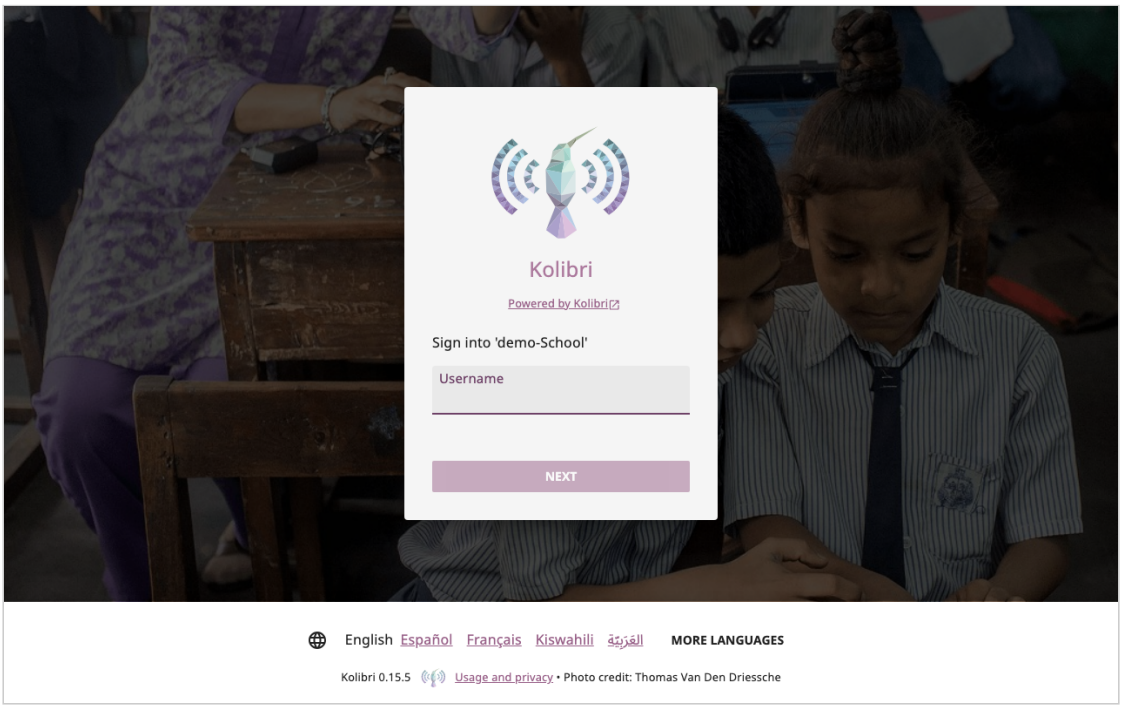


Figure 5. Kolibri opened from the Console

Good to know:

- A card that is greyed out and says **Not available** is not running. Do not try to fix it: tell your school operator.
- **Known problem: apps that don't open.** If an IDEA box is switched off, or was once connected to yours and is now at another school, the Console still shows its apps as if they worked. **Open** then gives a page that doesn't load. There is no fix yet. Use the apps that are **on** your own school's IDEA box (the name on its sticker).
- **The classroom computers reset at every restart.** Anything saved on the computer itself is gone after a restart. Save work in Nextcloud (chapter 6).
- Do not type numbers (addresses or ports) into the address bar. Always start from the Console and use **Open**.
- A student computer that no longer works properly is not repaired: the operator resets its card (chapter 12). Students lose only what they saved on the computer itself, so let them save their work in Nextcloud (chapter 6).
- The person icon at the top right (**Operator Login**) is for the operator (Part B). Teachers do not need it.

> **Tip.** You can bookmark the page that opens. If the bookmark stops working, go back to the Console and use **Open** again.

3. Kolibri for teachers

About the examples in this chapter. The screens in this chapter come from the IDEA demo setup. That setup is switched on (activated) on the demo IDEA box. It creates a Kolibri school named **IDEA Demo School**, with the class **Grade 5** and the demo accounts. The steps work the same at your school, but your school, class and account names will be different. On the kit, this demo setup is the app **Kolibri Demo**: open the **Kolibri Demo** card in the Console, not the **Kolibri** card. The demo accounts are listed in Appendix: demo accounts at the end of this tutorial.

In the demo, you sign in as `teacher1` (or `teacher2`). The demo school has three channels from the IDEA-DEMO disk: **Maths**, **Grade 5 Science** and **Early Reading**. The demo lesson uses **Maths > Fractions**. The quiz in 3.5 was made on a demo box with **Khan Academy (English - US curriculum) > Math by grade > 5th grade > Add and subtract fractions**.

3.1 Signing in

Open **Kolibri Demo** from the Console (2.3). Type your username and click **NEXT**, then type your password.

3.2 Create a class

Creating classes needs a Kolibri admin account (in the demo school, `admin`, password `admin911!`; `teacher1` is the coach of Grade 5).

1. Open **Facility > CLASSES**. You see the list of classes.

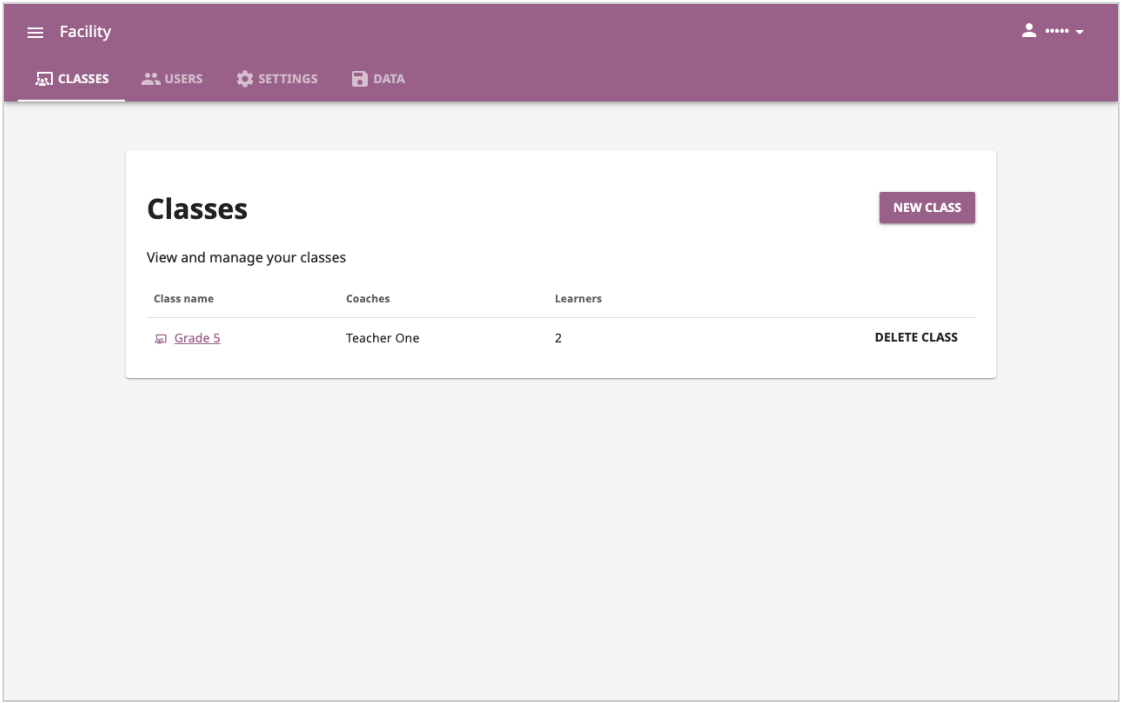


Figure 6. Classes list

2. Click **NEW CLASS**, type the class name, click **SAVE**. (In the demo we pressed **CANCEL** because `Grade 5` already exists.)

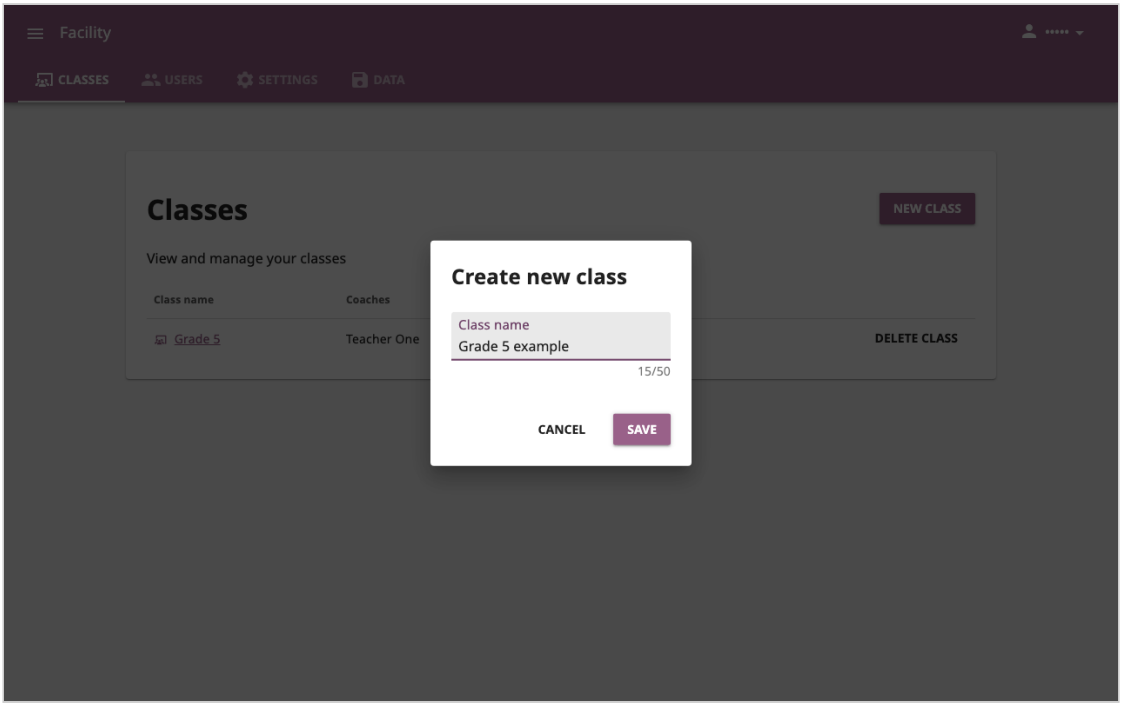


Figure 7. Create new class

3.3 Enroll learners

1. Click the class name. On the class page (with **Coaches** and **Learners**), click **ENROLL LEARNERS**. The **Enroll learners** page opens.

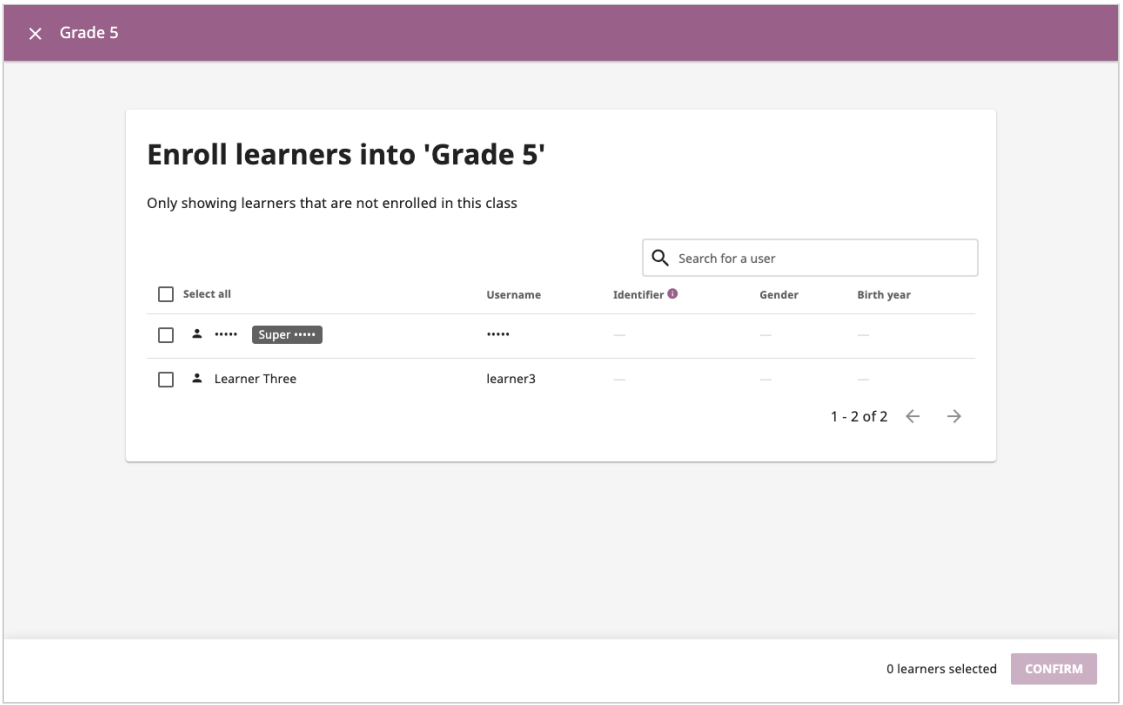


Figure 8. Enroll learners page

2. The list only shows learners who are not yet in the class. Tick each learner (here: Learner Three), then click **CONFIRM**.

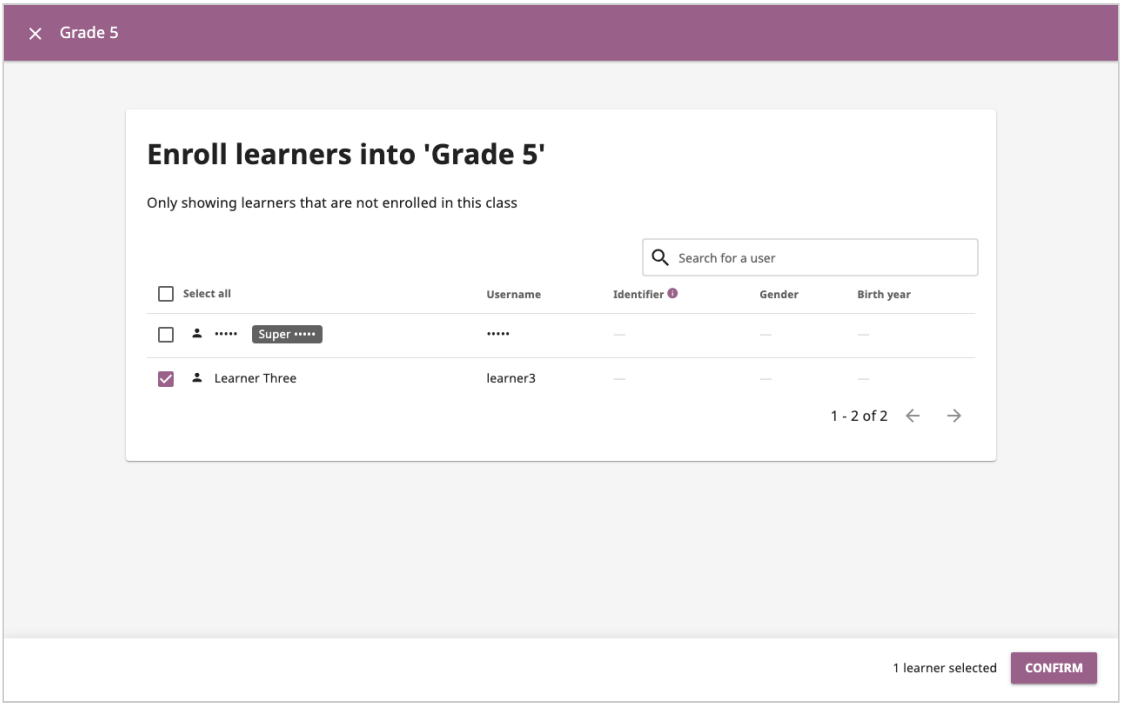


Figure 9. Learner ticked

3. The learner now appears under Learners.

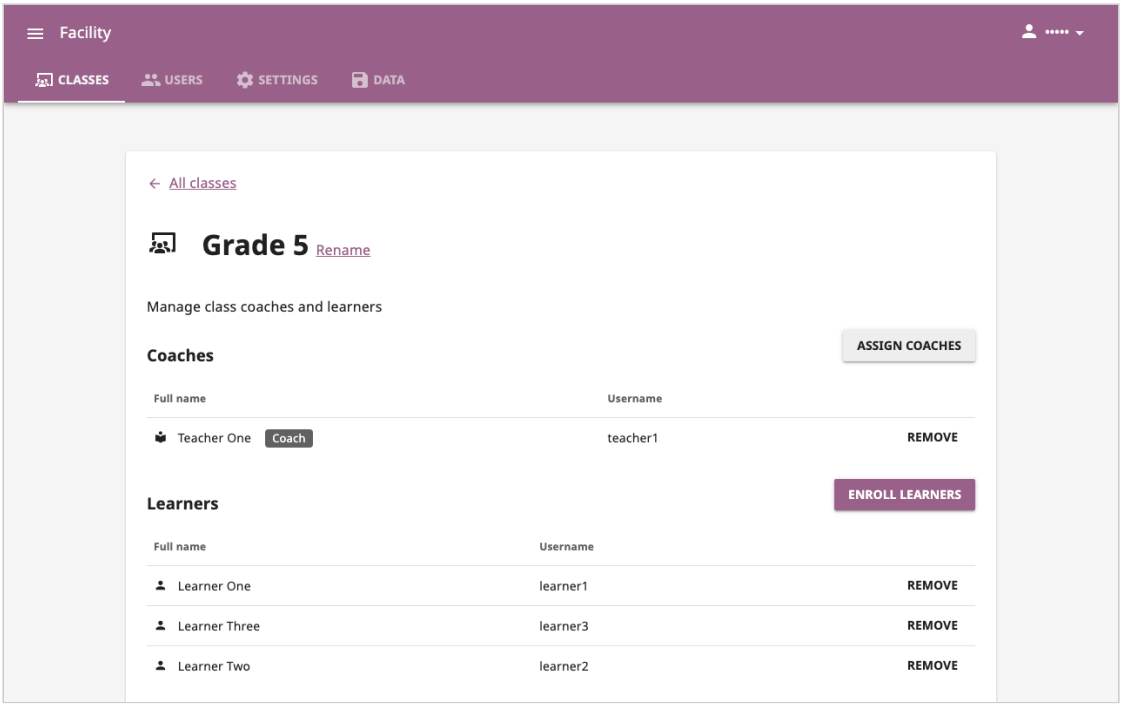


Figure 10. Learner enrolled

3.4 Make a lesson and show it to the class

A good lesson mixes resources that explain (videos, activities) with resources that let learners practise (exercises).

1. Open **Coach**, choose the class, click **PLAN > LESSONS > NEW LESSON**.

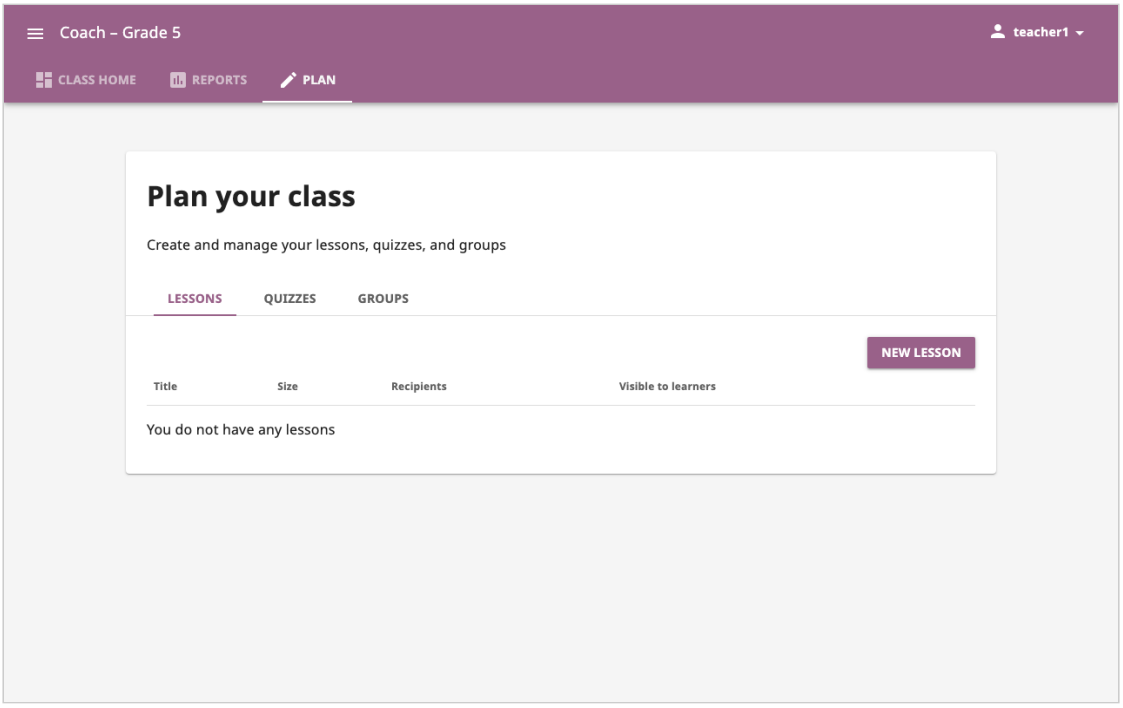


Figure 11. Plan your class

2. Type a title (`Fractions lesson`). Under **Recipients** keep **Entire class**. Click **SAVE CHANGES**.

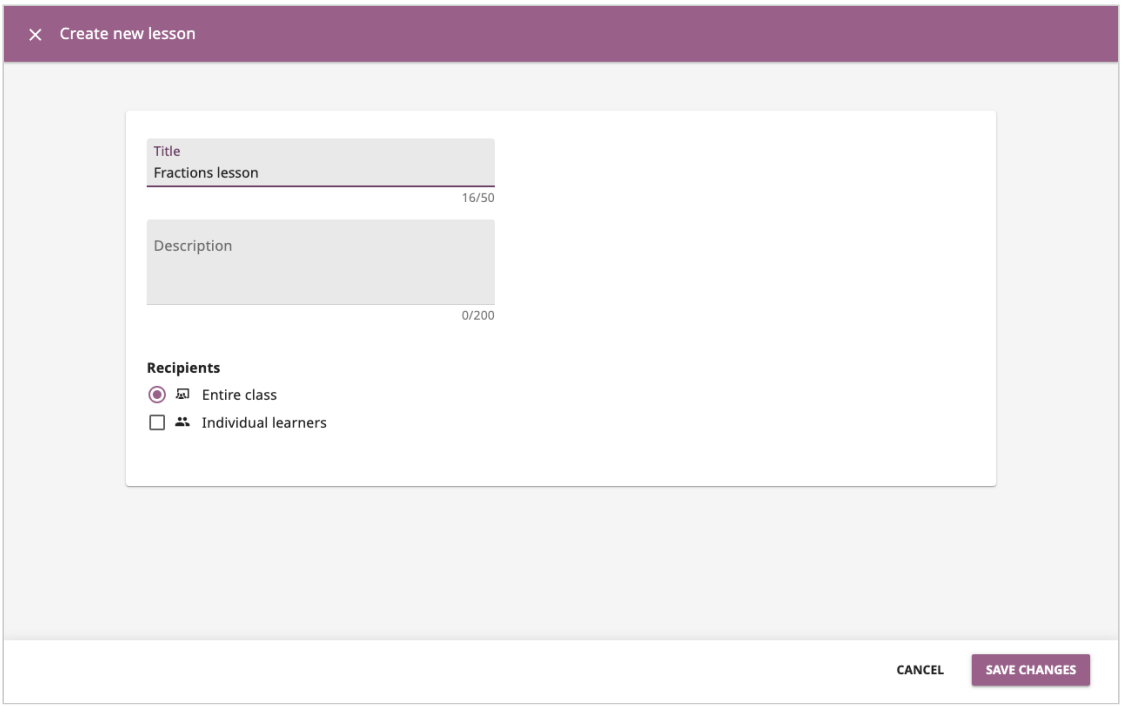


Figure 12. New lesson

3. On the lesson page, click **MANAGE RESOURCES**.

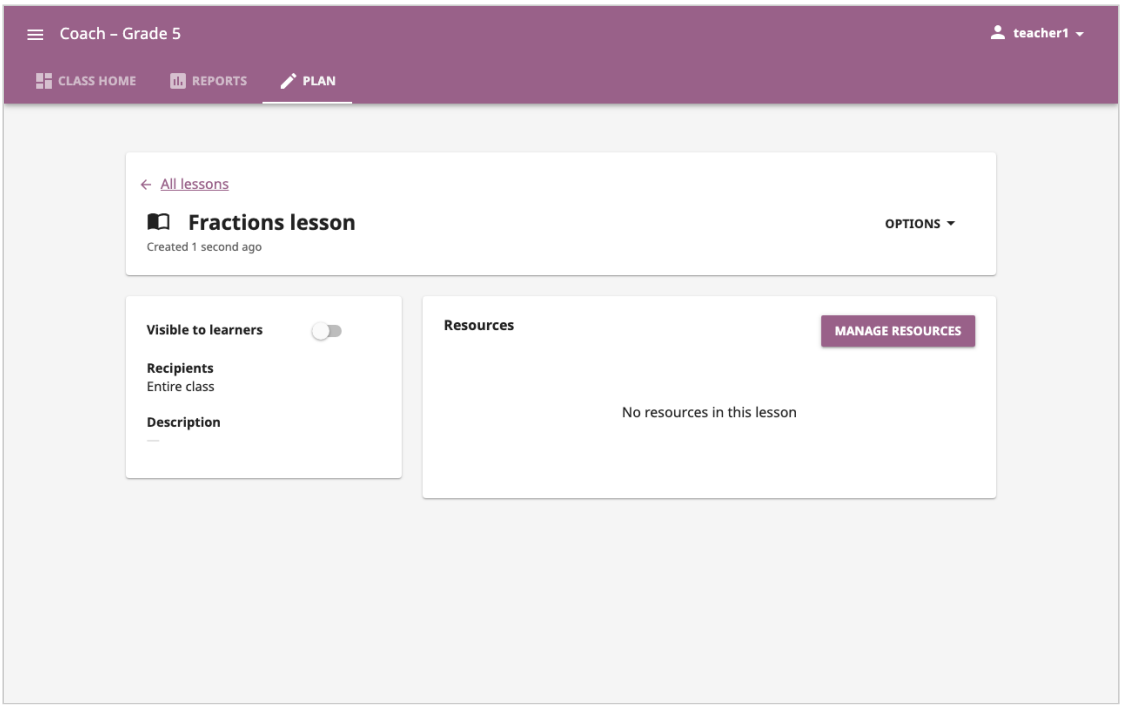


Figure 13. Empty lesson

4. Choose the channel.

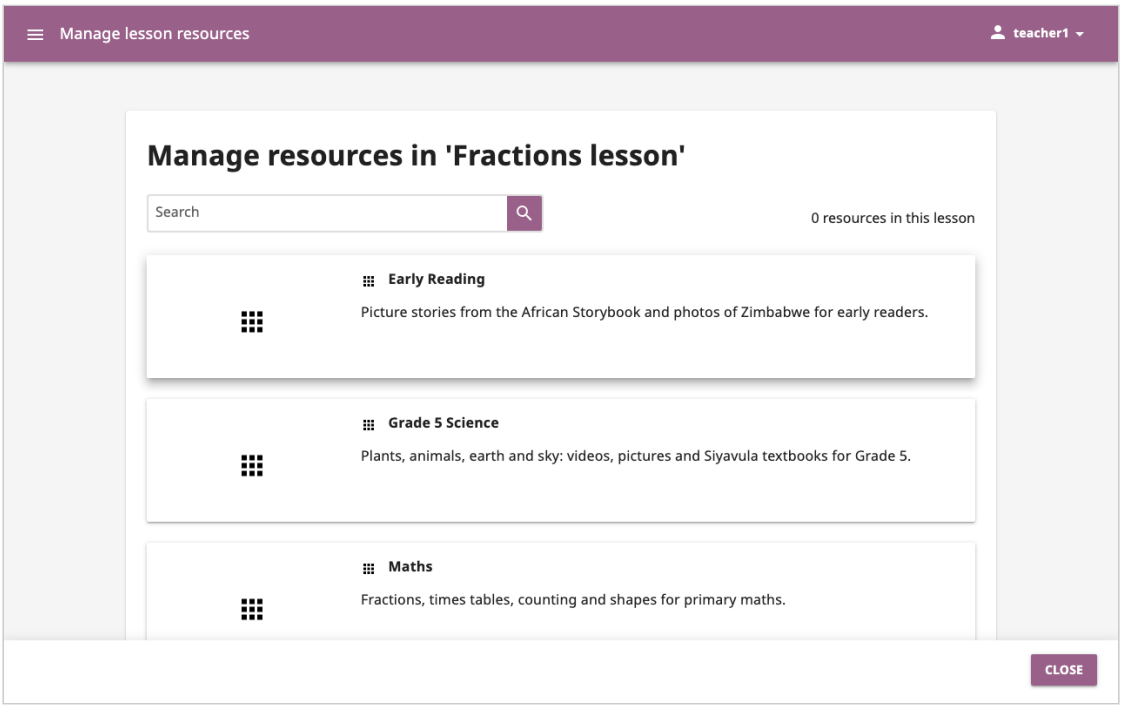


Figure 14. Channels

5. Click the channel (**Maths**). You see its topics.

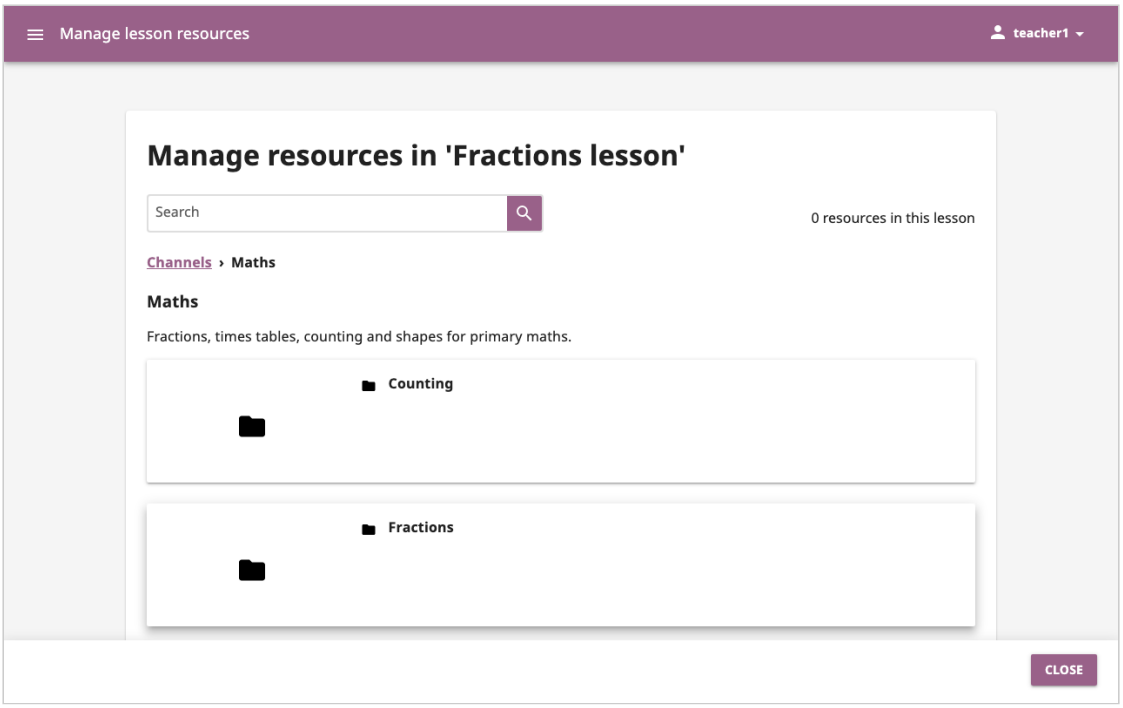


Figure 15. Topic

6. Open the topic **Fractions** and tick the resources you want (here all 3: Cake in quarters, Fraction circles, Paper fraction pies). Each tick is saved straight away; the count at the top right goes up. Click **SAVE CHANGES** (or **CLOSE**) when done.

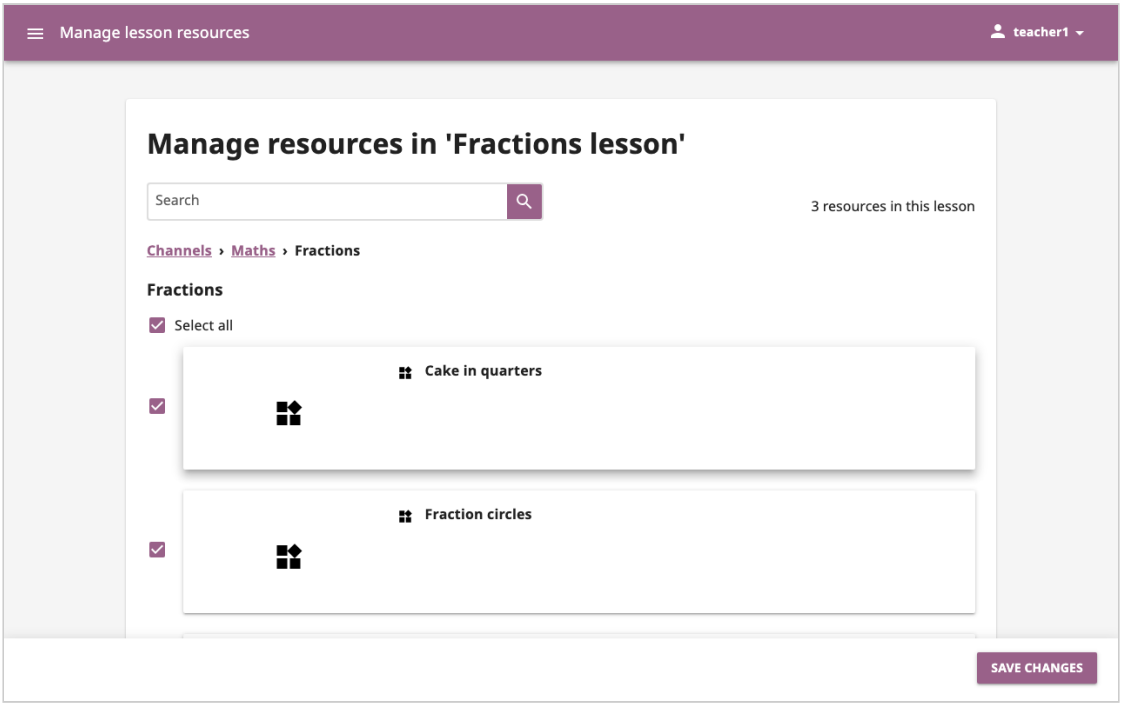


Figure 16. Resources ticked

7. Back on the lesson page, check the resources and switch on **Visible to learners**.

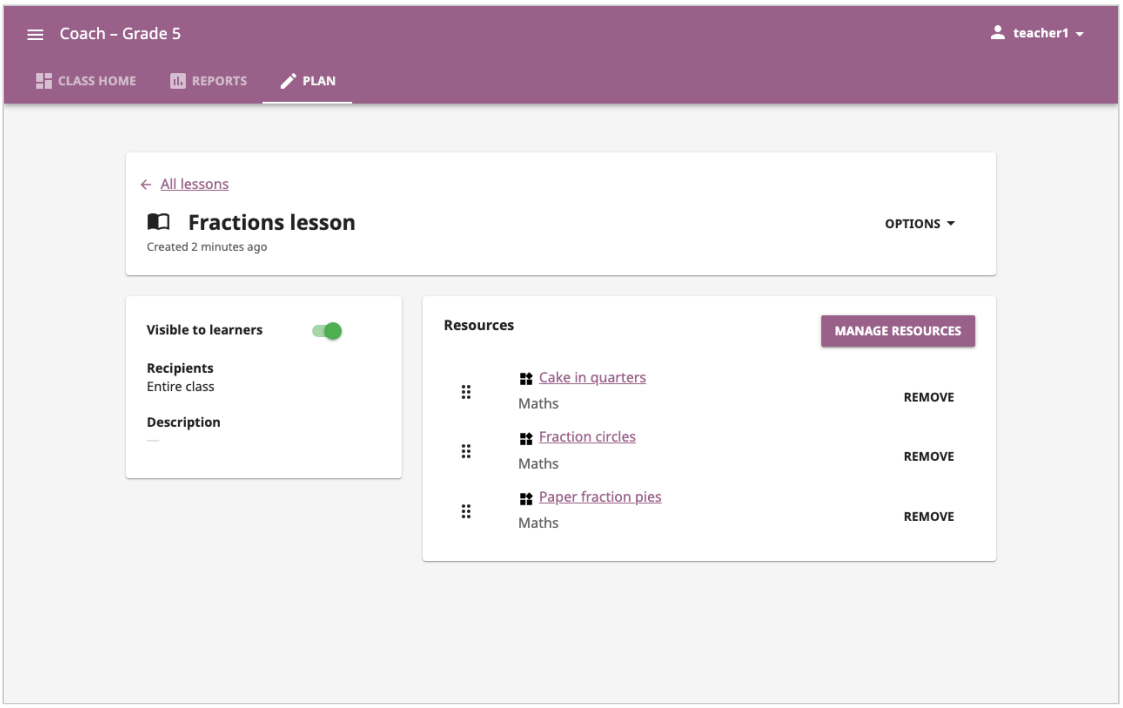


Figure 17. Lesson with its resources, visible

3.5 Make a quiz

Quizzes are built from exercises only.

On the kit. Khan Academy is not on the kit, and the three channels of **Kolibri Demo** have videos, pictures and documents but no exercises, so you can't make this quiz in Kolibri Demo. The steps are the same in the school's own **Kolibri**, with a channel that has exercises. The screenshots below come from the demo box.

1. Click **PLAN > QUIZZES > NEW QUIZ**.

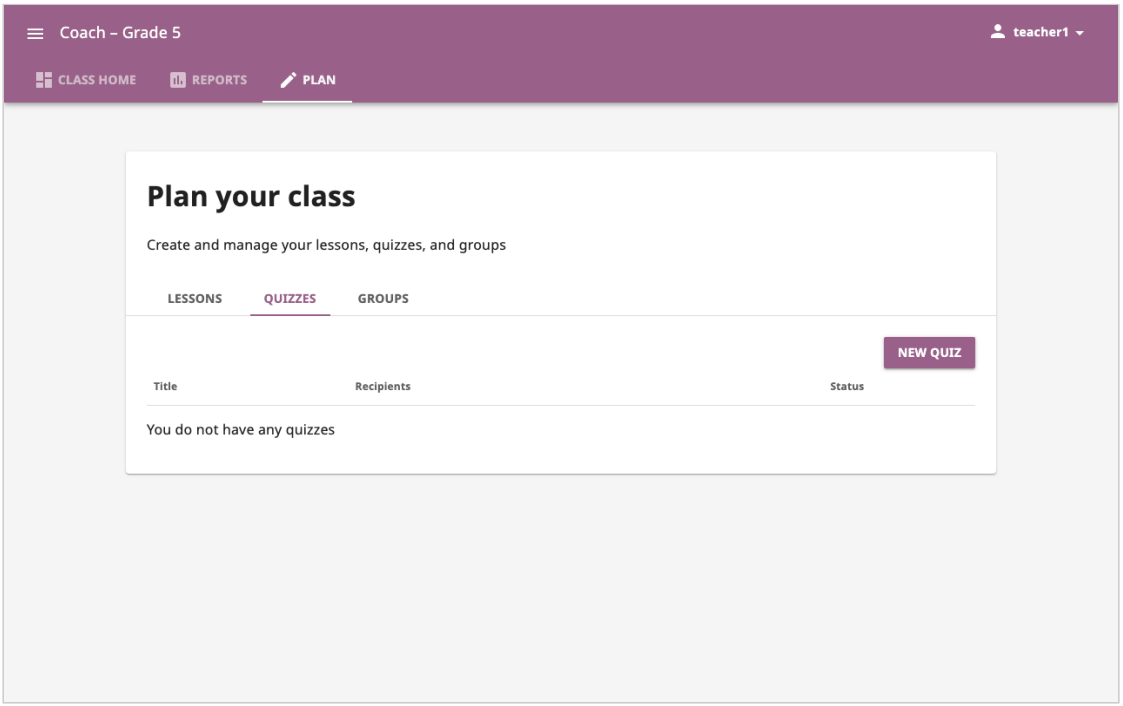


Figure 18. Quizzes tab

2. Type a title (`Fractions quiz`) and the **Number of questions** (6). Browse to **Khan Academy (English - US curriculum) > Math by grade > 5th grade > Add and subtract fractions** and tick a folder (here **Adding and subtracting fractions with unlike denominators**, 3 exercises) or single exercises. Click **CONTINUE**.

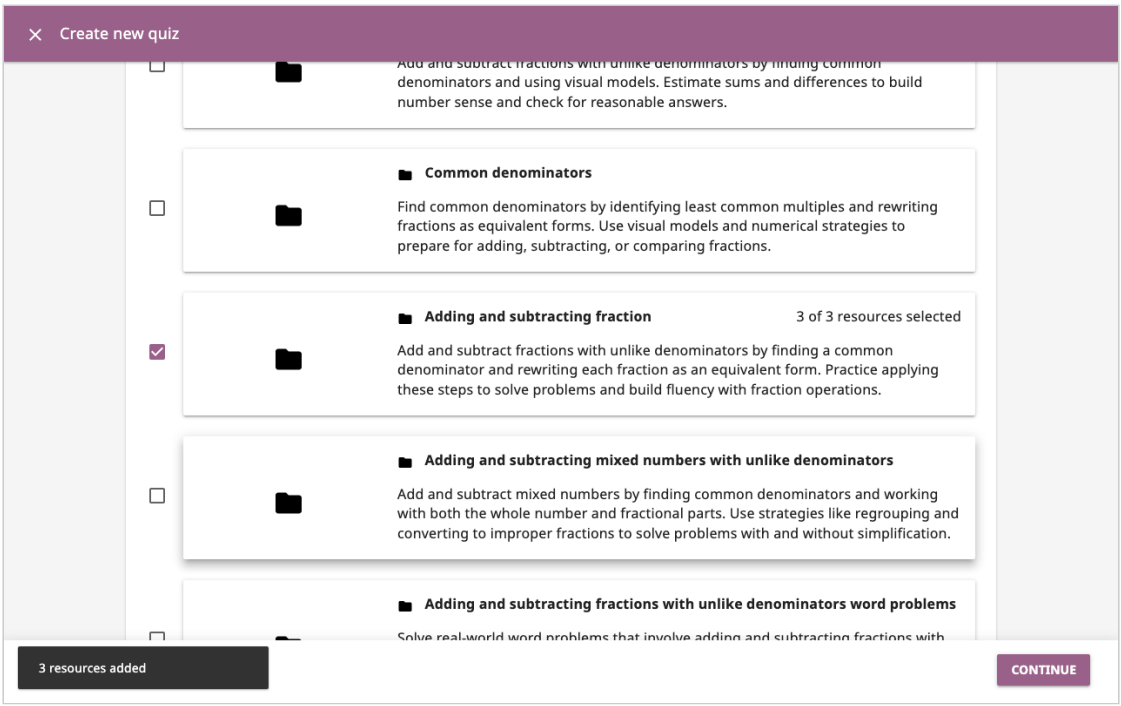


Figure 19. Create new quiz

3. **Preview quiz**: check the questions, choose **Randomized** or **Fixed** order, click **FINISH**.

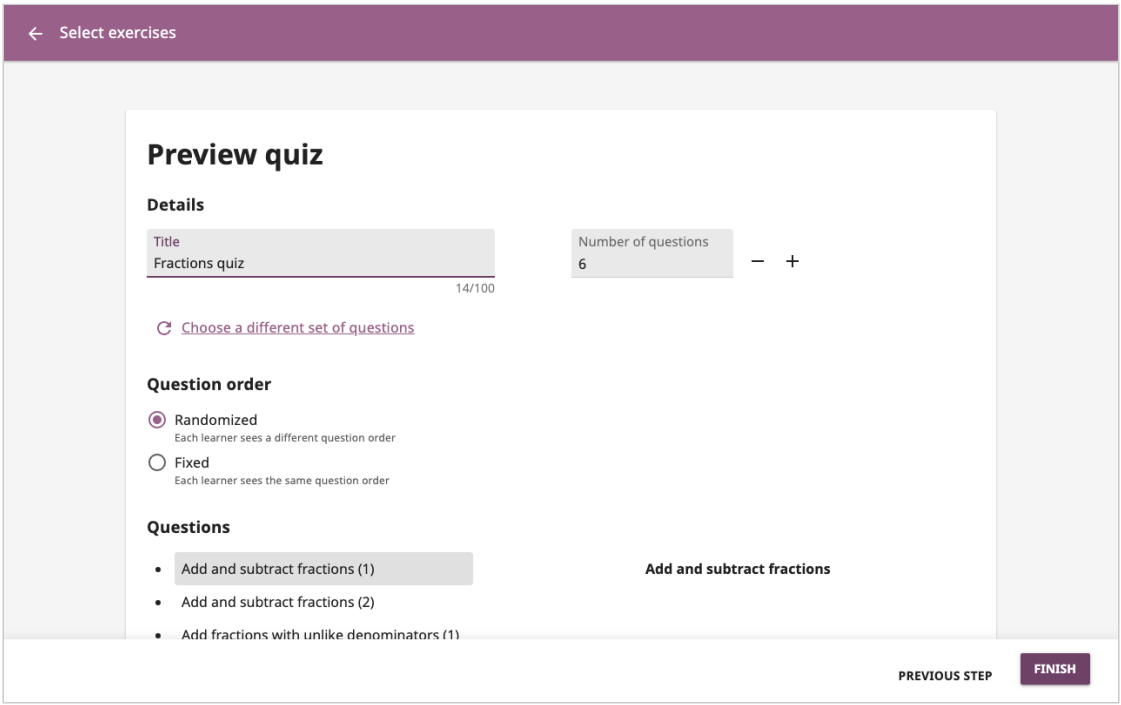


Figure 20. Preview quiz

4. The quiz is not open yet. When the class is ready, click **START QUIZ** and then **CONTINUE**. When the class has finished, open the quiz in **Coach** again and end it there, so no more answers come in.

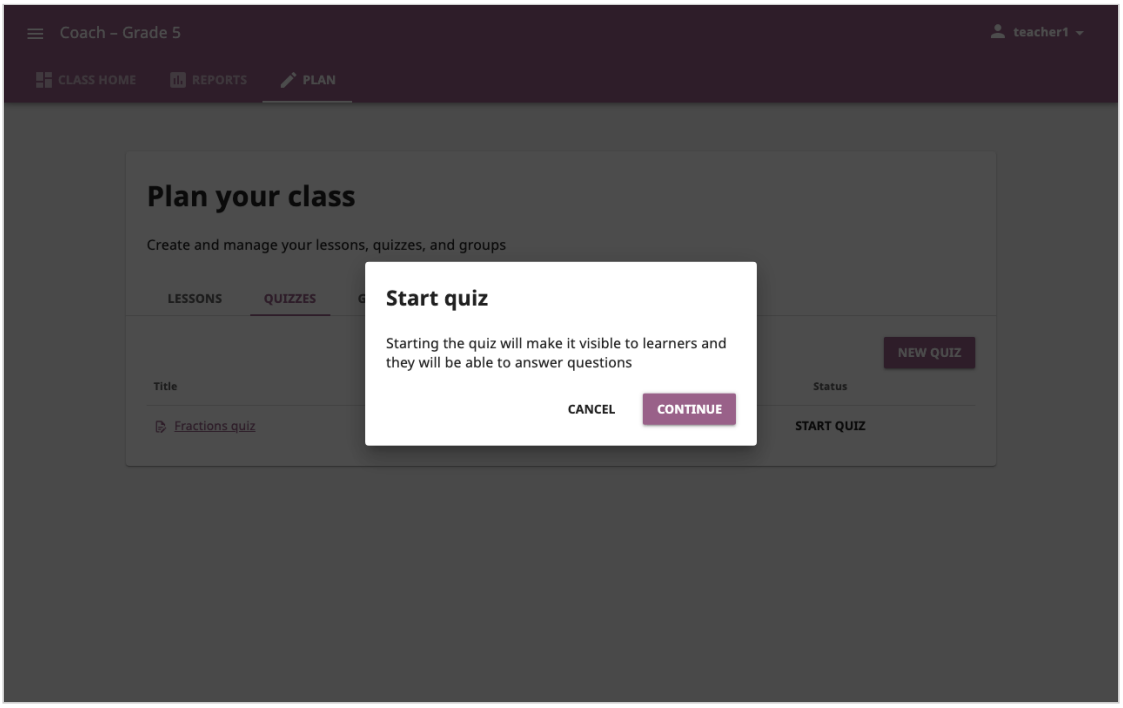


Figure 21. Start quiz

3.6 See progress (Reports)

Do this after learners have worked (chapter 4).

1. Click **REPORTS**. Tabs: **LESSONS**, **QUIZZES**, **GROUPS**, **LEARNERS**.

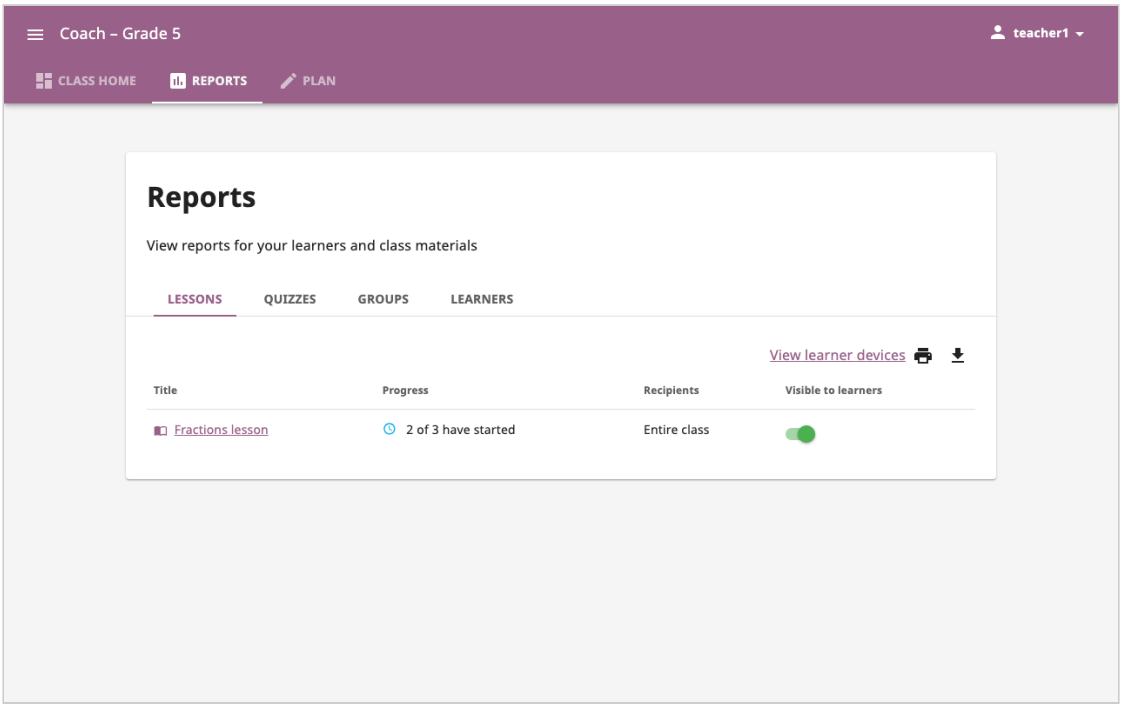


Figure 22. Reports

2. **LESSONS > (the lesson) > LEARNERS**: who completed, started or did not start.

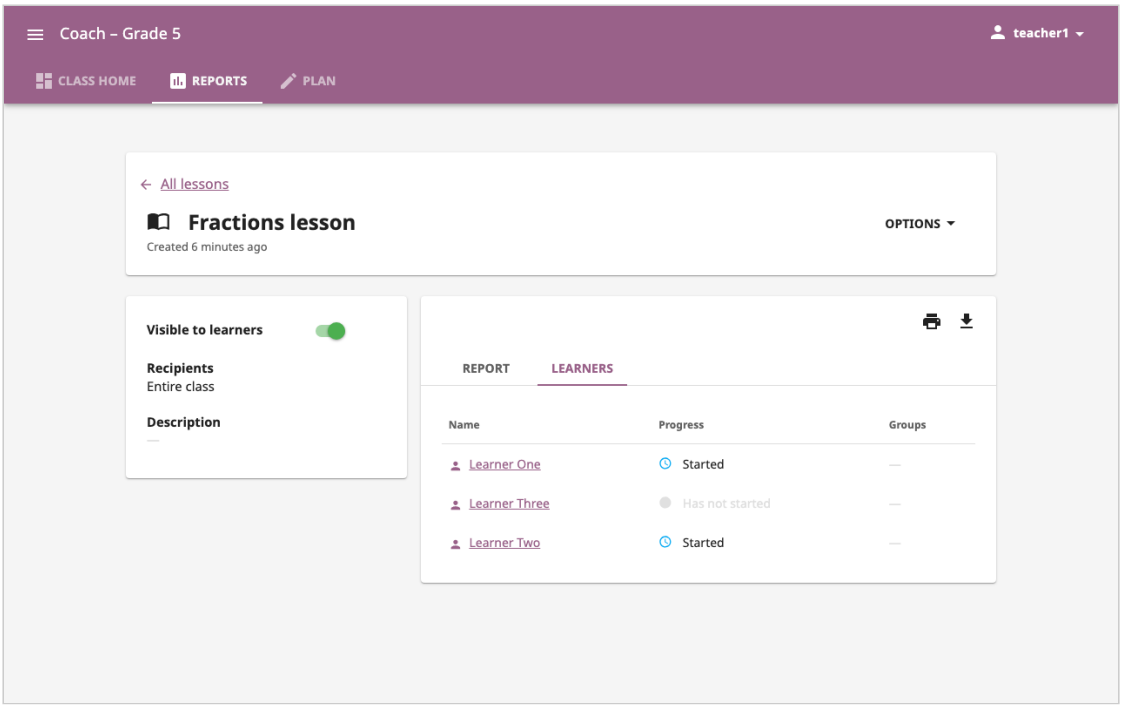


Figure 23. Lesson report

3. **QUIZZES > (the quiz)**: status and score per learner; **DIFFICULT QUESTIONS** shows what to re-teach.

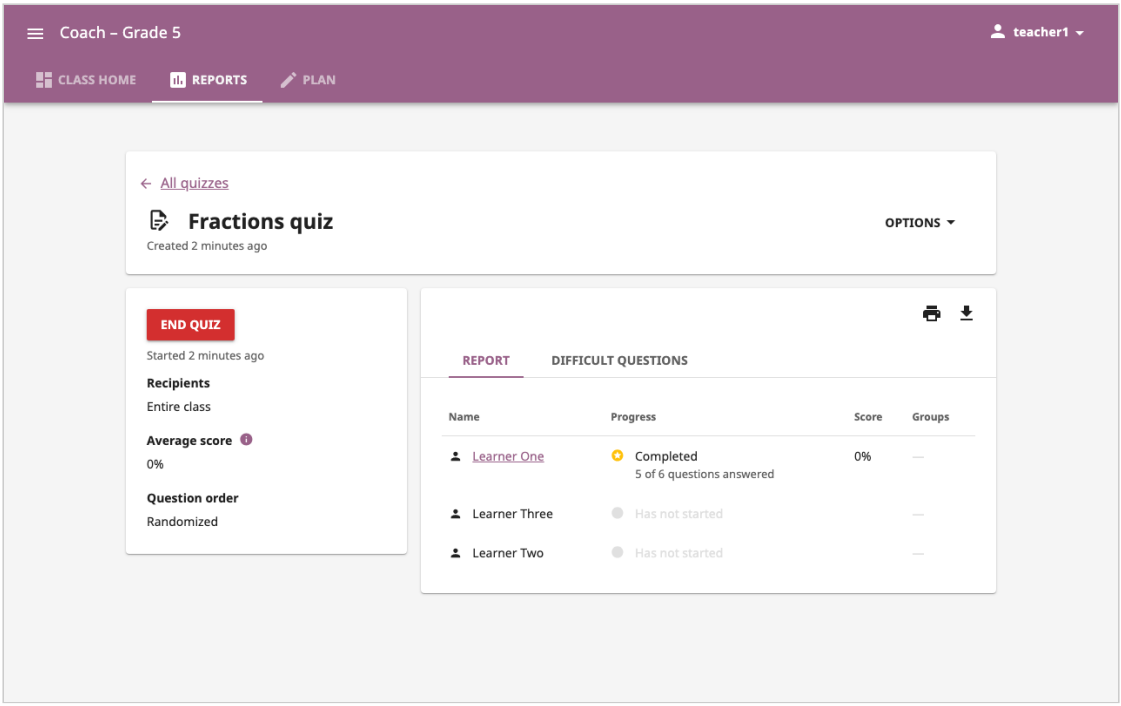


Figure 24. Quiz report

4. **LEARNERS > (a name)**: that learner's lessons and quizzes.

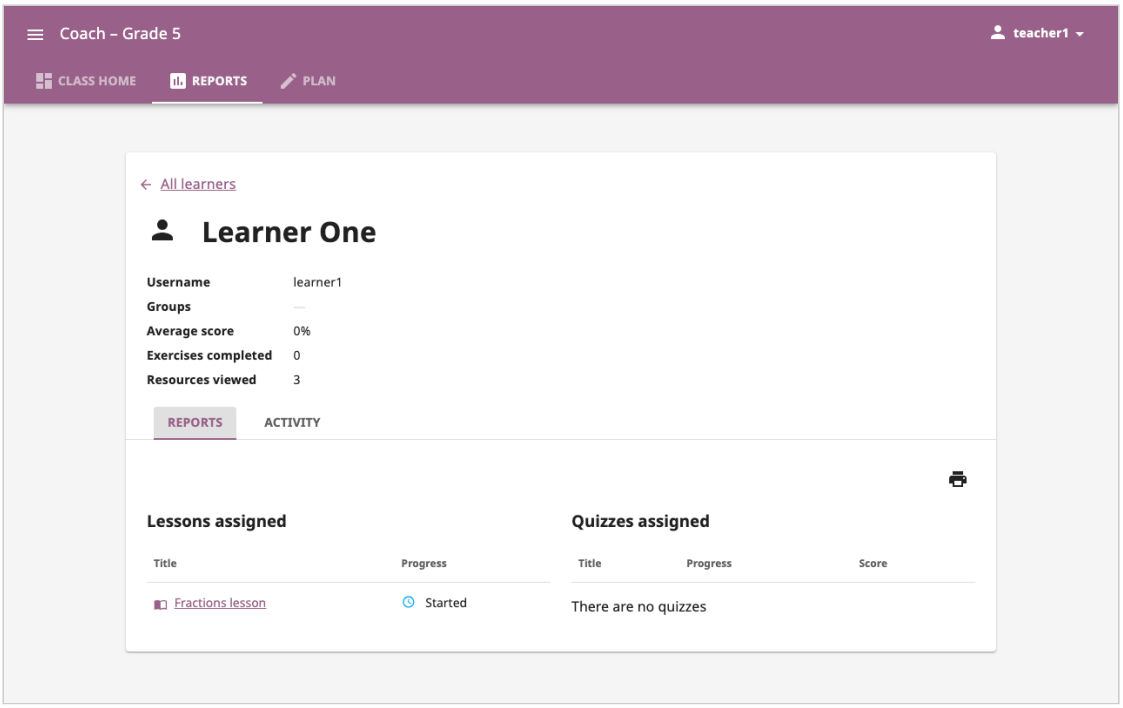


Figure 25. Learner report

Reports can be printed or exported as CSV (icons top right).

> **Tip.** Do not chase 100%. Look for the question most learners got wrong and teach it again.

4. Kolibri for learners (try it yourself)

Your learners open Kolibri the same way you do: on a classroom computer they double-click **IDEA** on the desktop and click **Open** on the Kolibri card in the Console (2.2.1). Show them this once. To try it yourself with the demo learners, open **Kolibri Demo**.

Sign in as a learner in a second browser window (private window) to see exactly what your class sees. Learners have no password in Kolibri. They sign in with only their username, for example `learner1`.

> **On the kit**, both **Kolibri** and **Kolibri Demo** already let learners sign in without a password. **At an existing school**, a teacher with admin rights (or the Kolibri admin) turns this on in Kolibri: open **Facility > SETTINGS**, untick **Require password for learners**, then click **SAVE CHANGES**.

4.1 Do a lesson

1. Sign in as `learner1`. **HOME** shows **Your classes**, **Recent lessons** and **Recent quizzes**.

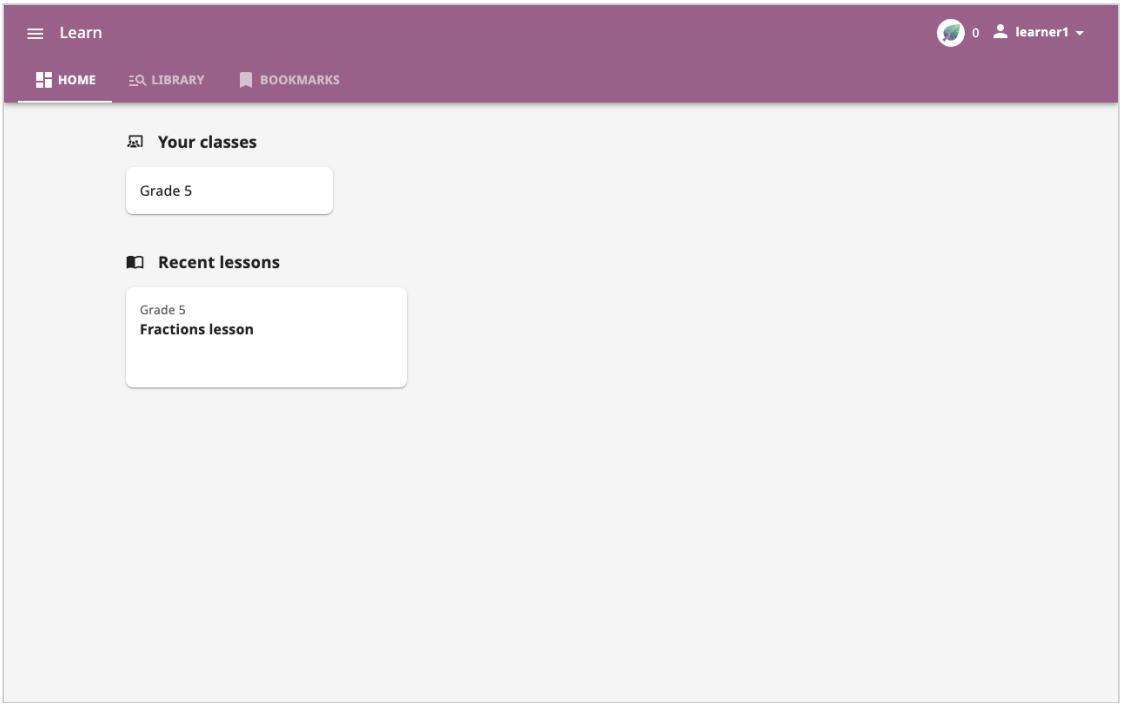


Figure 26. Learner home

2. Click **Fractions lesson**. Its items are listed in order.

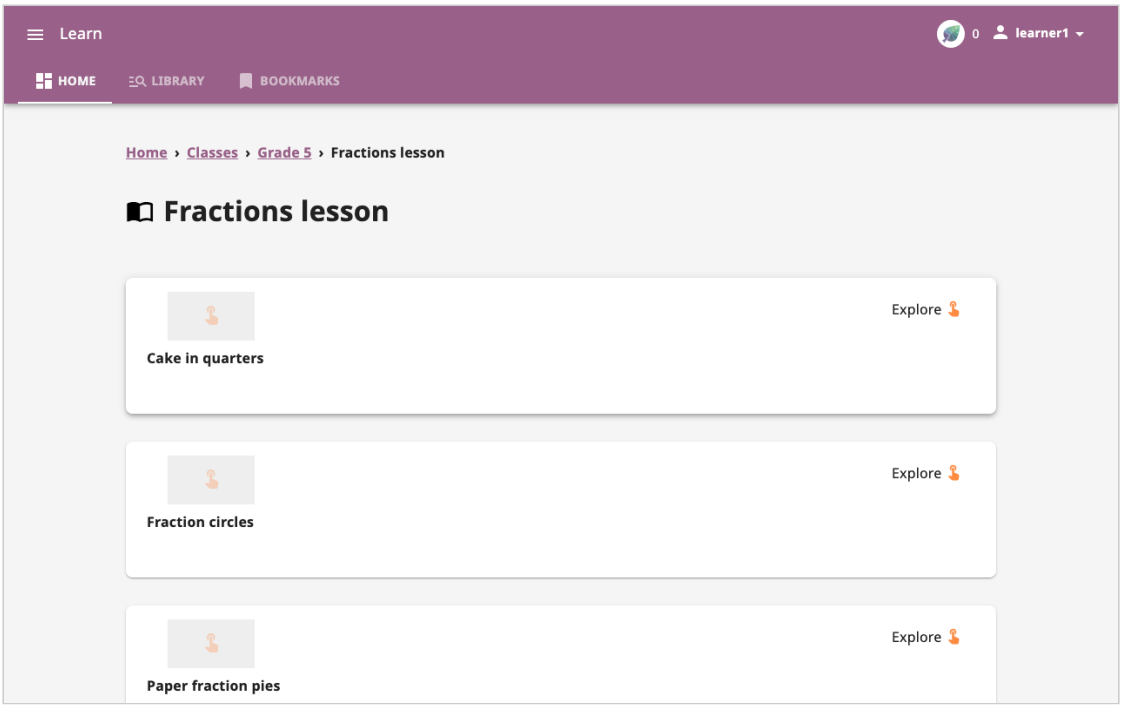


Figure 27. Lesson

3. Open the first item (**Cake in quarters**, a picture activity). Videos open the same way.

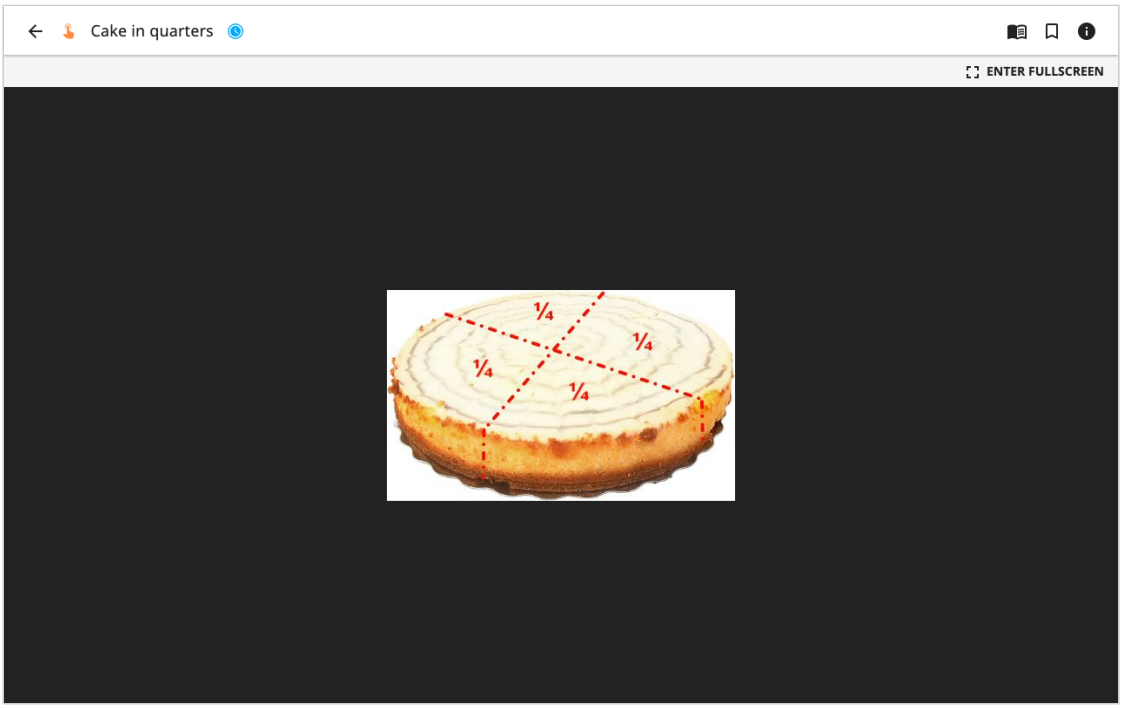


Figure 28. Video

4. Exercises work like this (here the Khan Academy exercise **Add fractions with unlike denominators**, from the demo box; Kolibri Demo on the kit has no exercises): type an answer, click **CHECK**. Keep going until the goal at the top ("Get 5 correct") is reached.

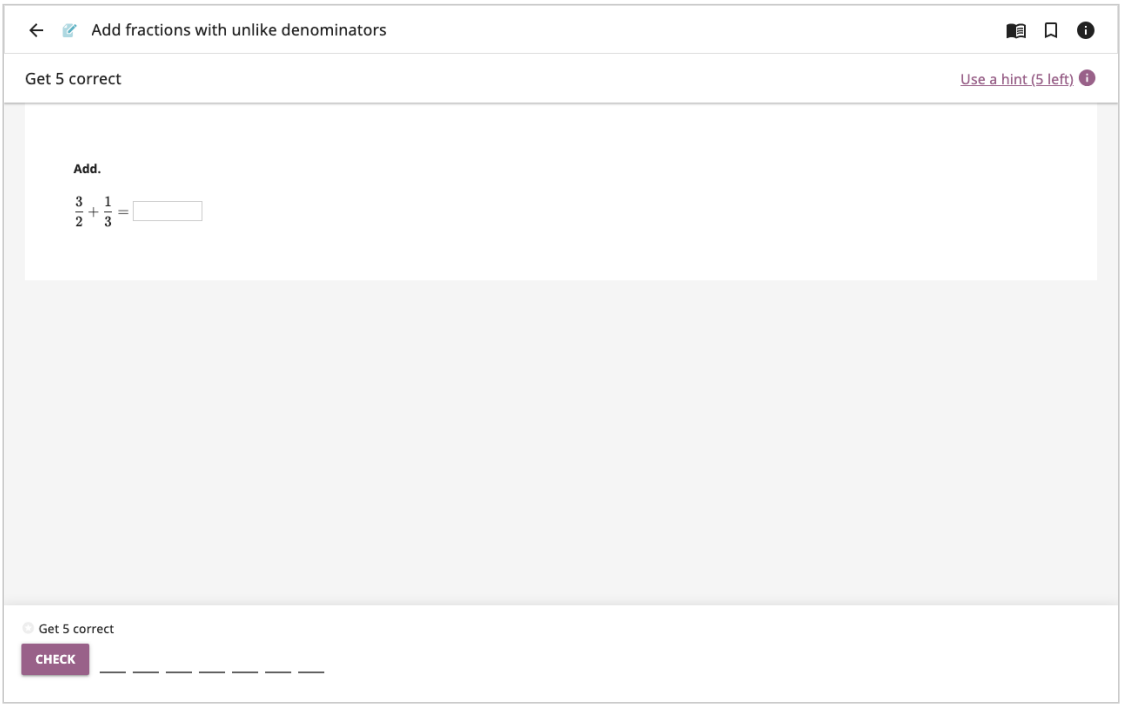


Figure 29. Exercise

5. Sign out. Sign in as `learner2` and open only one item, so the report shows a difference.

4.2 Take a quiz

This needs the quiz from 3.5, so on the kit it works in the school's own Kolibri, not in Kolibri Demo.

1. Open the quiz from **Recent quizzes** (or the class page).

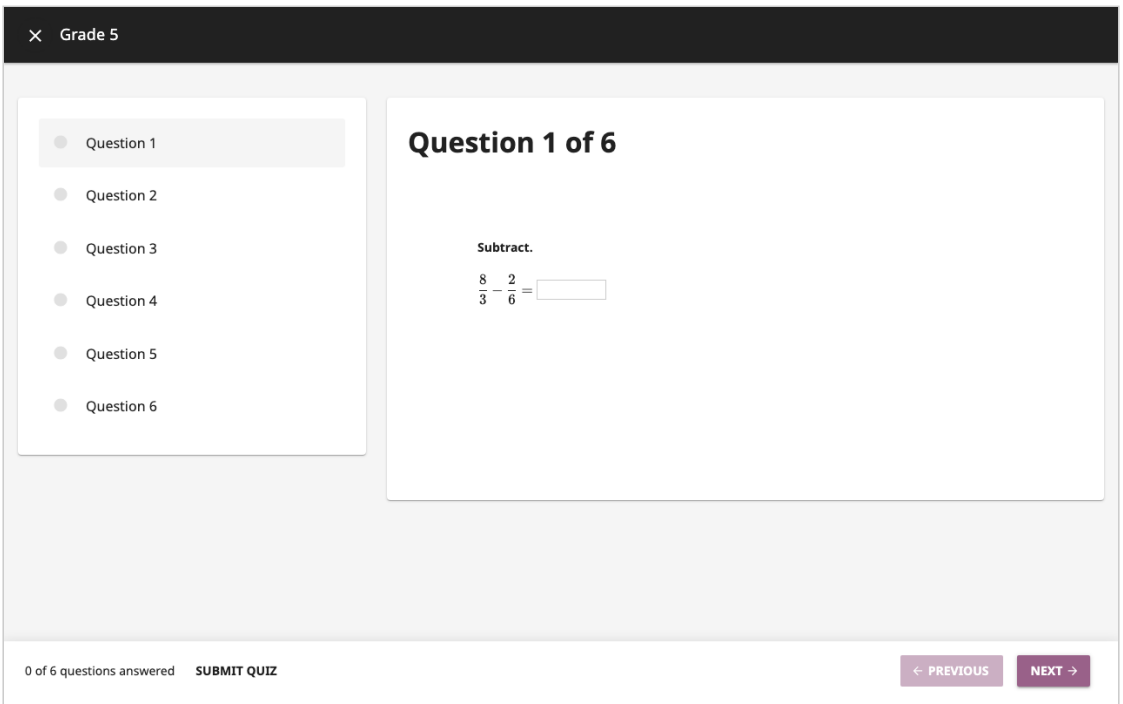


Figure 30. Quiz

2. Answer each question and use **NEXT** / **PREVIOUS**. The list on the left shows answered questions.

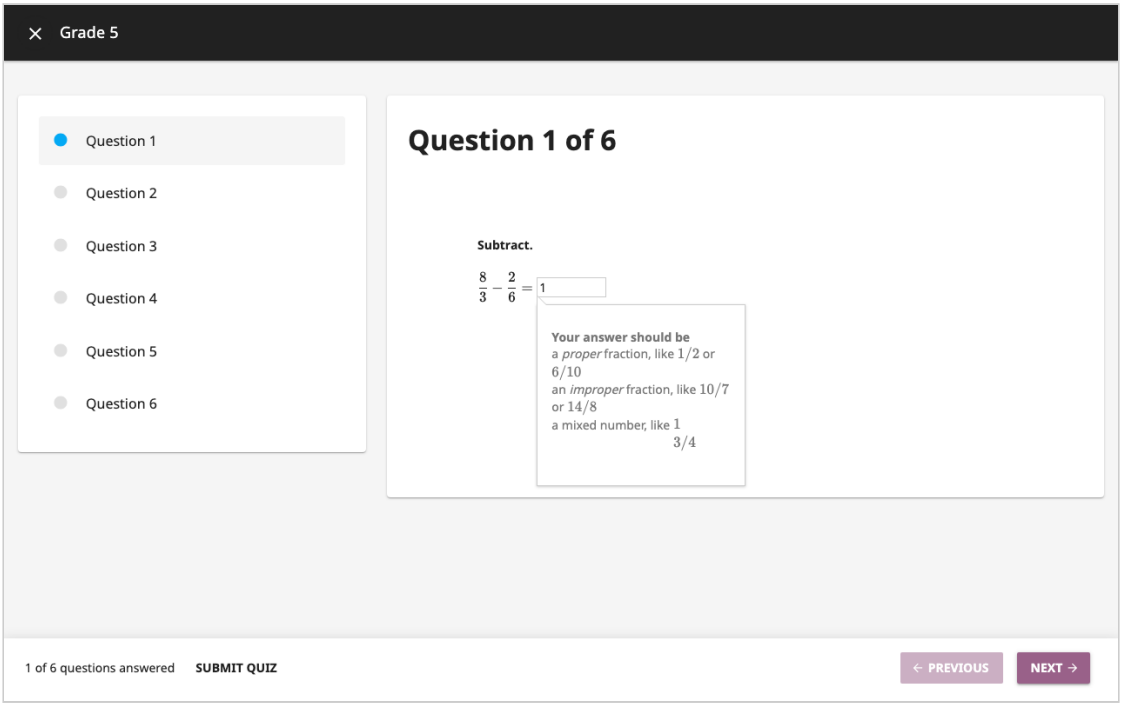


Figure 31. Question

3. Click **SUBMIT QUIZ** and confirm. After submitting, answers cannot be changed.

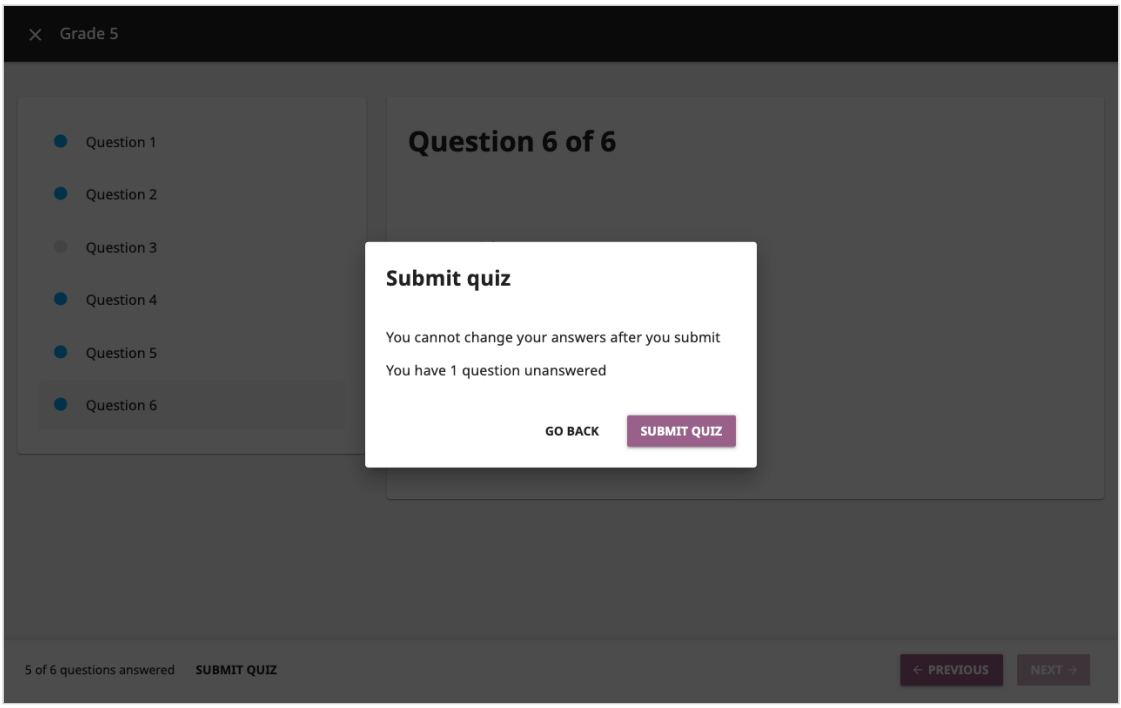


Figure 32. Submit

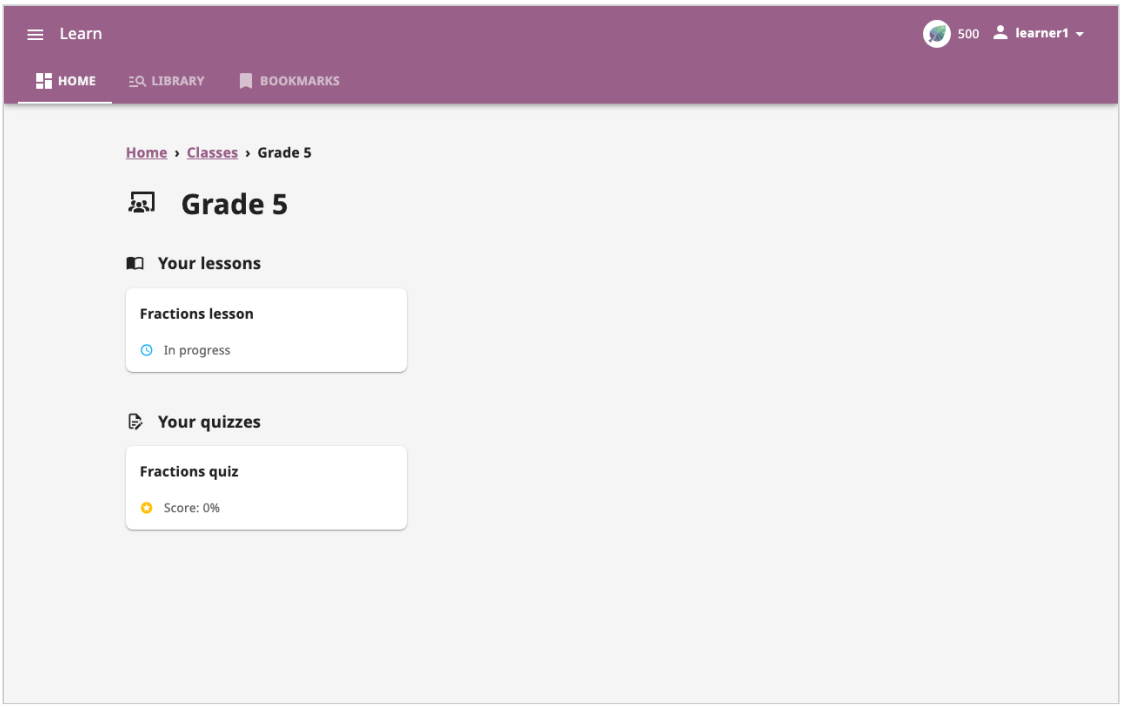


Figure 33. After submit

Now go back to the teacher window and look at Reports (3.6).

5. Nextcloud for teachers

About the examples in this chapter. On the kit, use the app **Nextcloud Demo**. It has the demo teachers `teacher1` and `teacher2` (group `teachers`), the demo learners `learner1` to `learner3` (group `learners`), and in each teacher's files the folders `Worksheets` and `Teacher Notes`. The screens in this chapter come from the demo IDEA box, where the class group was called `Grade 5`: on the kit, pick `learners` wherever the steps say the class group. The steps work the same at your school, but your names will be different. The demo accounts are listed in Appendix: demo accounts.

Office documents open in **Nextcloud Office**.

Open **Nextcloud Demo** from the Console (chapter 2) and sign in as `teacher1`. Click **Files** at the top.

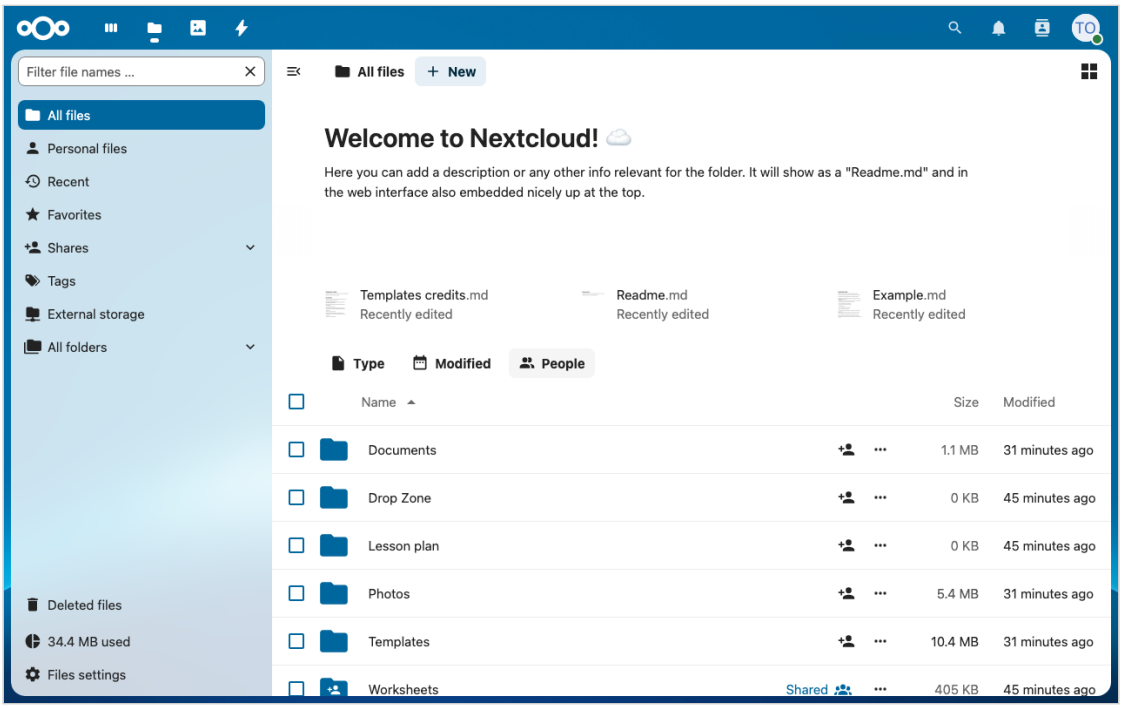


Figure 34. Files

5.1 Share worksheets with the class (view only)

1. Next to the folder `Worksheets`, click the share icon (**Show sharing options**). The right panel opens on **Sharing**.

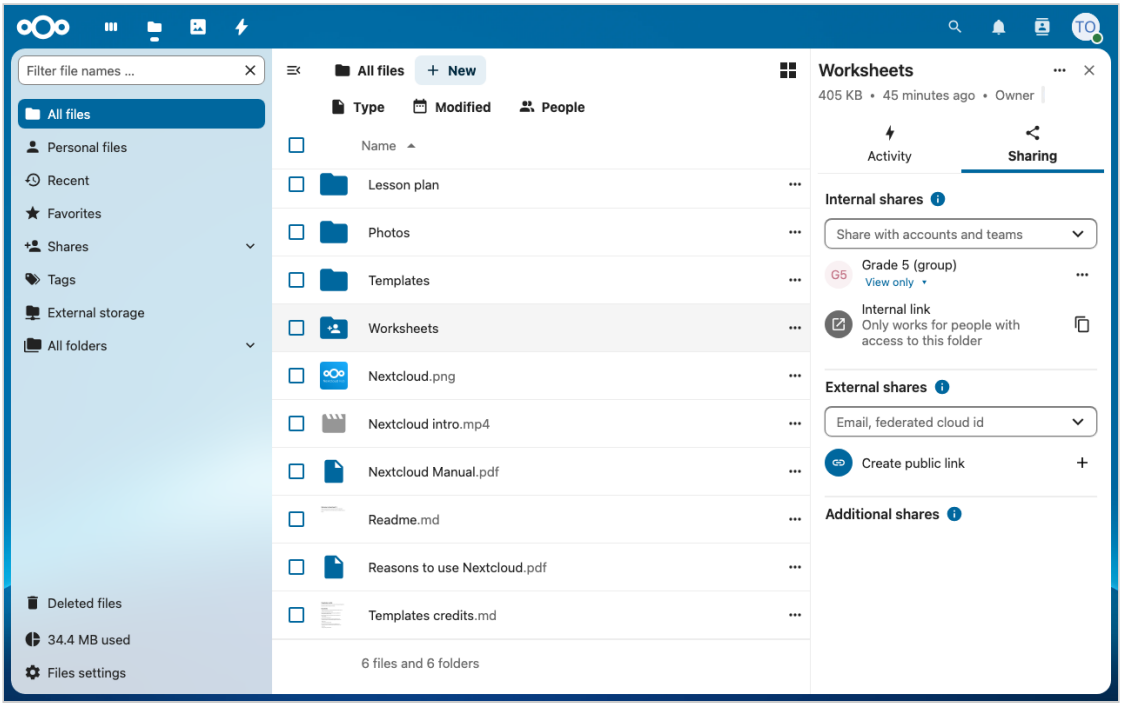


Figure 35. Sharing panel

2. In **Internal shares** ("Share with accounts and teams"), type the class group (on the kit: `learners`) and pick the group.
3. Set the permission to **View only**. The group then appears under Internal shares.

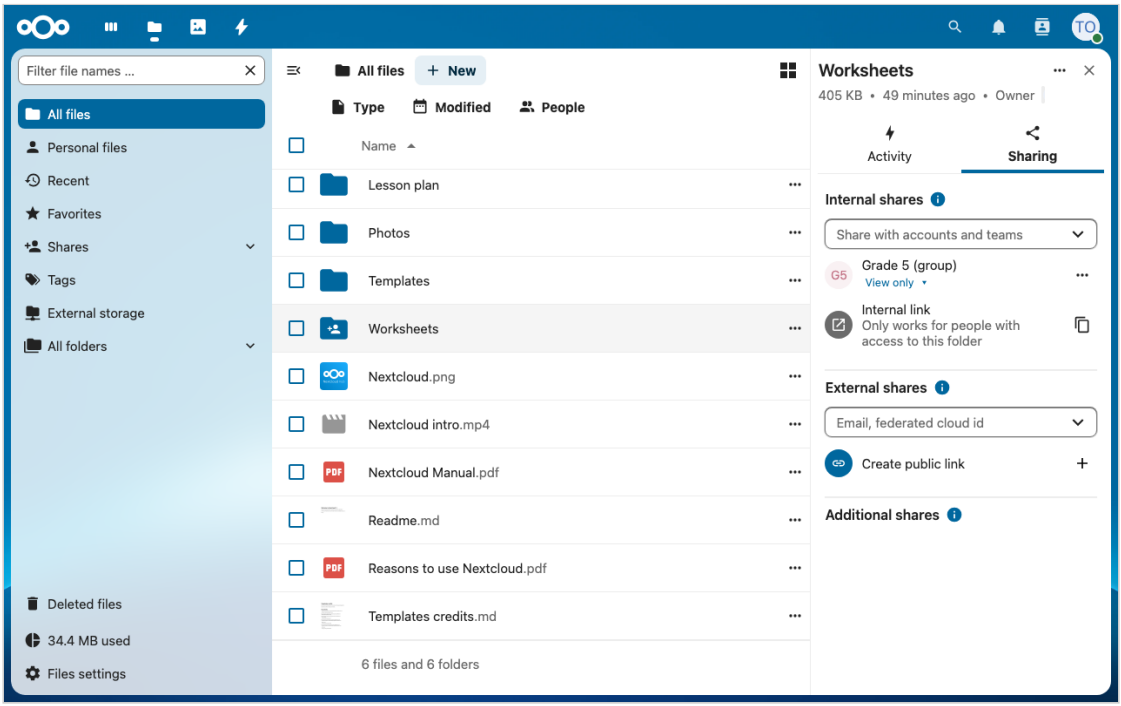


Figure 36. Shared with group

Learners now see the folder but cannot change or delete files.

5.2 Write a document together

1. Click **+ New > New document**.

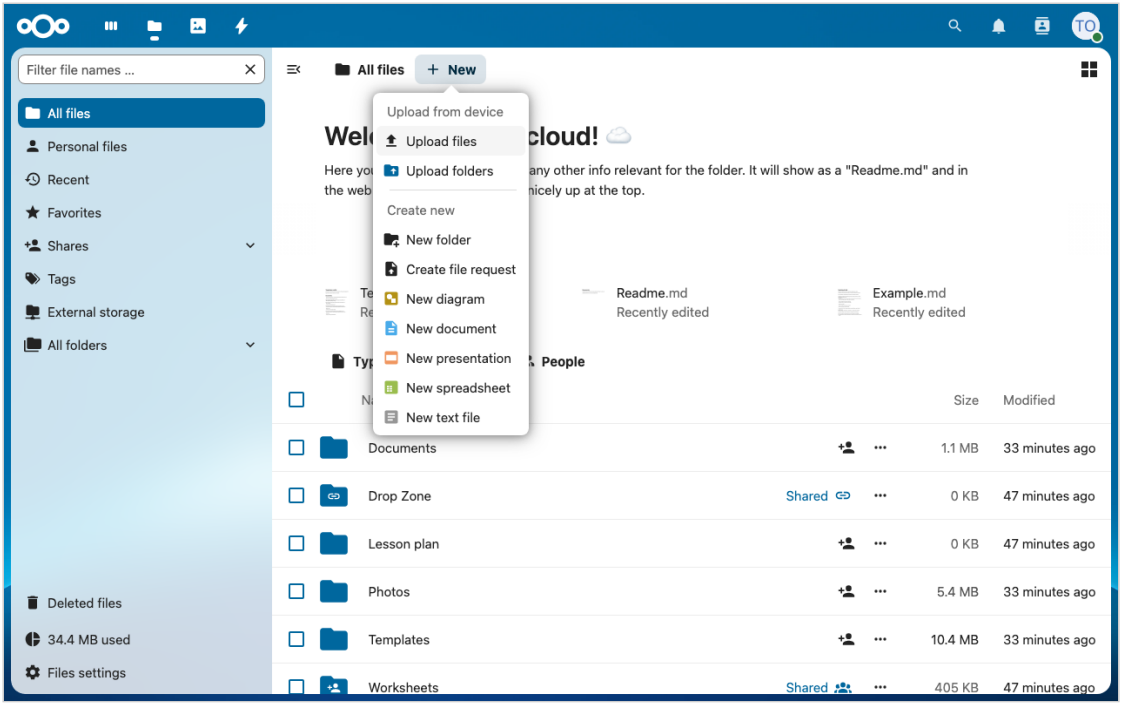


Figure 37. New menu

2. Type a name (`Lesson plan`) and click **Create**.

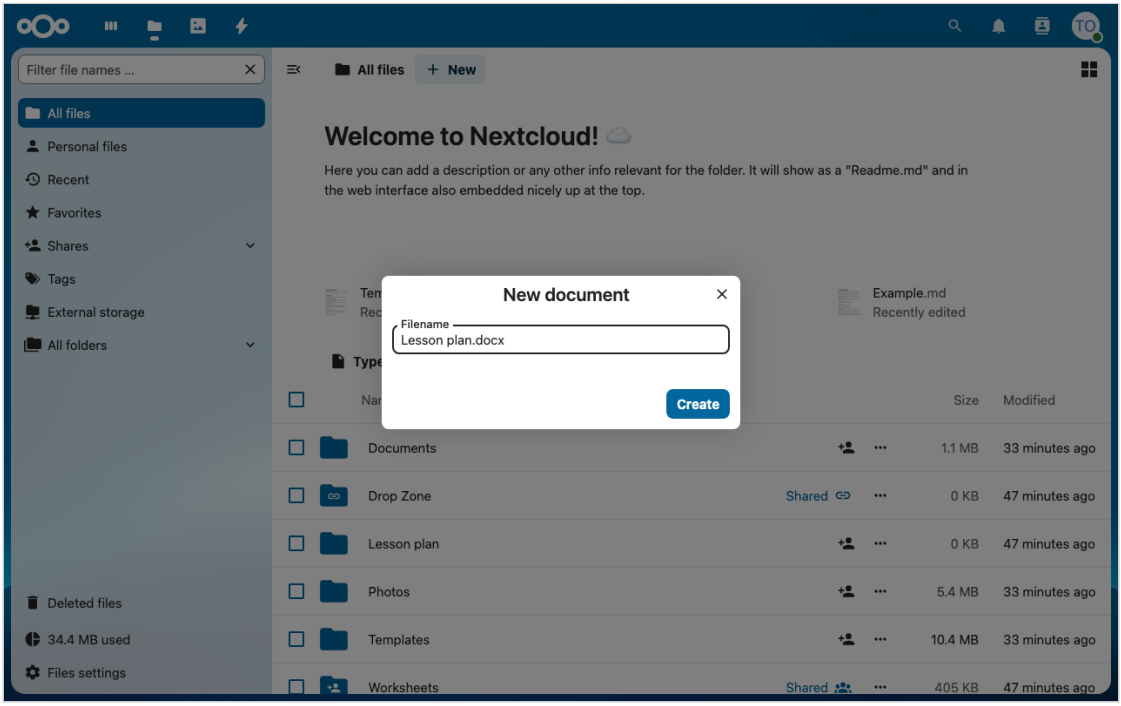


Figure 38. Name

3. Pick a template (Blank is fine) and click **Create**.

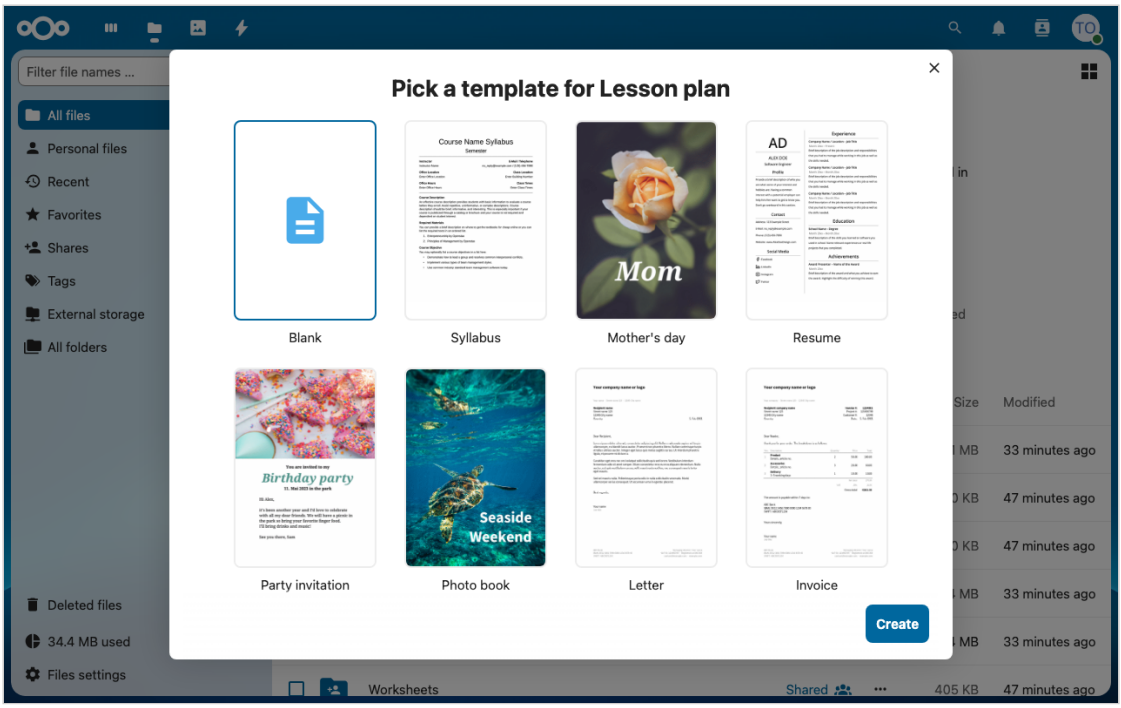


Figure 39. Template

4. Share the file with `teachers` and/or your class group (on the kit: `learners`), with editing allowed.

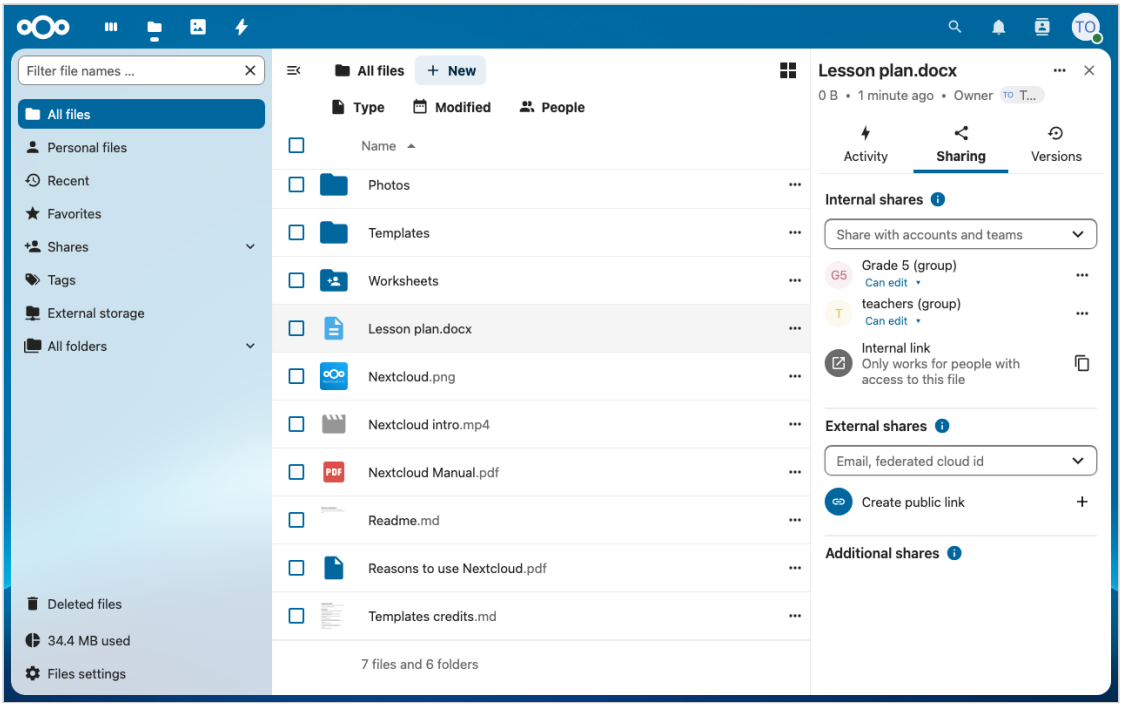


Figure 40. Shared document

5. The document opens in Nextcloud Office. When others open it at the same time, you see their names and cursors.

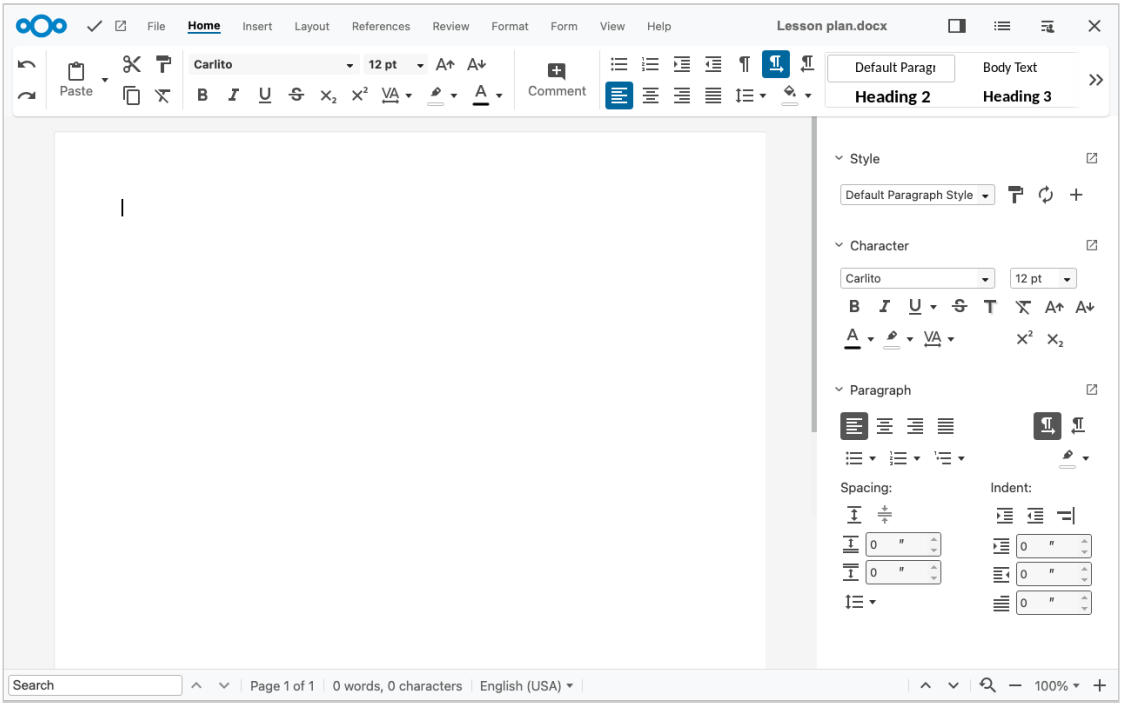


Figure 41. Nextcloud Office

5.3 Collect homework (File drop)

Learners upload work, but cannot see each other's work.

1. Make a folder for the homework: **+ New > New folder**, name it `Drop Zone`. Then open sharing for the folder `Drop Zone`.

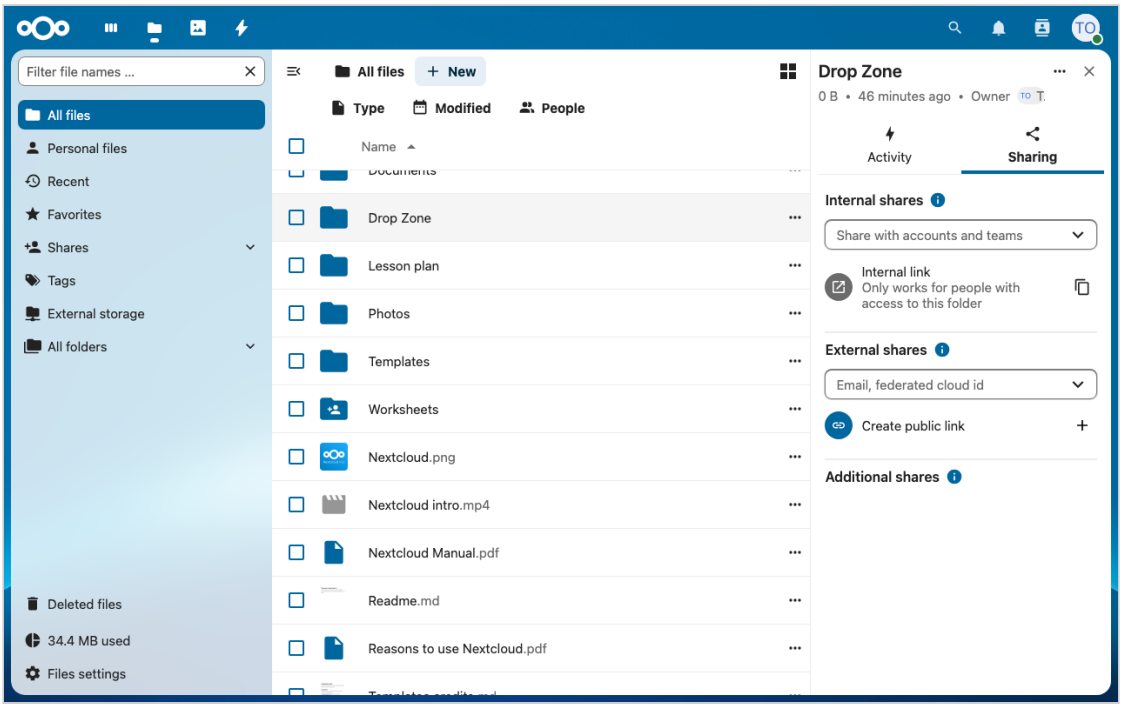


Figure 42. Sharing for Drop Zone

2. Under **External shares**, click **+** next to **Create public link** and choose **File request** (upload only). Copy the link.

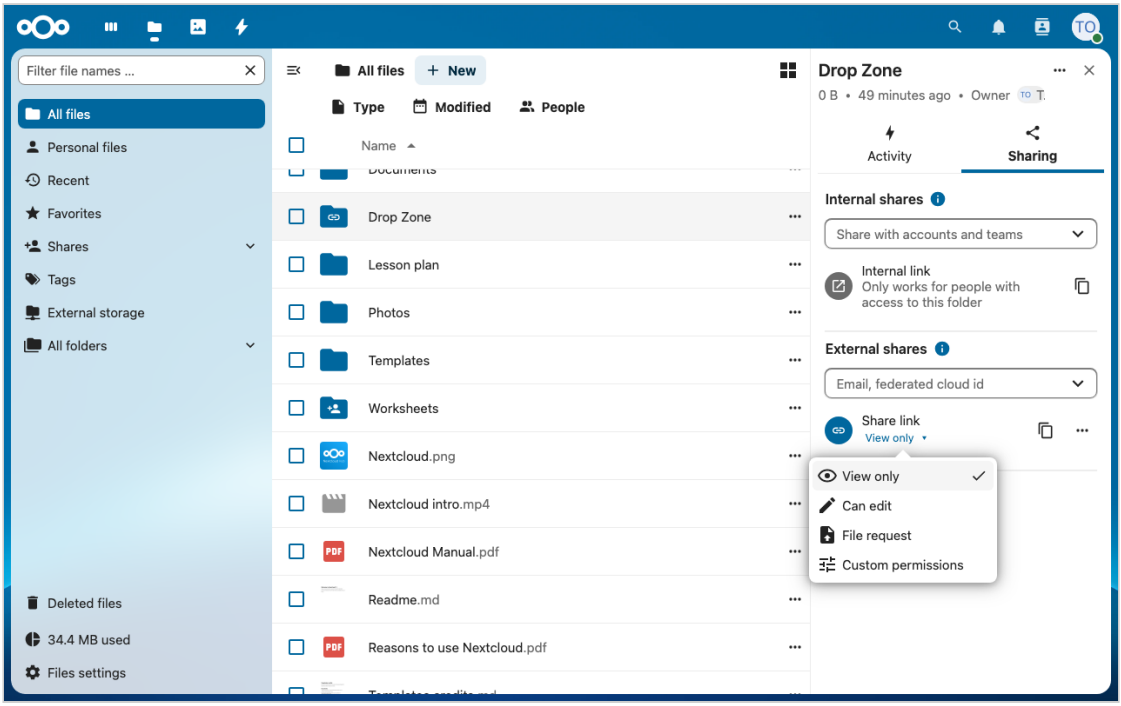


Figure 43. Create link

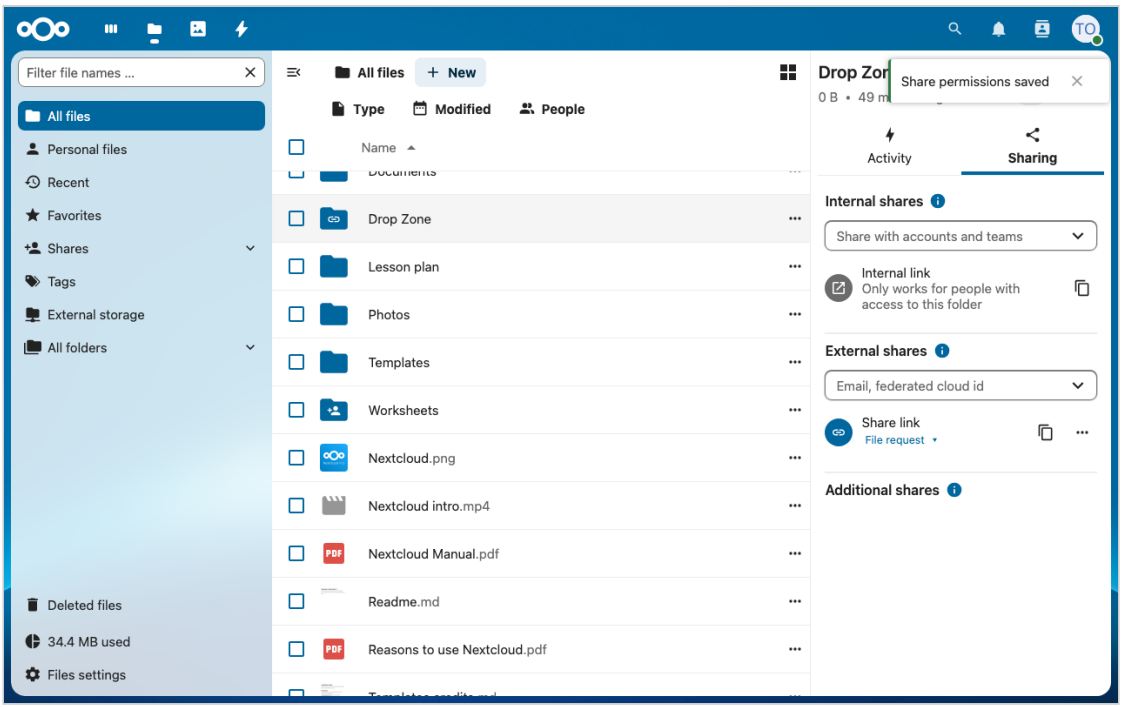


Figure 44. Link created

3. Give learners the link (for example, write it on the board). Uploaded files appear in your `Drop Zone` folder.

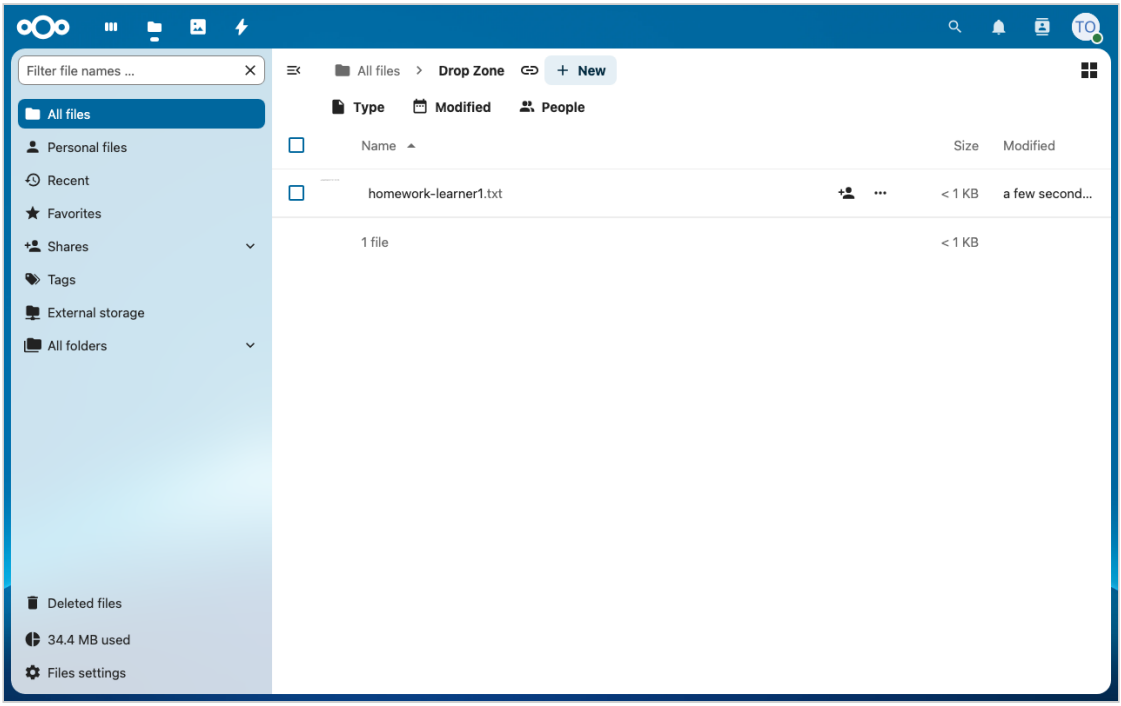


Figure 45. Received homework

6. Nextcloud for learners (try it yourself)

Learners open Nextcloud from the Console, like Kolibri: **IDEA** on the desktop, then **Open** on the Nextcloud card (2.2.1). They have no personal Nextcloud app on the classroom computer, so they always work in the browser.

6.1 Open class files and co-edit

1. Sign in to **Nextcloud Demo** as `learner1`, password `learner1911!` (private window). Shared folders and files appear in **Files** with the teacher's name.

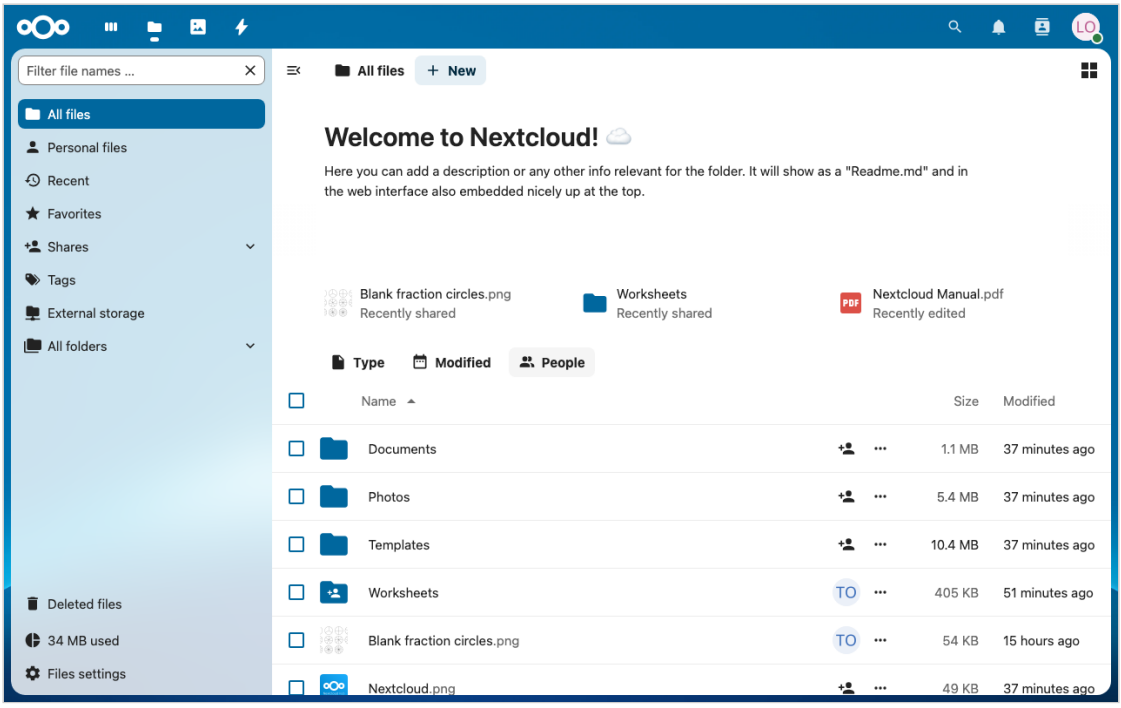


Figure 46. Learner files

2. Open `Worksheets`: the learner can open and download the worksheets, not change them.

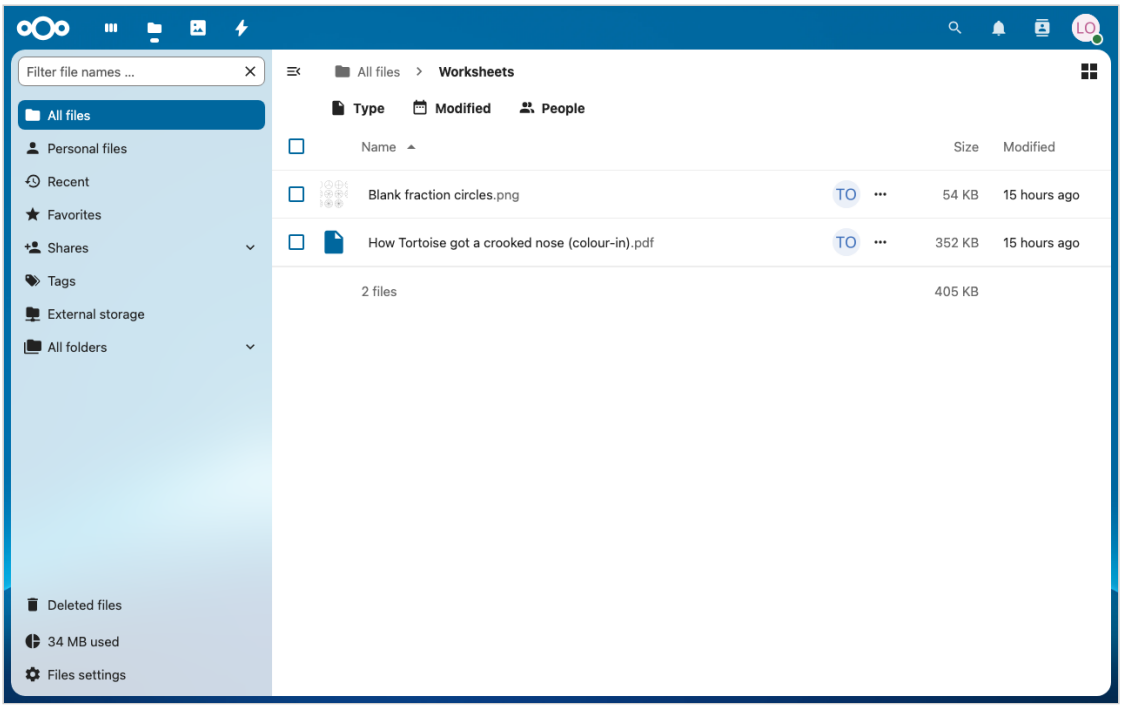


Figure 47. Worksheets

3. Open `Lesson plan.docx`: it opens in Nextcloud Office and the learner can type.

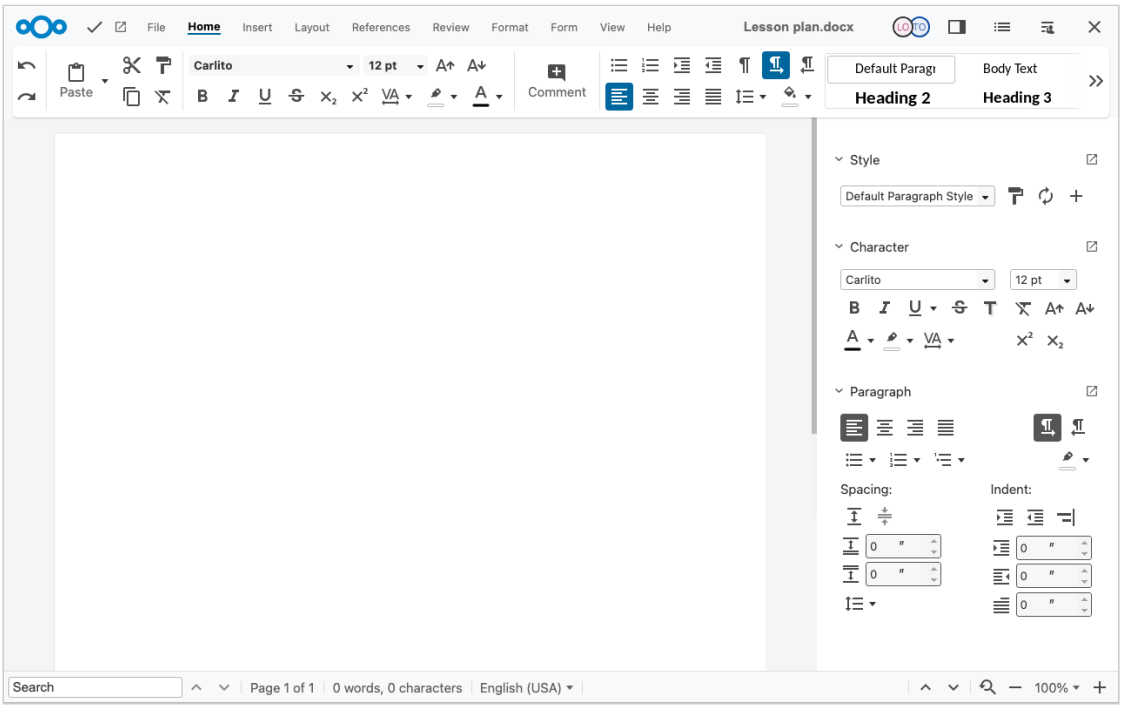


Figure 48. Learner in document

6.2 Hand in homework

1. Open the File drop link from 6.3. The page says **File drop** and shows an **Upload** button; the learner cannot see other files.

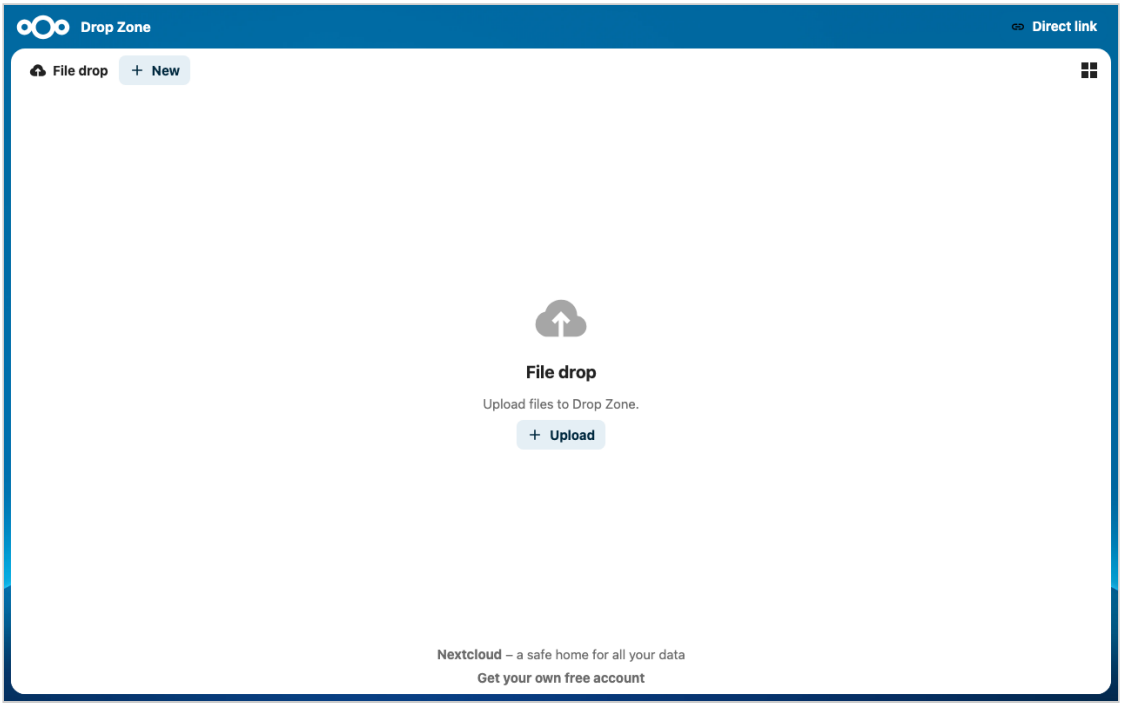


Figure 49. File drop

2. Click **Upload** and choose the file (e.g. a photo of written work). If asked, type your name.

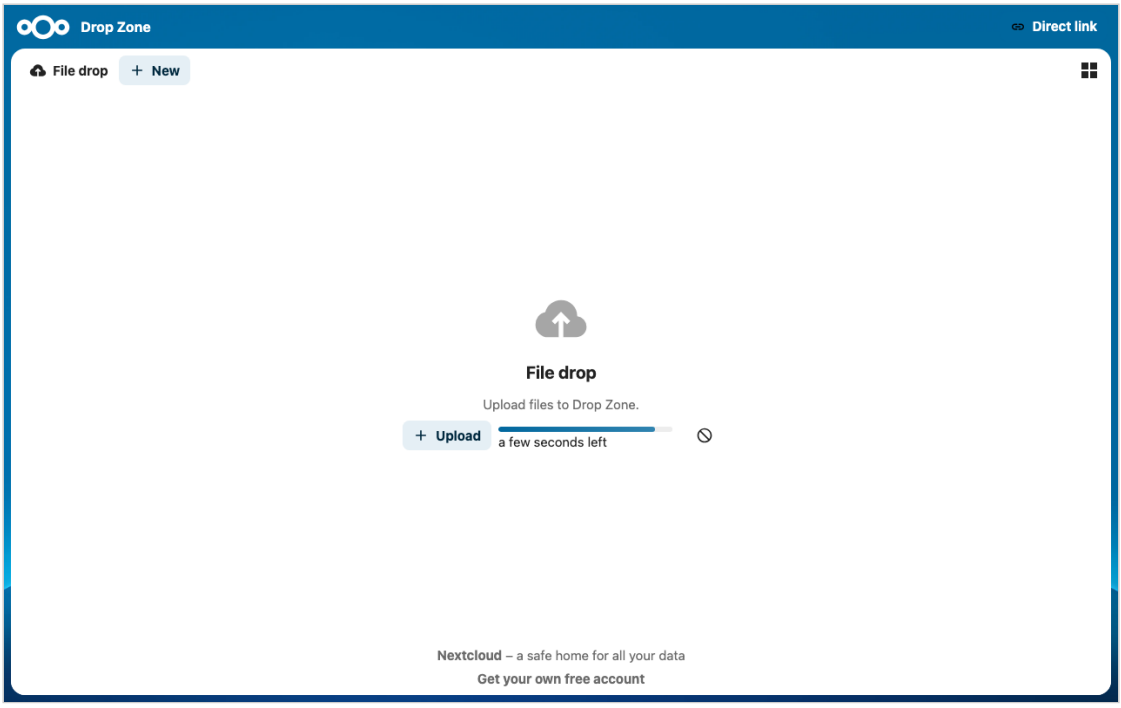


Figure 50. Uploading: the bar shows the progress. When it is gone, the file is in the teacher's folder

The teacher sees the file in `Drop Zone` (5.3).

7. Wikipedia offline

Every IDEA box of the kit has two Wikipedia apps. Both work without internet.

Console card	What it has
Wikipedia	The full English Wikipedia.
Wikipedia (Demo)	**Wikipedia in simple English**: shorter articles in easier English. The steps below use it.

> **Two cards called Wikipedia.** On a computer screen the Console's **Apps** page can show both cards with the title **Wikipedia**. Open one: the library page names the book, and **Wikipedia in simple English** is the Demo.

7.1 Look something up and use it in class

1. Open **Wikipedia (Demo)** from the Console (chapter 2). The library page shows the book.

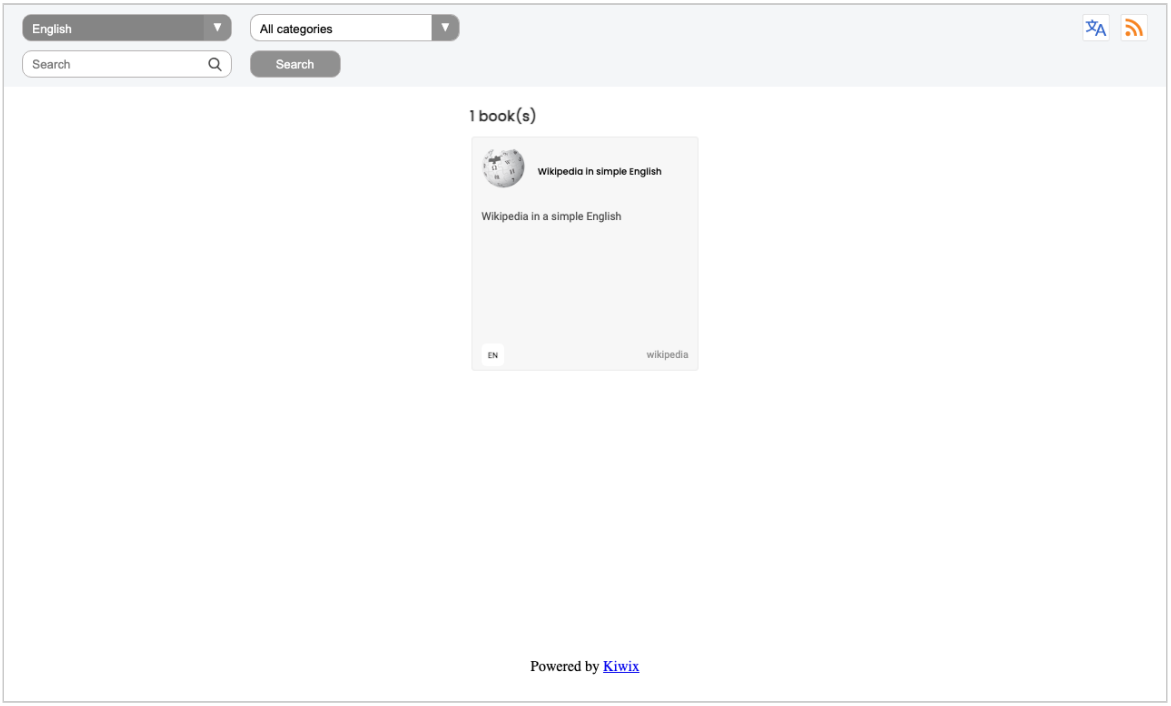


Figure 51. The library page

2. Click **Wikipedia in simple English**. Its welcome page opens.



Figure 52. Wikipedia home

3. Type in the search box at the top, e.g. `Fraction`. Pick a result.

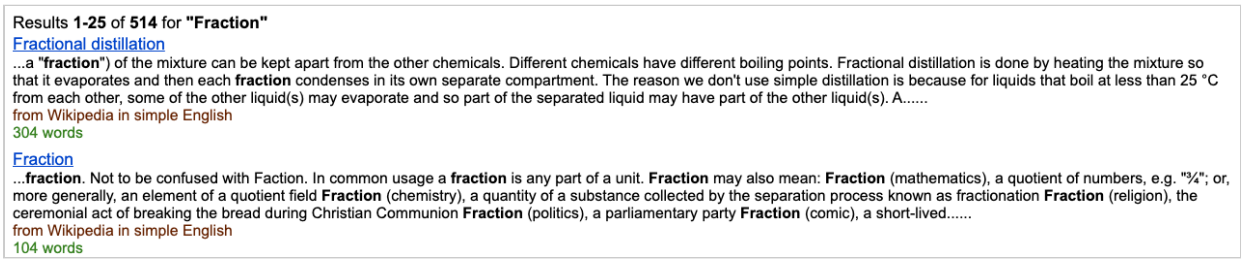


Figure 53. Search results

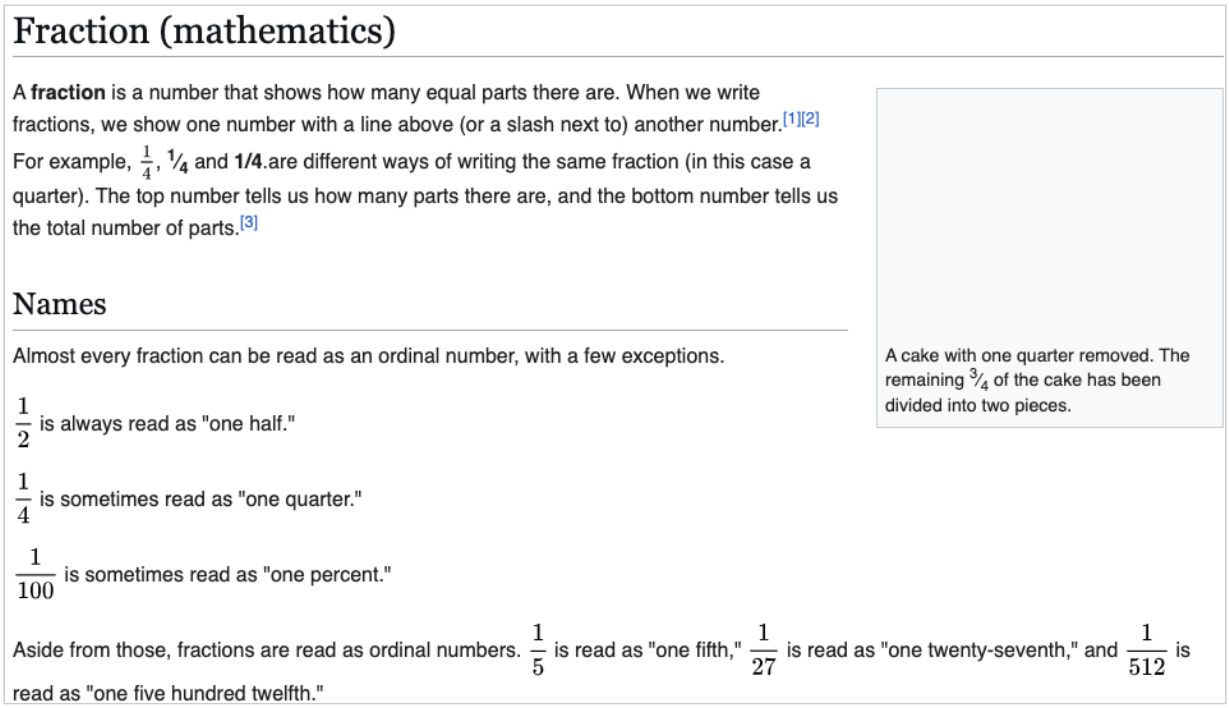
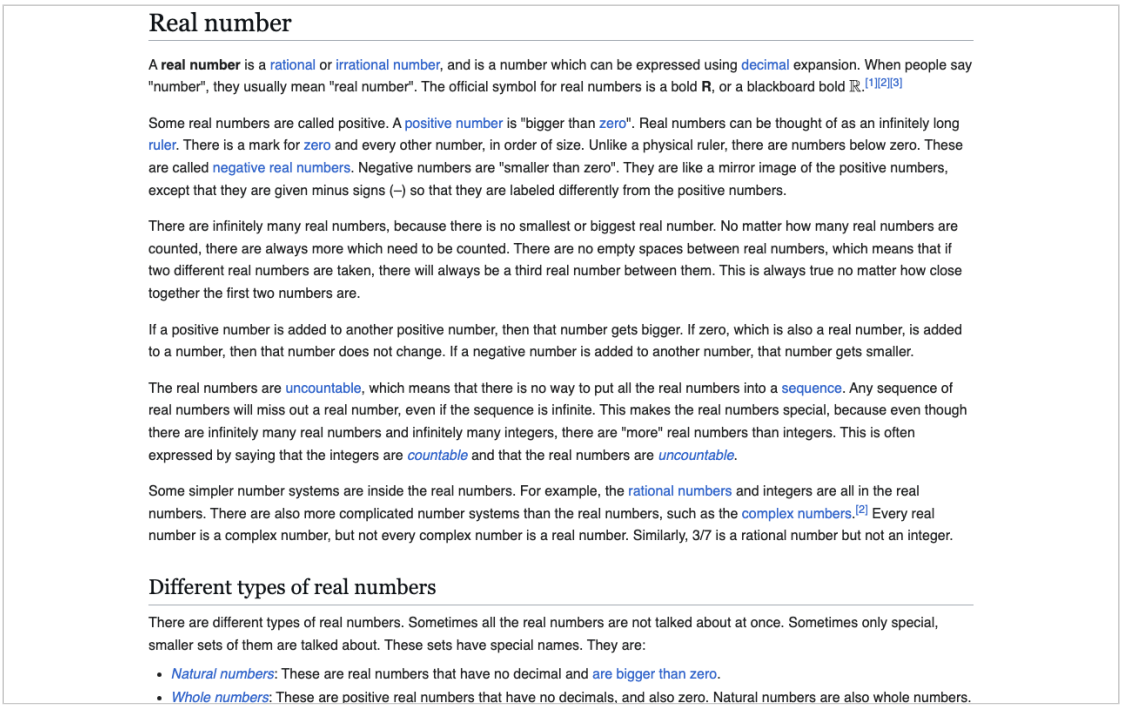


Figure 54. Fraction article

4. Follow blue links to related articles.



Real number

A **real number** is a rational or irrational number, and is a number which can be expressed using decimal expansion. When people say "number", they usually mean "real number". The official symbol for real numbers is a bold **R**, or a blackboard bold $\mathbb{R}$.[1][2][3]

Some real numbers are called positive. A positive number is "bigger than zero". Real numbers can be thought of as an infinitely long ruler. There is a mark for zero and every other number, in order of size. Unlike a physical ruler, there are numbers below zero. These are called negative real numbers. Negative numbers are "smaller than zero". They are like a mirror image of the positive numbers, except that they are given minus signs (–) so that they are labeled differently from the positive numbers.

There are infinitely many real numbers, because there is no smallest or biggest real number. No matter how many real numbers are counted, there are always more which need to be counted. There are no empty spaces between real numbers, which means that if two different real numbers are taken, there will always be a third real number between them. This is always true no matter how close together the first two numbers are.

If a positive number is added to another positive number, then that number gets bigger. If zero, which is also a real number, is added to a number, then that number does not change. If a negative number is added to another number, that number gets smaller.

The real numbers are uncountable, which means that there is no way to put all the real numbers into a sequence. Any sequence of real numbers will miss out a real number, even if the sequence is infinite. This makes the real numbers special, because even though there are infinitely many real numbers and infinitely many integers, there are "more" real numbers than integers. This is often expressed by saying that the integers are *countable* and that the real numbers are *uncountable*.

Some simpler number systems are inside the real numbers. For example, the rational numbers and integers are all in the real numbers. There are also more complicated number systems than the real numbers, such as the complex numbers.[2] Every real number is a complex number, but not every complex number is a real number. Similarly, 3/7 is a rational number but not an integer.

Different types of real numbers

There are different types of real numbers. Sometimes all the real numbers are not talked about at once. Sometimes only special, smaller sets of them are talked about. These sets have special names. They are:

- *Natural numbers*: These are real numbers that have no decimal and are bigger than zero.
- *Whole numbers*: These are positive real numbers that have no decimals, and also zero. Natural numbers are also whole numbers.

Figure 55. Linked article

> **Classroom idea.** Give learners 3 questions to answer from one article; they read on their own device.

8. Your own files: the Files Disk

A **Files Disk** is an **IDEA Disk** that the operator prepares in the Console (chapter 11). Its content appears in Nextcloud as the folder **School Files**. Use it to bring in material you downloaded elsewhere, for example worksheets you copied at another school. On the kit, every IDEA box has its own Files Disk.

Good to know:

- **School Files** is only in **Nextcloud**, not in Kolibri.
- Only teachers see the **School Files** folder. Learners get only the files you share with them.
- **The Backup Disk does not save School Files.** Keep your own copy of important files.

8.1 Open School Files

1. Open Nextcloud from the Console and sign in. In **Files** you see the folder **School Files** (marked as **External storage** in the left menu).

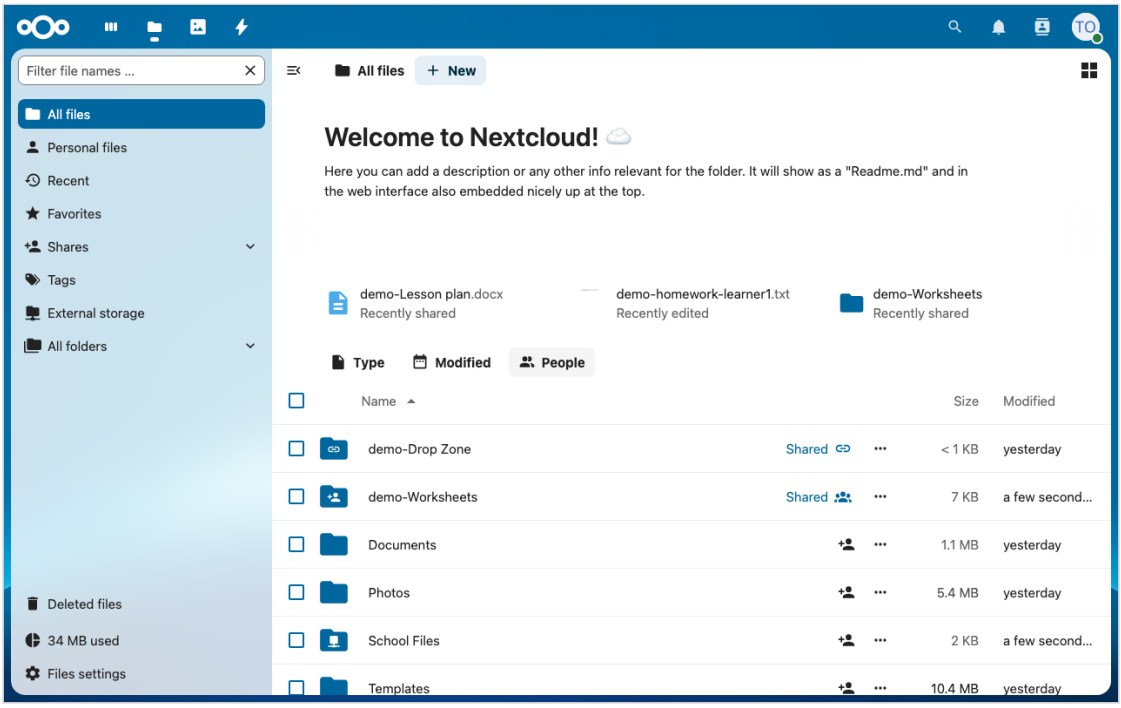


Figure 56. School Files in Nextcloud

2. Open **School Files** to see what is on the disk (here `Fractions-Worksheet-1.pdf`).

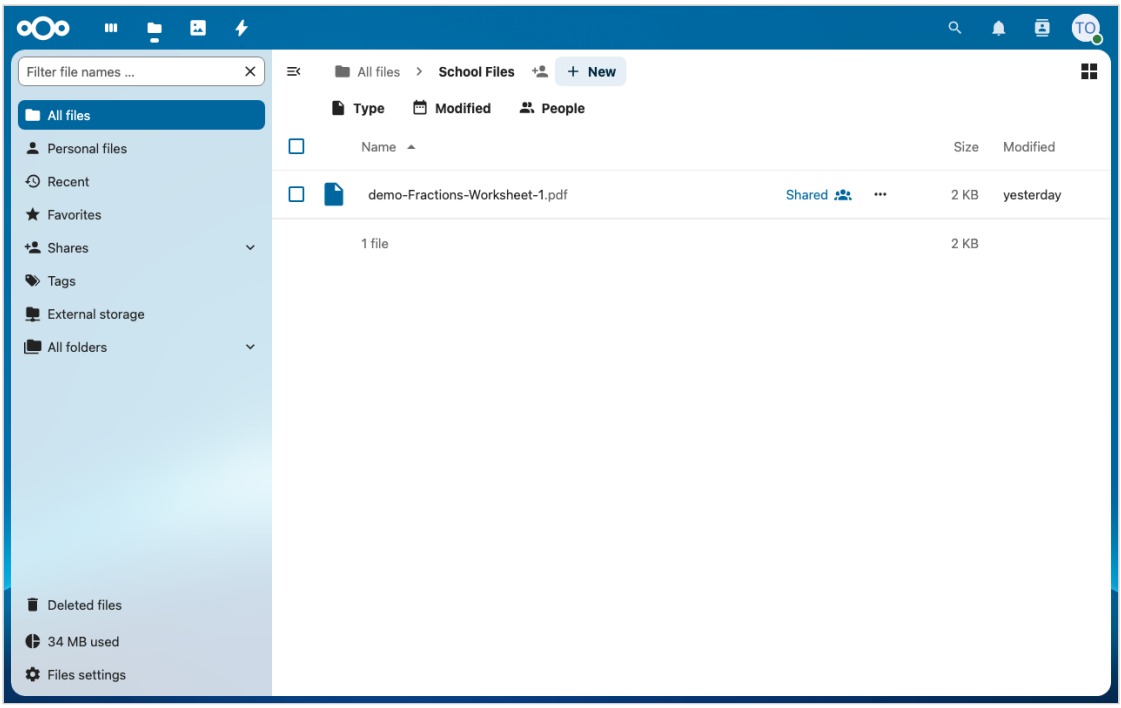


Figure 57. School Files content

8.2 Share a file from School Files with your class

1. On the file's row click **··· (Actions) > Show sharing options**, or click the share icon.
2. In **Internal shares**, type the class group name (here `Grade 5`; in Nextcloud Demo on the kit, `learners`) and choose it from the list. Keep **View only**. The group then appears under **Internal shares**.

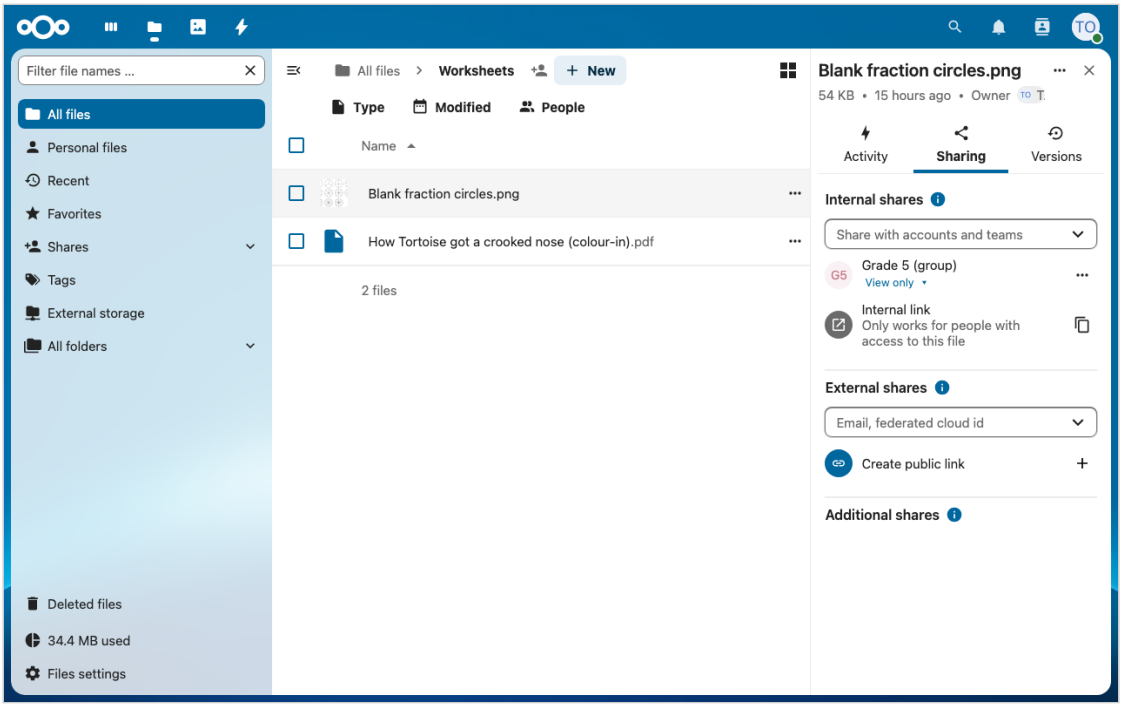


Figure 58. A file shared with Grade 5 (here a worksheet)

3. A learner in that group signs in and finds the file in **All files** and under **Recently shared**.

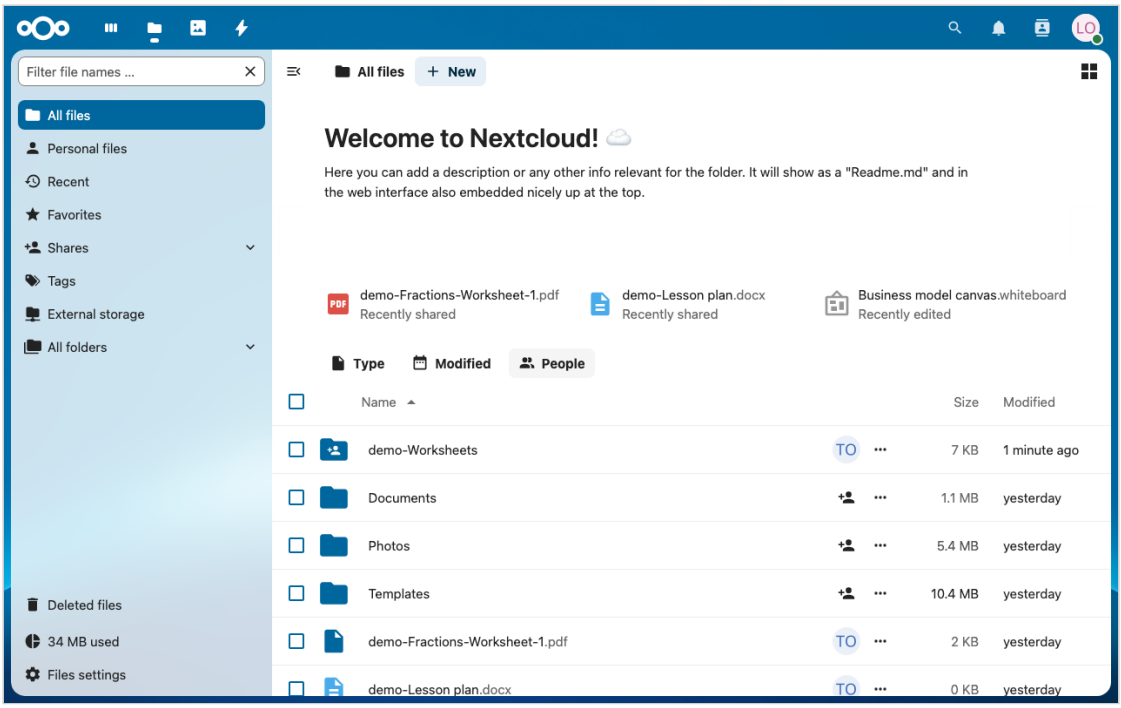


Figure 59. A learner sees the shared file

9. Keeping data safe: the Backup Disk

The operator looks after backups (chapter 11). As a teacher you should know:

- A **Backup Disk** is a separate **IDEA Disk** with the **backup** badge in the Console.
- It saves **every app in full**: Kolibri (classes, learners, progress, and the channels), Nextcloud (accounts, and the files inside Nextcloud) and Wikipedia. On the kit that is about 164 GB per IDEA box, which fits on its 240 GB Backup Disk.
- It does **not** save the **School Files** folder (chapter 8).
- **During its backup, an app is stopped and can't be opened.** It starts again by itself when its backup is done. The first backup of all apps takes about 2 h 15 min, so backups are planned outside lesson time.
- How often backups run (on request, each time the disk is plugged in, or on a schedule) is set by your IDEA contact.
- If the IDEA box breaks, the operator restores the latest backup onto a spare disk. Only the work done after the last backup is lost.
- Never unplug a disk while a backup is running.

Part B: For the school operator

10. The Console for the operator

This chapter is for the school operator.

10.1 Operator Login

1. In the Console, click the person icon (top right). The **Operator Login** panel opens with a username and a password field (the eye icon shows or hides the password).

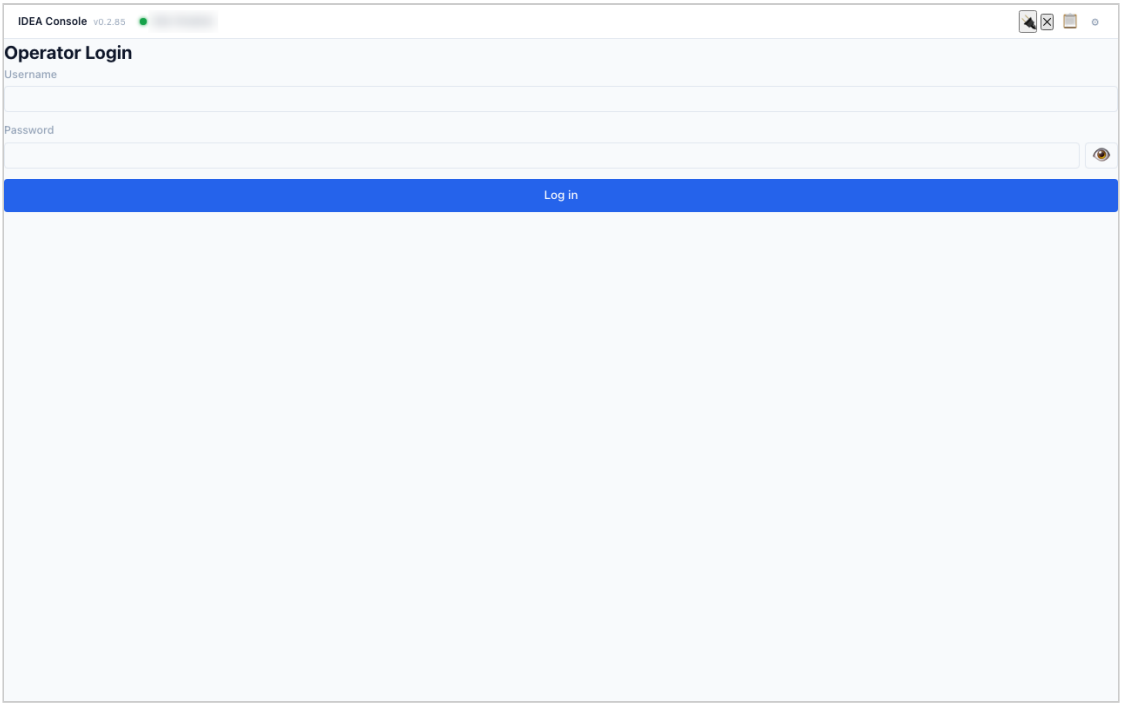


Figure 60. Operator Login

2. Type the operator username and password and click **Log in**. The **Account** panel shows your name, the role **Operator**, **Change Password**, **Manage Operators** and **Log out**.

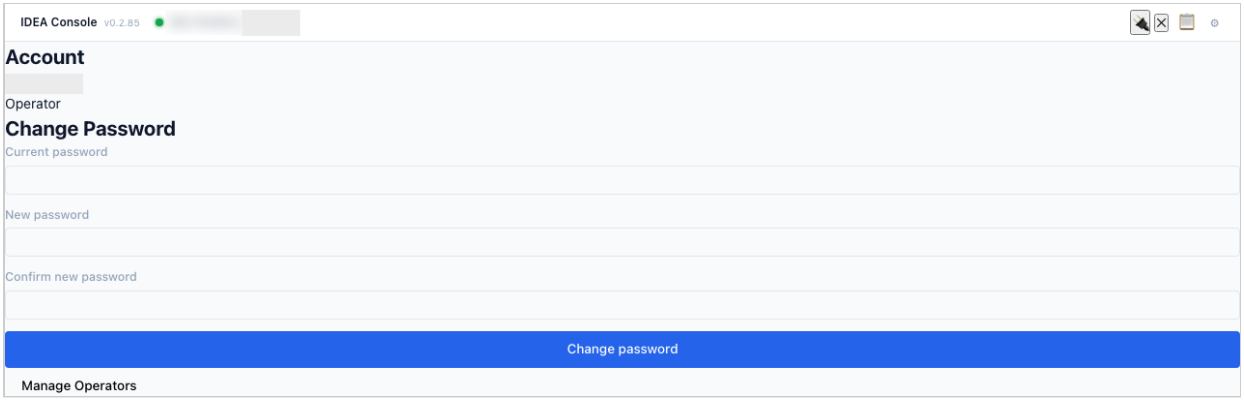


Figure 61. Account panel

3. Close the panel with ×. Use **Log out** in the same panel when you are done.

10.2 The Network view: the IDEA box and its disks

1. After login the left side shows **NETWORK**: **ALL APPS**, then the IDEA box with a green dot (online) and its disks. A box with a **red dot** is offline: switched off, or at another school.

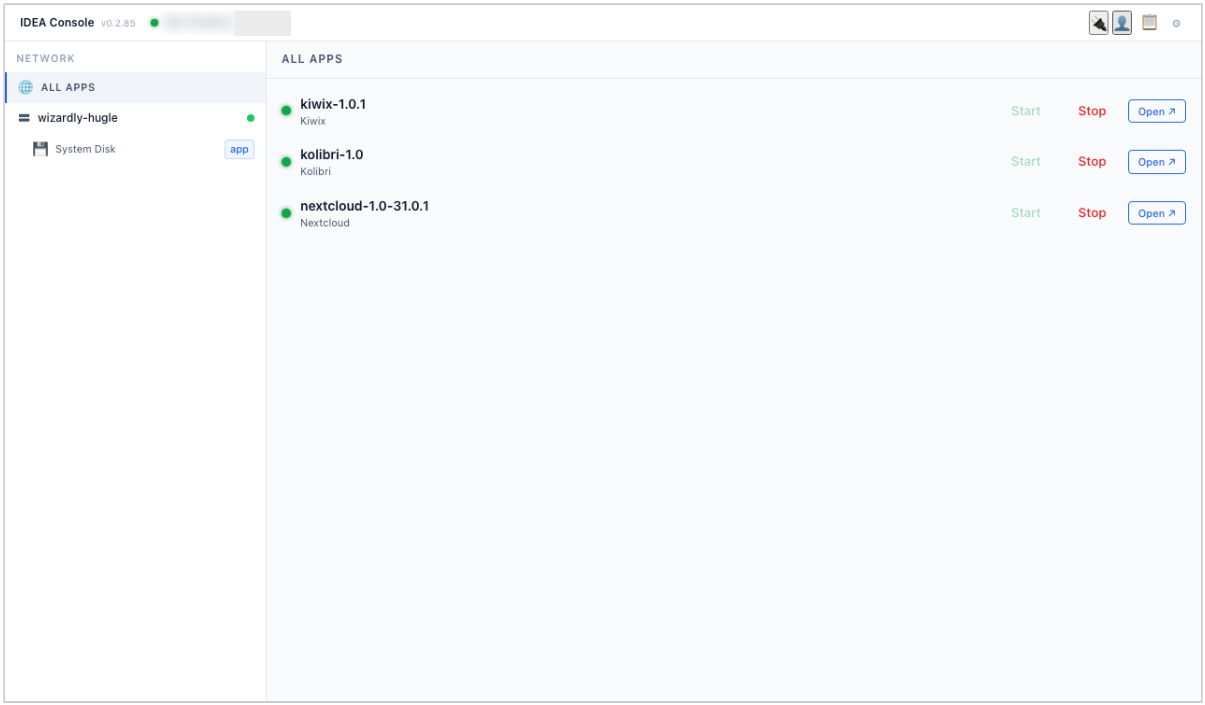


Figure 62. All apps

2. Click the IDEA box name to select it, then click a disk (here **System Disk**). Each disk shows a role badge: **app**, **backup**, **files** or **empty**. The right side lists the apps on that disk.

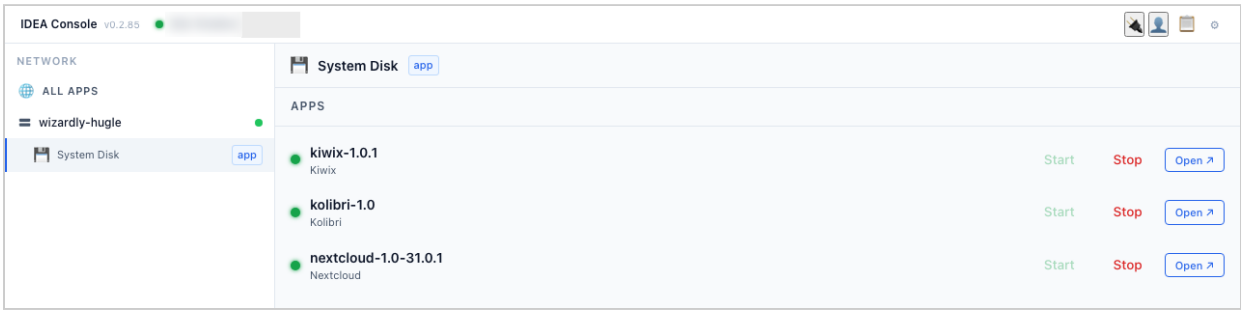


Figure 63. Disk view

10.3 App status

1. Each app row shows the app name and version, a coloured status dot, and the buttons **Start**, **Stop** and **Open ↗**.

2. Click the status dot (**Show details**) to see **STATUS** (for example **Running**), **VERSION**, **PORT**, **CREATED**, **LAST STARTED**, **LAST BACKUP** and the description.

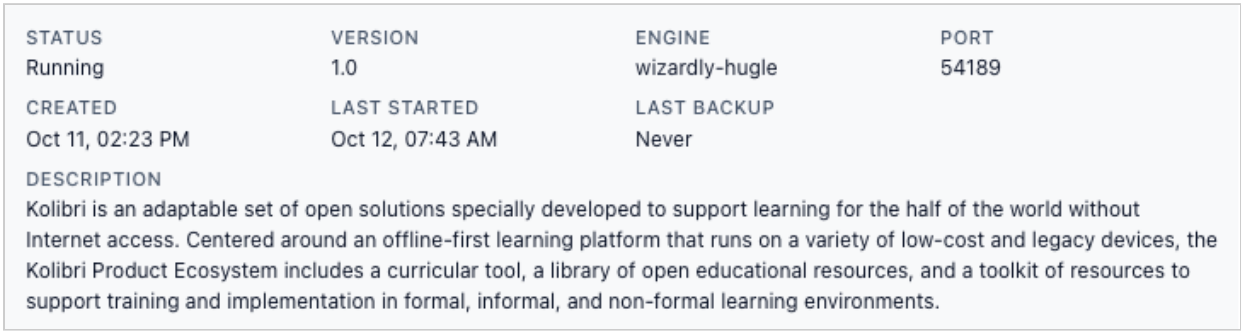


Figure 64. App details

3. To check that an app really works, click **Open ↗**: the app's page must load.
4. Only start or stop an app when your IDEA contact asks you to. Do not touch apps your IDEA contact did not mention.
5. **Known problem.** The apps of an IDEA box with a red dot still show as running, in **ALL APPS** and on the teachers' **Apps** page, but they won't open. There is no fix yet. Do not try to start or stop them: just tell the teachers to use the apps of your own box.

10.4 Command History

The clipboard icon (top right) lists recent Console commands with ✓ (done) or ✗ (failed) and how long ago. Read it out to your IDEA contact when you report a problem.

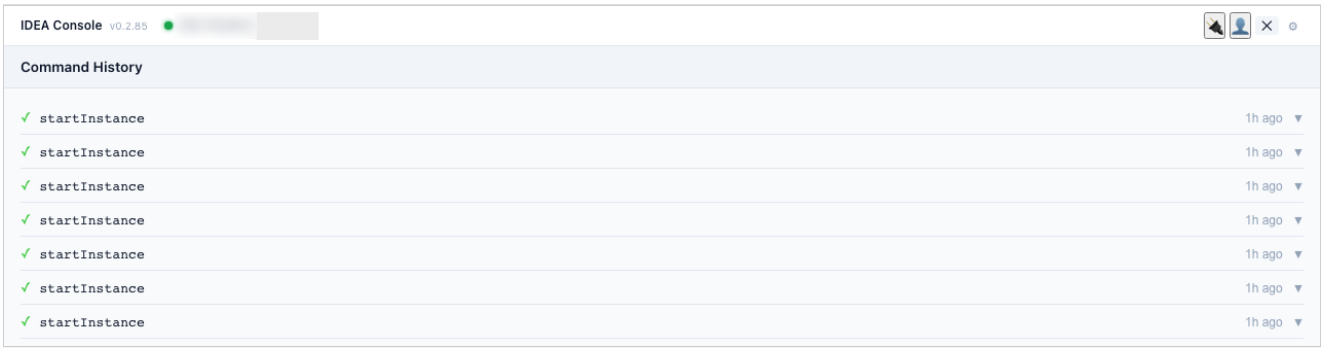


Figure 65. Command History

10.5 What the buttons do

- **Start** and **Stop** start or stop an app. A stopped app cannot be opened.
- **Install App** puts a new app on a disk.
- **Make this a Files Disk** turns an empty IDEA Disk into a Files Disk for Nextcloud (chapter 11).
- **Make this a Backup Disk** turns an empty IDEA Disk into a Backup Disk (chapter 11).
- **Restore** brings an app back from the Backup Disk onto a spare disk (chapter 11).
- **Erase this disk…** deletes everything on a disk. The System Disk has no Erase button.
- The small ↺ next to the IDEA box name is **Reboot**. It restarts the whole IDEA box, so all apps are away for a few minutes. Use it outside lesson time.

11. Disks: Files Disk and Backup Disk

This chapter is for the school operator.

11.1 A new disk (not yet an IDEA Disk)

1. A new disk appears under the IDEA box with its model and size and an **UNFORMATTED** badge.

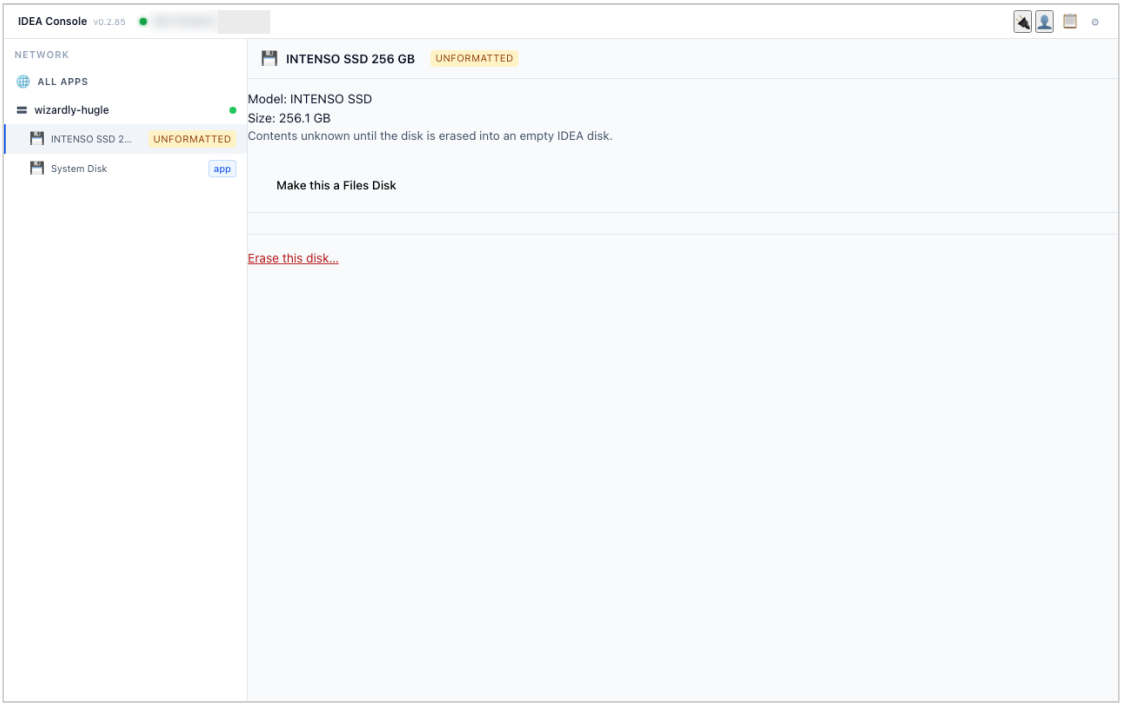


Figure 66. Unformatted disk

2. Its view shows the model, the size and two buttons: **Make this a Files Disk** and **Erase this disk…**.

USB sticks. A teacher's FAT32 or exFAT stick shows as **USB stick: (name) (read-only)**. Do not erase it: **Erase this disk…** turns the stick into an IDEA Disk and deletes the teacher's files, so only do it when asked.

11.2 Erase a disk

1. Click **Erase this disk…**. The confirmation shows the disk name, model, size and contents, and the red warning "This erases everything on the disk…".
2. Check the **size and contents**: several disks can have the same name (for example **IDEA Disk**). Make sure it is not the Backup Disk.

3. The box above the text field says **TYPE IDEA DISK TO CONFIRM** (the disk name in capitals). Type the name as it is written on the disk, for example `IDEA Disk`. The **Erase this disk** button stays grey until the name is right. Press **Cancel** if anything looks wrong.

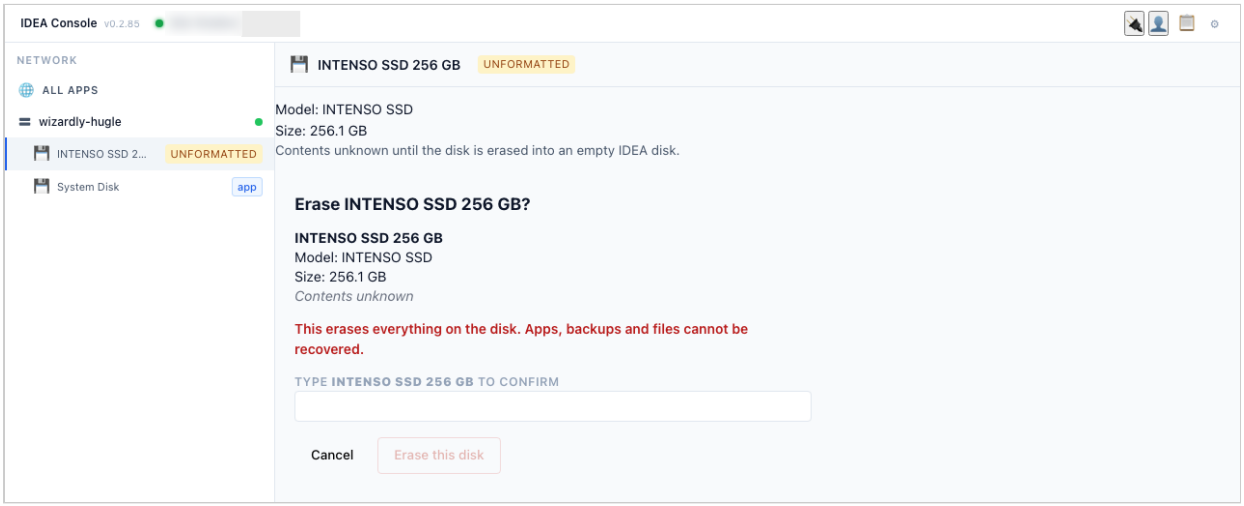


Figure 67. Erase confirmation

4. After the erase, the disk shows as an **IDEA Disk** with the **empty** badge.

> **A Files Disk that Nextcloud is using cannot be erased.** Ask your IDEA contact to stop Nextcloud first.

11.3 Make a Files Disk

1. Select the **IDEA Disk** marked **empty**. The view asks "What would you like to do with this disk?": **Make this a Files Disk**, **Make this a Backup Disk** or **Install App**.

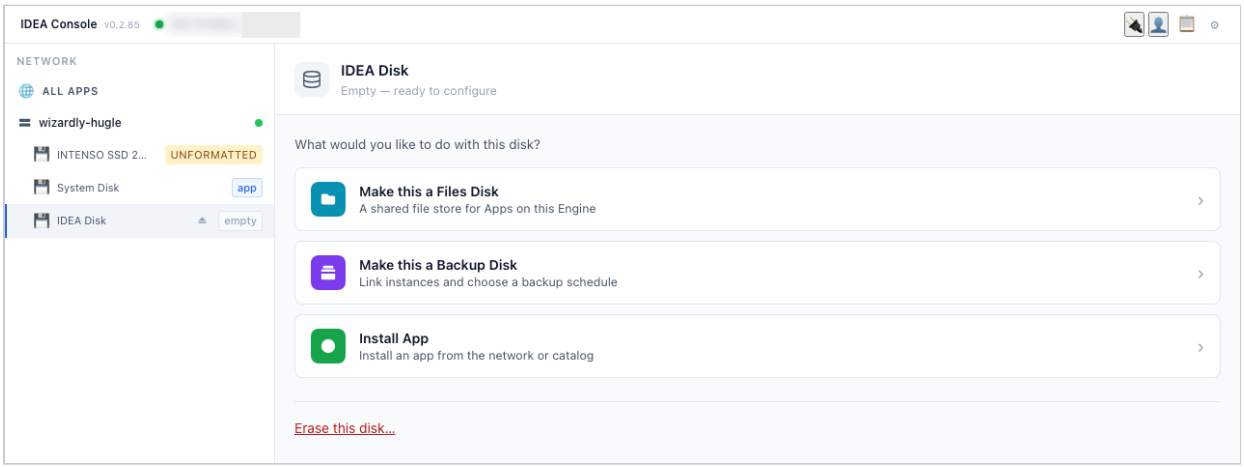


Figure 68. Empty IDEA Disk

2. Click **Make this a Files Disk**. The panel says "Nothing on the disk is erased", with **SHARE NAME** filled in as **School Files**. Click **Make this a Files Disk**.

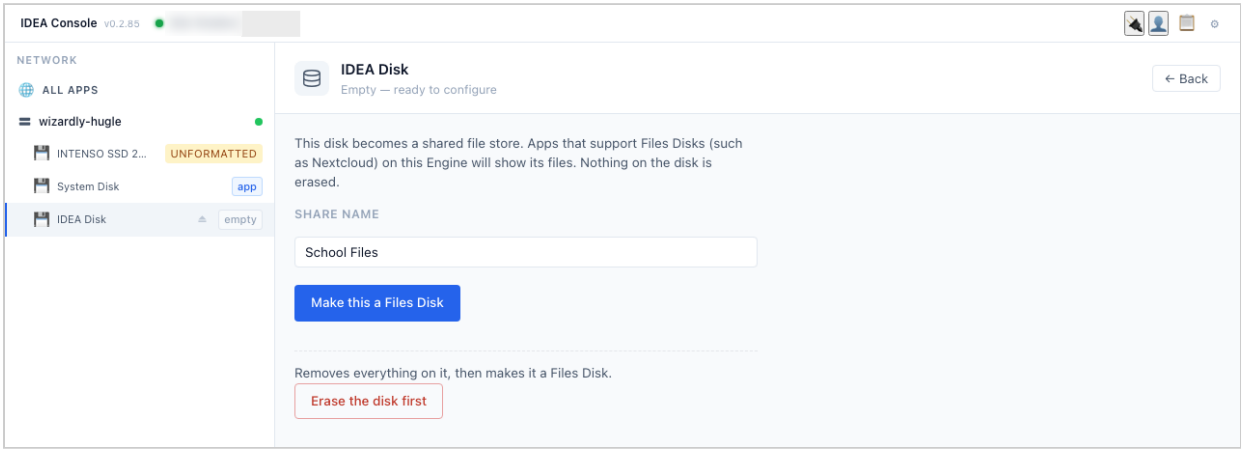


Figure 69. Make this a Files Disk

3. After a short wait the disk shows the **files** badge, "Share name: School Files", the free space, an **Eject** button and **Erase this disk…**. Once your IDEA contact has linked it, the view also says **Available in: Nextcloud**. Teachers then find it in Nextcloud (chapter 8).

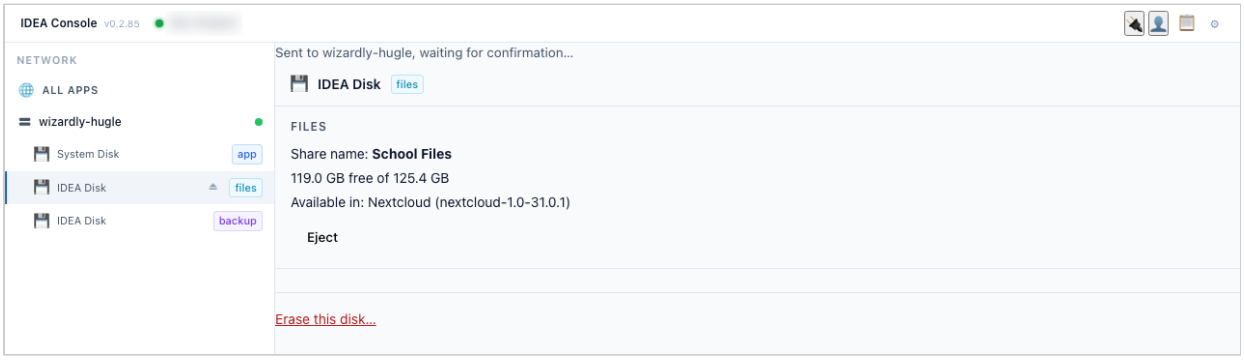


Figure 70. Files Disk available in Nextcloud

11.4 Eject a disk

An IDEA Disk has a small eject icon next to its name in the Network list and an **Eject** button in its disk view. The System Disk has none. Never eject or unplug a disk during a backup.

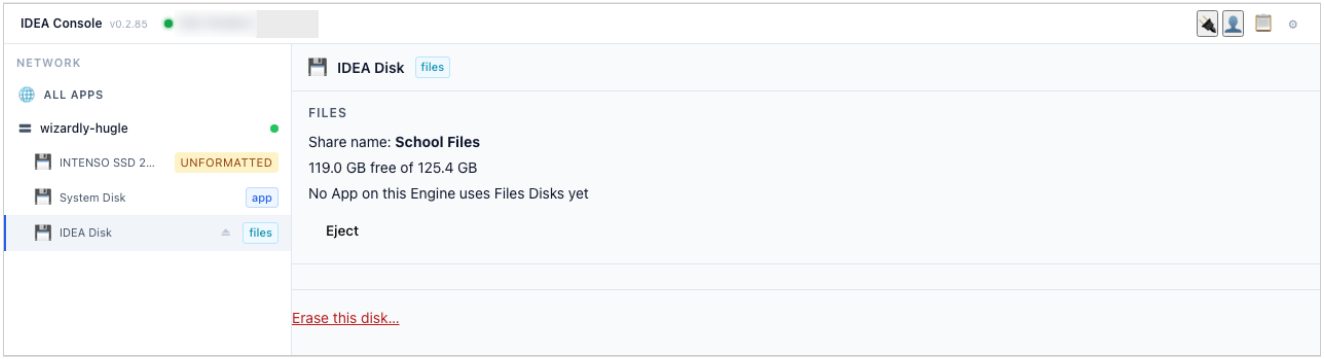


Figure 71. Eject button on a Files Disk

11.5 Make a Backup Disk

1. Select an **IDEA Disk** marked **empty** and click **Make this a Backup Disk**.

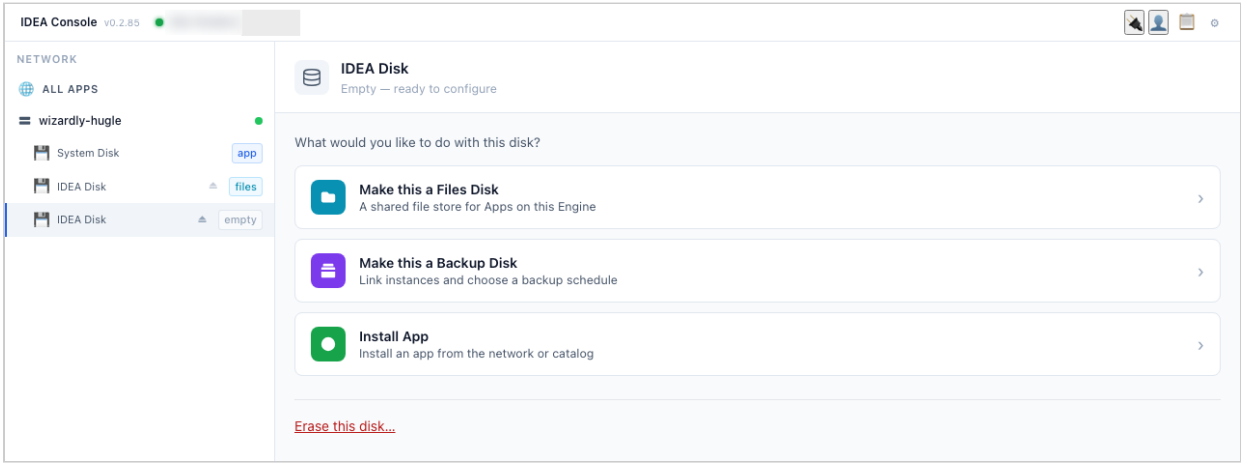


Figure 72. Empty disk options

2. Choose when backups run: **On demand** (only when you press **Back up**), **Immediate** (as soon as the disk is plugged in) or **Scheduled**. Your IDEA contact tells you which.

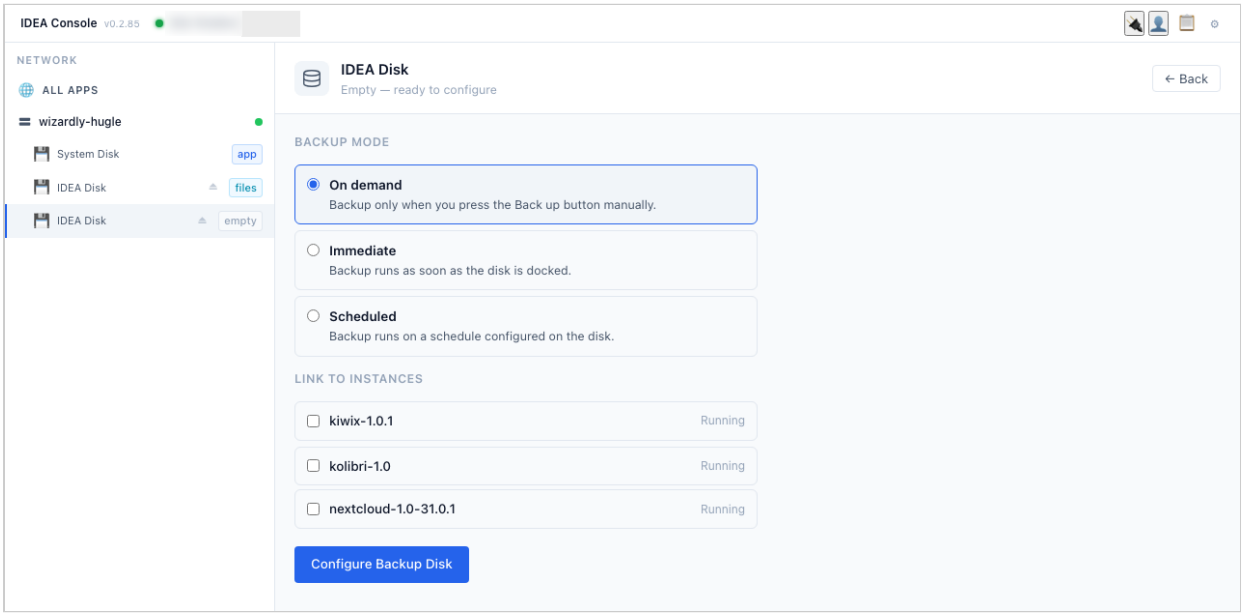


Figure 73. Backup mode

3. Under **LINK TO INSTANCES**, tick every app of this IDEA box (on the kit: all six). Backups are full: they include the Kolibri channels and Wikipedia, so a restore needs no internet.

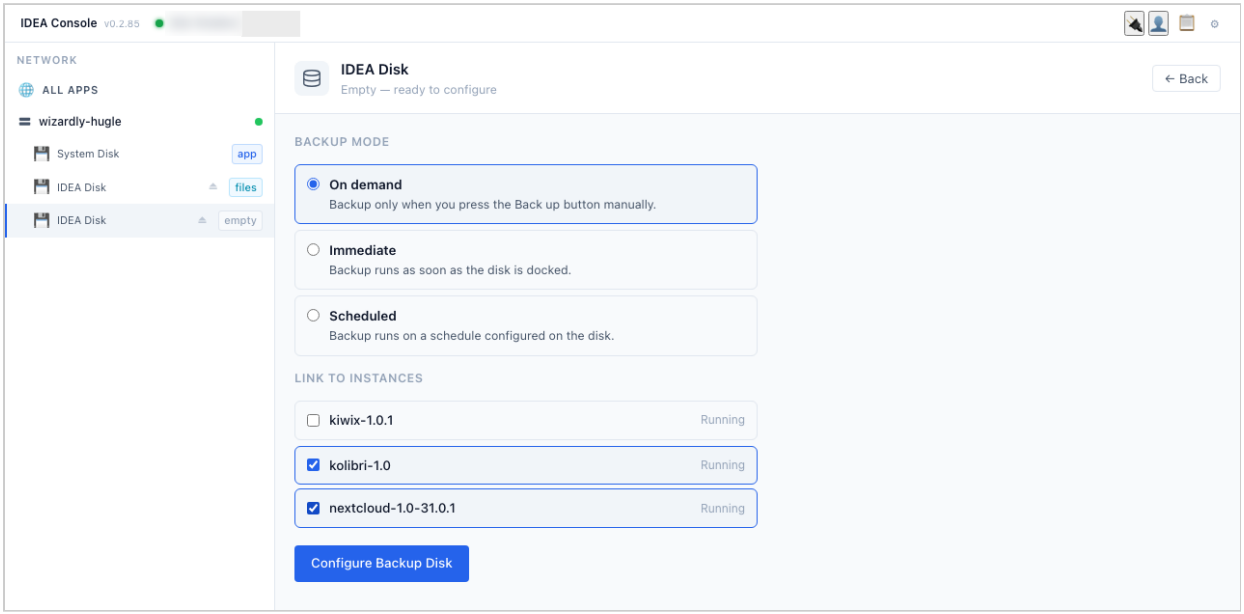


Figure 74. Linked apps

4. Click **Configure Backup Disk**. The disk gets the **backup** badge and a **Backups** list with each linked app.

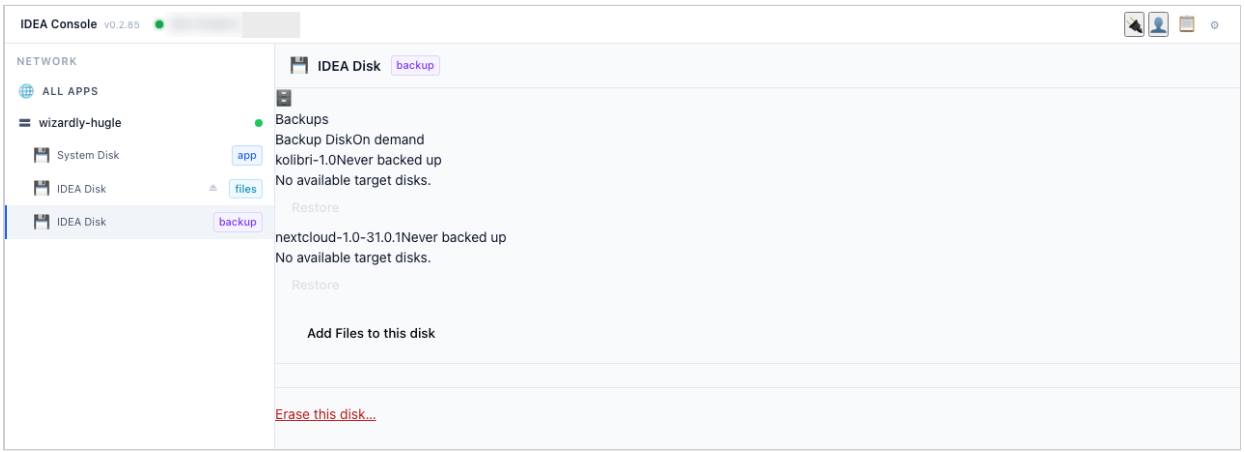


Figure 75. Backup Disk configured

11.6 Run a backup

1. In **ALL APPS**, each linked app row has a **Back up** button next to Start, Stop and Open. Click it for Kolibri. The row shows **Backing up…** with a progress bar, and the **OPERATIONS** box at the top shows the same job.

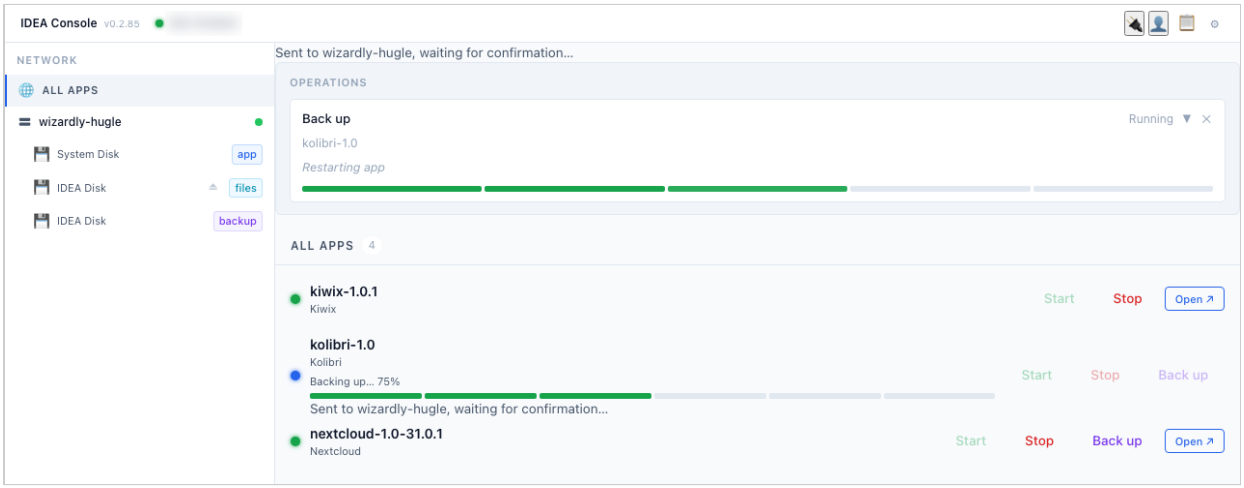


Figure 76. Back up clicked

2. The app is stopped while it is copied, then starts again by itself. When the job is finished the OPERATIONS box shows **Done**; close it with ×. A small app takes a few minutes, but the first backup of a big one takes much longer: on the kit about 37 minutes for Kolibri, an estimated 1 h 30 for the full Wikipedia and about 2 h 15 for all apps together. Do this outside lesson time.
3. When it is done, the Backup Disk's **Backups** list shows **Last backup** with the date and time for each app. Repeat for each other app.

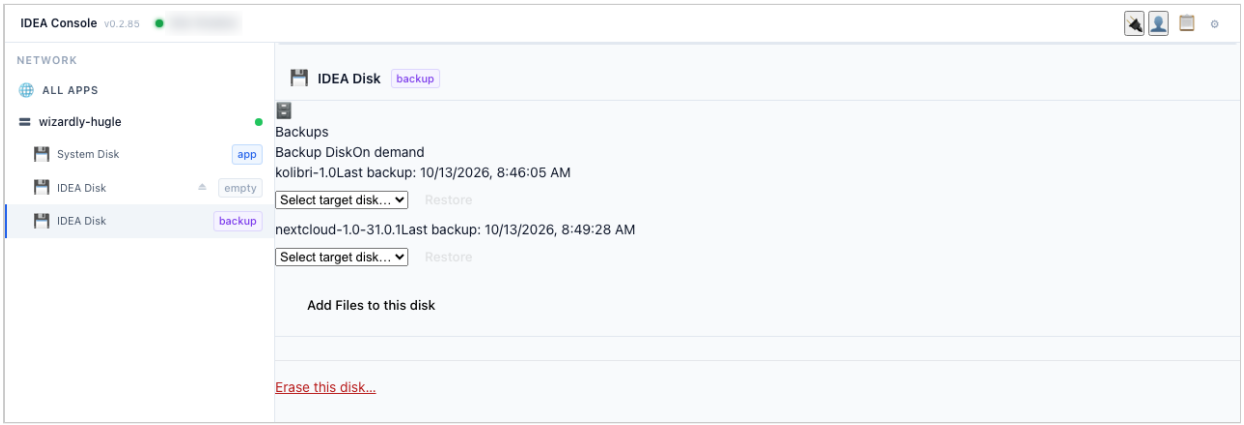


Figure 77. Last backup for both apps

11.7 Restore after a breakdown

You need a **spare disk** (an empty IDEA Disk) to restore onto.

1. Open the **Backup Disk**. Each app in the **Backups** list shows **Last backup**, a **Select target disk…** list and a **Restore** button (grey until a target is chosen).

2. In **Select target disk…** choose the spare disk. Only spare disks are offered here. Click **Restore**.

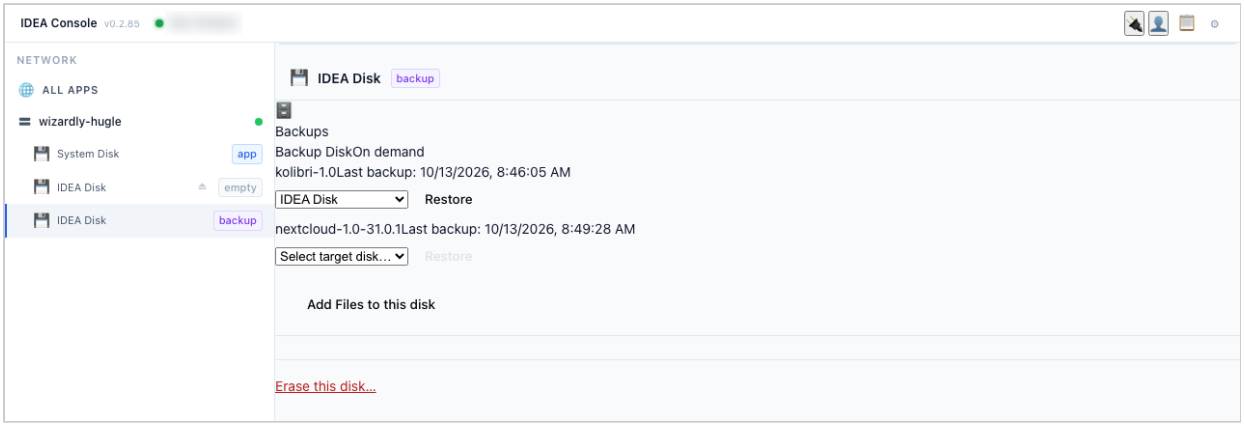


Figure 78. Target chosen

3. The Console asks: "Are you sure? This will overwrite **kolibri-1.0** on **IDEA Disk**." Click **Confirm Restore**.

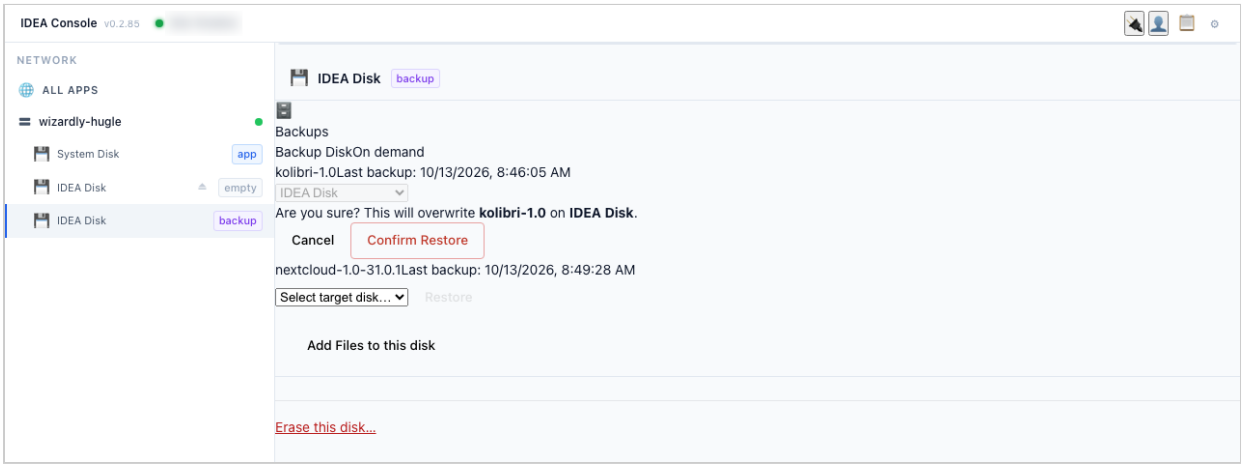


Figure 79. Confirm Restore

4. When the restore is finished, the spare disk shows the **app** badge and lists the restored app. A small app takes about a minute; Kolibri with its channels or Wikipedia takes much longer, so wait.

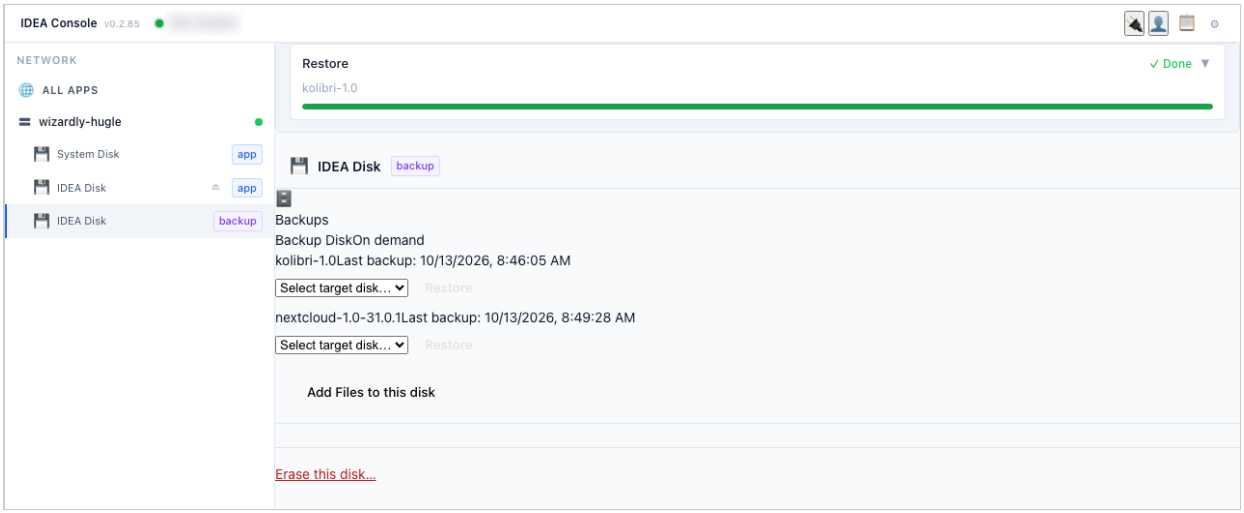


Figure 80. Restore done

5. The restore **moves** the app: it is no longer listed on its old disk, and the restored app is **stopped** (orange dot, **Start** button). Until you start it, the app cannot be opened. A restore is for a real breakdown, not for practice on a working school box.
6. Click **Start** on the restored app's row. After about 30 seconds it runs again and the row shows **Open ↗**.

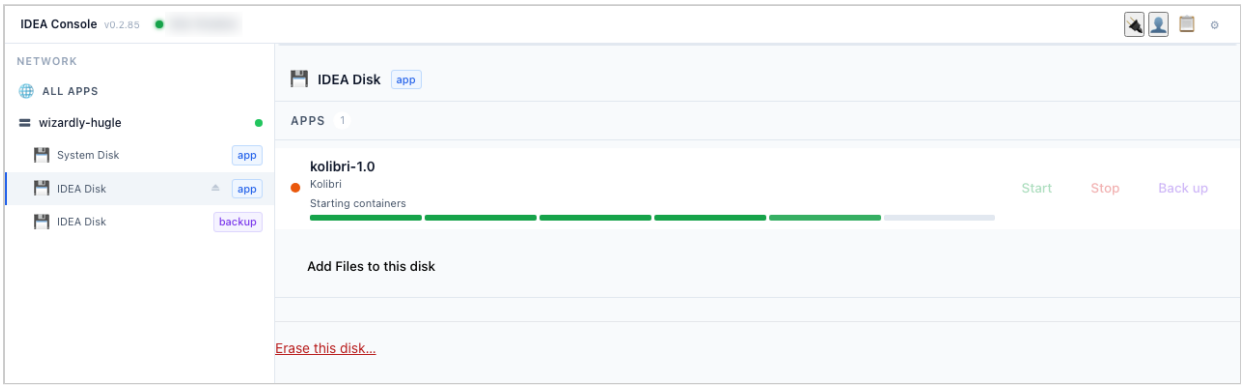


Figure 81. Start clicked

7. Open the app from its card on the **Apps** page (as in 2.3) and sign in as the teacher (coach).

8. Check that the data is back: classes, learners, lessons, quizzes and reports, as at the last backup.

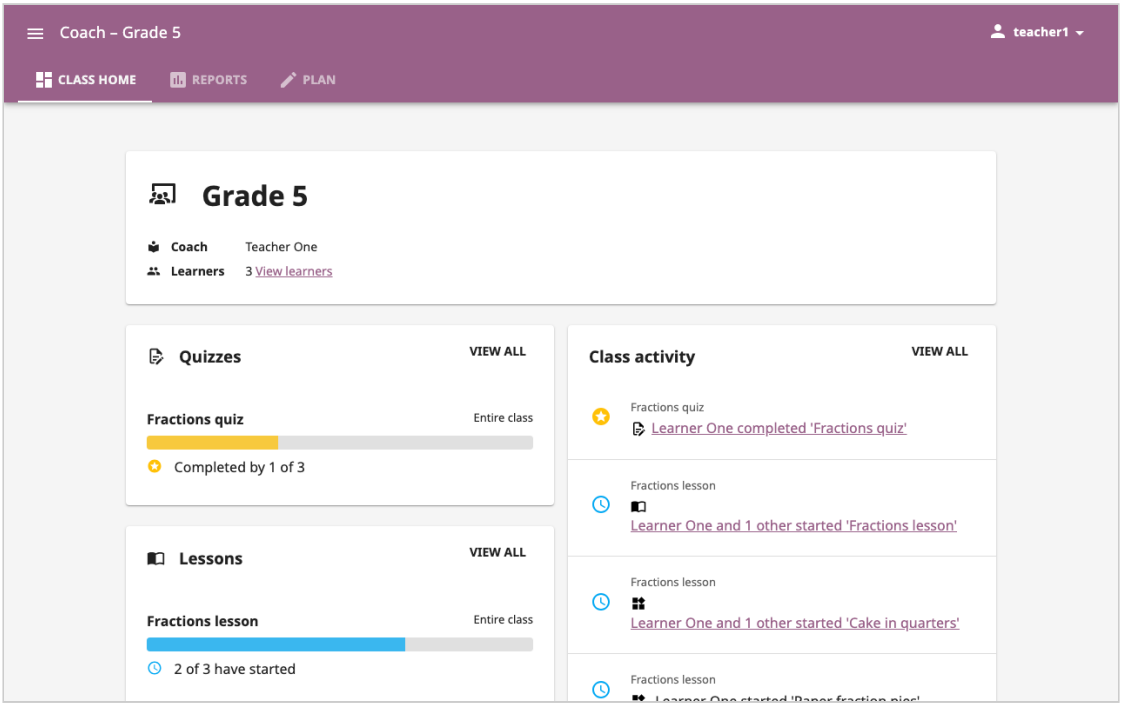


Figure 82. Class after restore

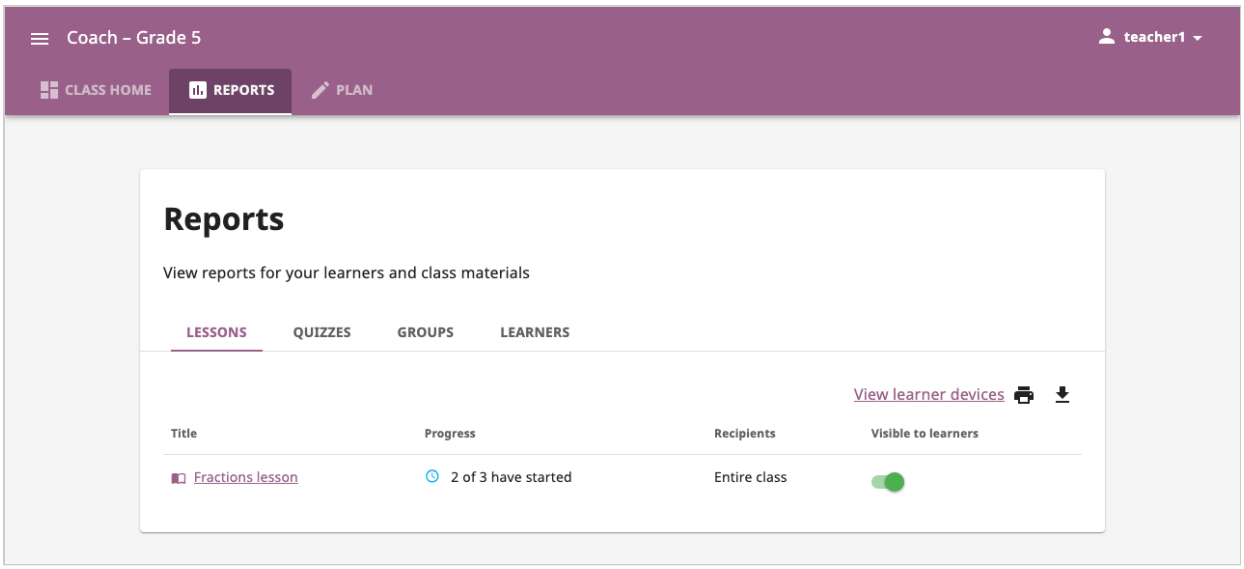


Figure 83. Reports after restore

9. Later, your IDEA contact can move the app back to the System Disk and prepare the spare disk again.

Important. Anything done after the last backup is lost. Make a new backup of the restored app as soon as it runs.

12. Student computers: reset a card from the Console

This chapter is for the school operator.

Every student computer (Raspberry Pi 400 or Pi 500) starts from a **microSD card** with the same IDEA student image. When a computer goes wrong (it no longer starts, the screen looks different, the card is full or a student changed settings), you do not repair it. You **reset the card**: the IDEA box writes a fresh copy of the student image onto it. This takes about 5 to 10 minutes.

Good to know:

- **Resetting wipes the card.** Anything saved on the computer itself (the local Documents folder) is gone. Students' work in **Nextcloud** stays: that is why students save there (chapter 6).
- Every card is the same, so a card works in **any** student computer, Pi 400 or Pi 500.
- The student computers **do not update themselves**. New software comes only with a new, tested image version, which you get by resetting the card.
- **A restart already undoes most problems.** Every restart puts the computer back as its card was written (files and settings saved on the computer are gone). Try a restart first; reset the card only when that doesn't help.
- You need a **USB card reader** plugged into the IDEA box. Any card reader works.

12.1 Before you start

1. Switch the student computer off and take the microSD card out. Note the computer's name from its sticker (for example `idea-pc-3f2a`).
2. Put the card into the card reader, and plug the card reader into a free USB port of the IDEA box.
3. Open the Console and sign in with **Operator Login** (chapter 10).

12.2 Find the card

1. Click the IDEA box in the list on the left. Scroll down to **CLIENT CARDS**. It shows the client image the box will write, and every card in a card reader: the reader's name, the card's size and what is on it now.

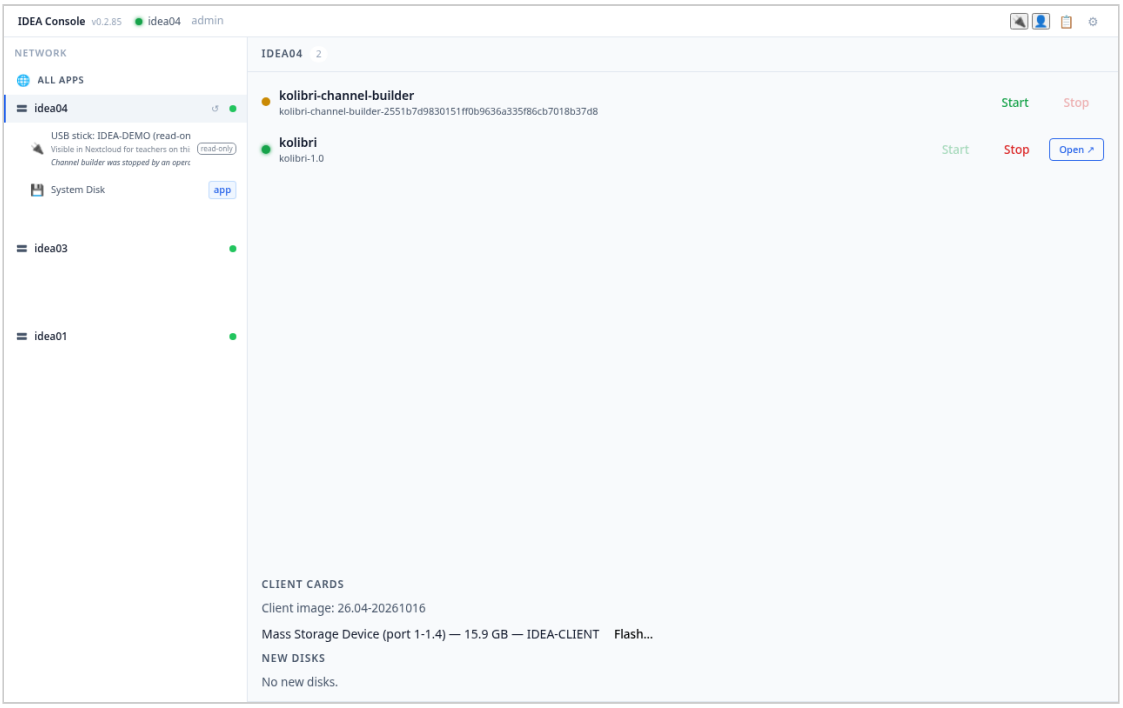


Figure 84. Client cards: one card in a reader, with Flash...

2. Check that the **size** matches your card. If the card is not listed, see 13.5.

> **Never pick an IDEA Disk.** The Files Disk, the Backup Disk and app disks are never offered here, but always check the name and size before you continue.

12.3 Reset (flash) the card

1. The card belongs to the IDEA box you flash it from: its **IDEA** shortcut tries that box first. Flash it from the IDEA box of the classroom it will be used in.
2. Click **Flash…** on the card's row.
3. Read the panel **FLASH CLIENT CARD**. It repeats the card, its size, what is on it now and the image. It says in red that everything on the card will be wiped. If anything looks wrong, click **Back**.

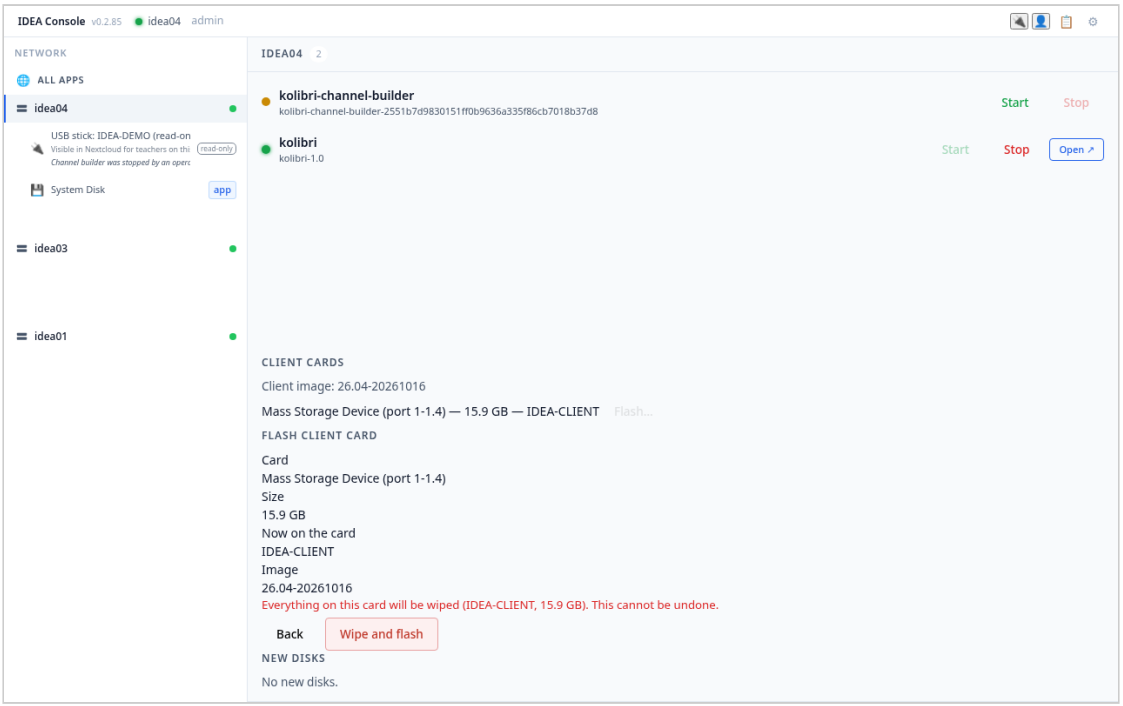


Figure 85. Flash client card: check the card, then Wipe and flash

4. Click **Wipe and flash**. The Console shows the progress in steps, for example **Step 3 of 6: Writing**.

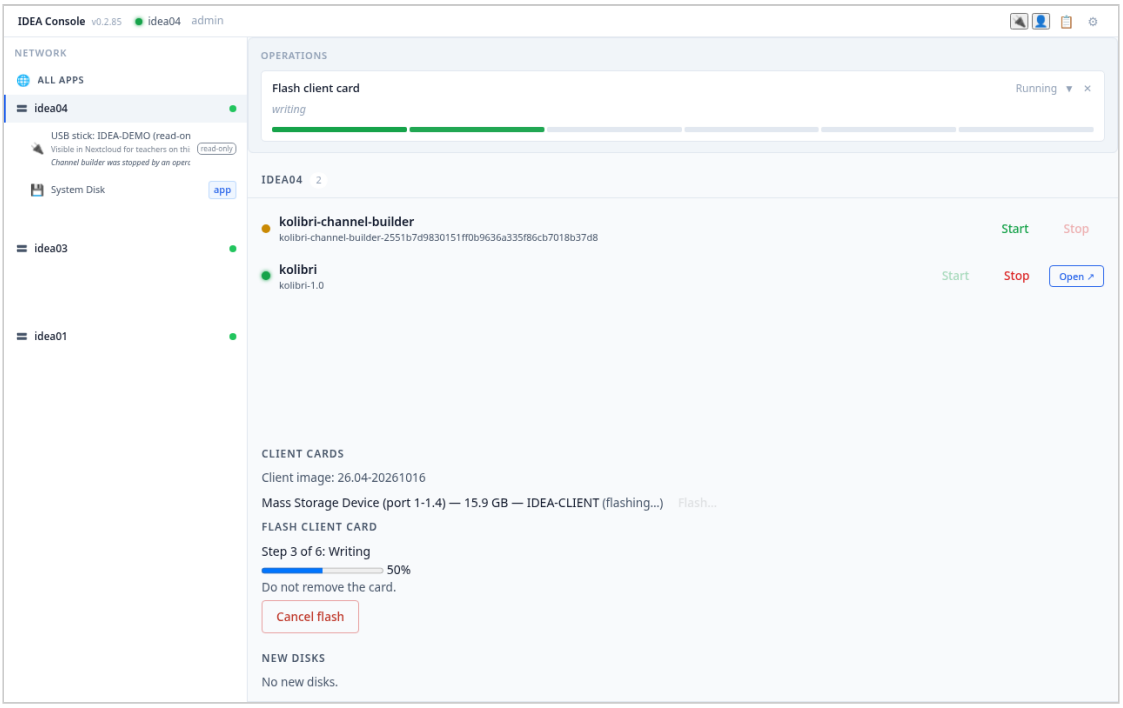


Figure 86. Flash in progress

5. Wait for **Done – safe to remove**. Do not pull the card reader out before that. Then click **Close**.

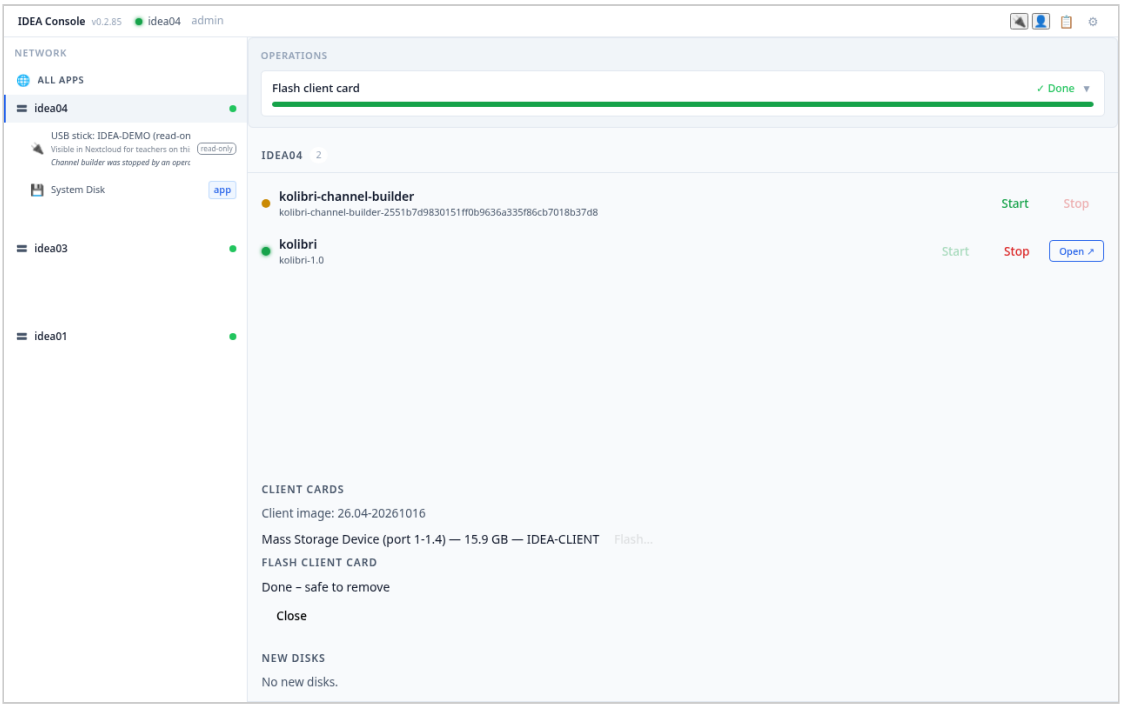


Figure 87. Done – safe to remove

6. Take the card out of the reader and put it back into the student computer. Switch the computer on.

12.4 After the reset

1. The first start takes a few minutes longer: the card sets itself up and names the computer after its Pi (the name on the sticker).
2. The computer opens the student desktop by itself, with no password. The desktop has one shortcut, **IDEA**.

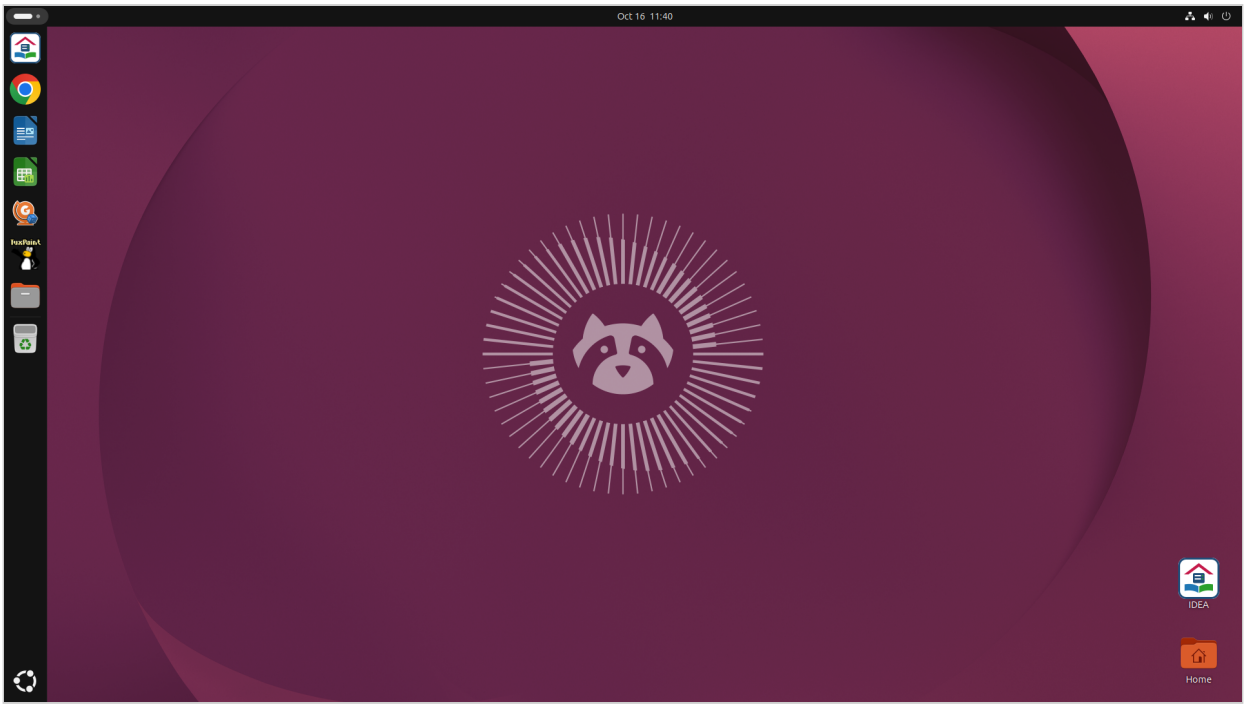


Figure 88. The student desktop after the first start, with the IDEA shortcut

3. Double-click **IDEA**. Chrome opens the start page, finds the IDEA box and opens the Console (chapter 2). Check that **Open** on the Kolibri card works.

12.5 Questions

Question	Answer
The card is not listed.	Check that the card is pushed fully into the reader and that the reader is plugged into **this** IDEA box. Try another USB port.
The Flash button is not there or the Console refuses.	The IDEA box never writes to an IDEA Disk, the system disk, or a disk that is in use. Check that you picked the card and not another disk. Otherwise ask your IDEA contact.
The flash stopped with an error.	Flash again. If it fails again, the card is probably worn out: use a new card.
The computer does not start after the reset.	Try the card in another student computer. If it starts there, the computer may need a bootloader update: ask your IDEA contact.
Can I install an extra program on one computer?	No. It would be lost at the next reset. Ask your IDEA contact to add it to the next image version.
The IDEA shortcut says No IDEA server found.	Check the network connection. Type the name on the IDEA box's sticker in the field on the start page (chapter 2). If it keeps happening, the card may belong to an IDEA box that has been switched off or moved: flash it again from the right box.
Can my IDEA contact do this remotely?	Yes. Put the card in the reader and tell them. They press Flash from the Console, and you put the card back when the Console says it is done.

Appendix: demo accounts

This appendix lists the demo accounts and the demo content that the demo setup creates. On the kit they are in the apps **Kolibri Demo** and **Nextcloud Demo** of every IDEA box. Use them to follow the tutorial.

The demo school in Kolibri Demo is **IDEA Demo School**. The same usernames work in Kolibri Demo and in Nextcloud Demo.

Passwords:

- In Kolibri, learners have no password. They sign in with only their username.
- Every other demo password is the account name followed by `911!`, for example `teacher1911!`.

Username	Password	App	Role	Used in
admin	admin911!	Kolibri Demo	Kolibri admin of IDEA Demo School	3.2, 3.3
teacher1	teacher1911!	Kolibri Demo	Coach of class Grade 5	3, 11.7
teacher2	teacher2911!	Kolibri Demo	Coach of class Grade 5	3
learner1	none (username only)	Kolibri Demo	Learner in Grade 5	4
learner2	none (username only)	Kolibri Demo	Learner in Grade 5	4
learner3	none (username only)	Kolibri Demo	Learner in Grade 5	3.3
admin	admin911!	Nextcloud Demo	Nextcloud administrator	
teacher1	teacher1911!	Nextcloud Demo	Teacher (group teachers)	5, 8
teacher2	teacher2911!	Nextcloud Demo	Teacher (group teachers)	5.2
learner1	learner1911!	Nextcloud Demo	Learner (group learners)	6, 8.2
learner2	learner2911!	Nextcloud Demo	Learner (group learners)	6.1
learner3	learner3911!	Nextcloud Demo	Learner (group learners)	

Demo content:

- Kolibri: the school **IDEA Demo School** with the class **Grade 5**.
- Kolibri Demo: the three channels from the IDEA-DEMO disk, **Maths**, **Grade 5 Science** and **Early Reading** (videos, pictures and documents; no exercises).
- Kolibri Demo: the **Fractions lesson** that you make yourself in chapter 3. (The **Fractions quiz** needs exercises: see 3.5.)
- Nextcloud Demo: the folders **Worksheets** and **Teacher Notes** for `teacher1` and `teacher2`. You make the **Drop Zone** folder and the document **Lesson plan** yourself in chapter 5.
- Wikipedia (Demo): Wikipedia in simple English.

The school's own apps (not demo) start almost empty: **Kolibri** has only `admin` (the school creates its teachers and learners), **Nextcloud** has `admin`, `teacher1` and `learner1`. The passwords follow the same rule. Learners need a password in Nextcloud, not in Kolibri.

The pictures on the IDEA-DEMO disk are from Zimbabwe for now. The file `CREDITS.txt` on the disk gives the source and licence of every file.